Organic Chemistry Part 1

CHAPTER 1

Classification and Nomenclature of Organic Compounds 1.1–1.38

1 Classification and Nomenclature of Organic Compounds

1.1 INTRODUCTION

Organic chemistry deals with the study of carbon compounds with a few exceptions (e.g., CO_2 and carbonate salts). Inorganic chemistry deals with the study of all other compounds.

Carbon has the unique property of catenation due to which it forms covalent bonds with other carbon atoms. It forms covalent bonds with other elements also such as H, O, N, S, P, and halogens. These resulting compounds are studied in organic chemistry. Organic compounds are vital for understanding life on earth and include complex molecules such as DNA and proteins which constitute essential compounds of our blood, muscle, and skin. Moreover, these compounds are important constituents of fabrics, polymers, fuels, dyes, and medicines.

The term 'organic' dates back to the year 1780, when all those compounds which were obtained from plants or animals, that is, from living organisms, were called organic compounds. The study of the compounds obtained from mineral sources, e.g., common salt and sulphur, is termed inorganic chemistry.

Swedish chemist Berzelius (1779–1846) proposed that a 'vital force' was responsible for the formation of organic compounds. The 'vital force' theory was rejected in 1828, when a German chemist, F. Wohler, synthesised an organic compound, urea, from an inorganic compound, ammonium cyanate.

$$\underset{\text{Ammonium cyanate}}{\overset{\oplus}{N}H_4\left[(N=C=O)^{\ominus}\right]} \xrightarrow{\Delta} \underset{\text{Urea}}{NH_2CONH_2}$$

It was only after Kolbe (1845) and Berthelot (1856), who synthesised acetic acid and methane, respectively, that the belief in the vital force theory was rejected and it was shown that organic compounds could be synthesised from inorganic sources in a laboratory.

1.2 PROPERTIES OF ORGANIC COMPOUNDS

In general, organic compounds

a. are much more in number than inorganic compounds. This is due to the catenation property of C atom. Carbon forms bonds with almost every other element (other than the noble gases), and bond to each other in single and multiple bonds, forming long chains as well as ring compounds. Moreover, C compounds exist in many isomers.

b. react more slowly and require high temperatures for reaction.

c. are less stable and sometimes decompose on heating to compounds of lower energy content.

d. undergo more complex reactions and produce more side reaction products.

e. are generally insoluble in water.

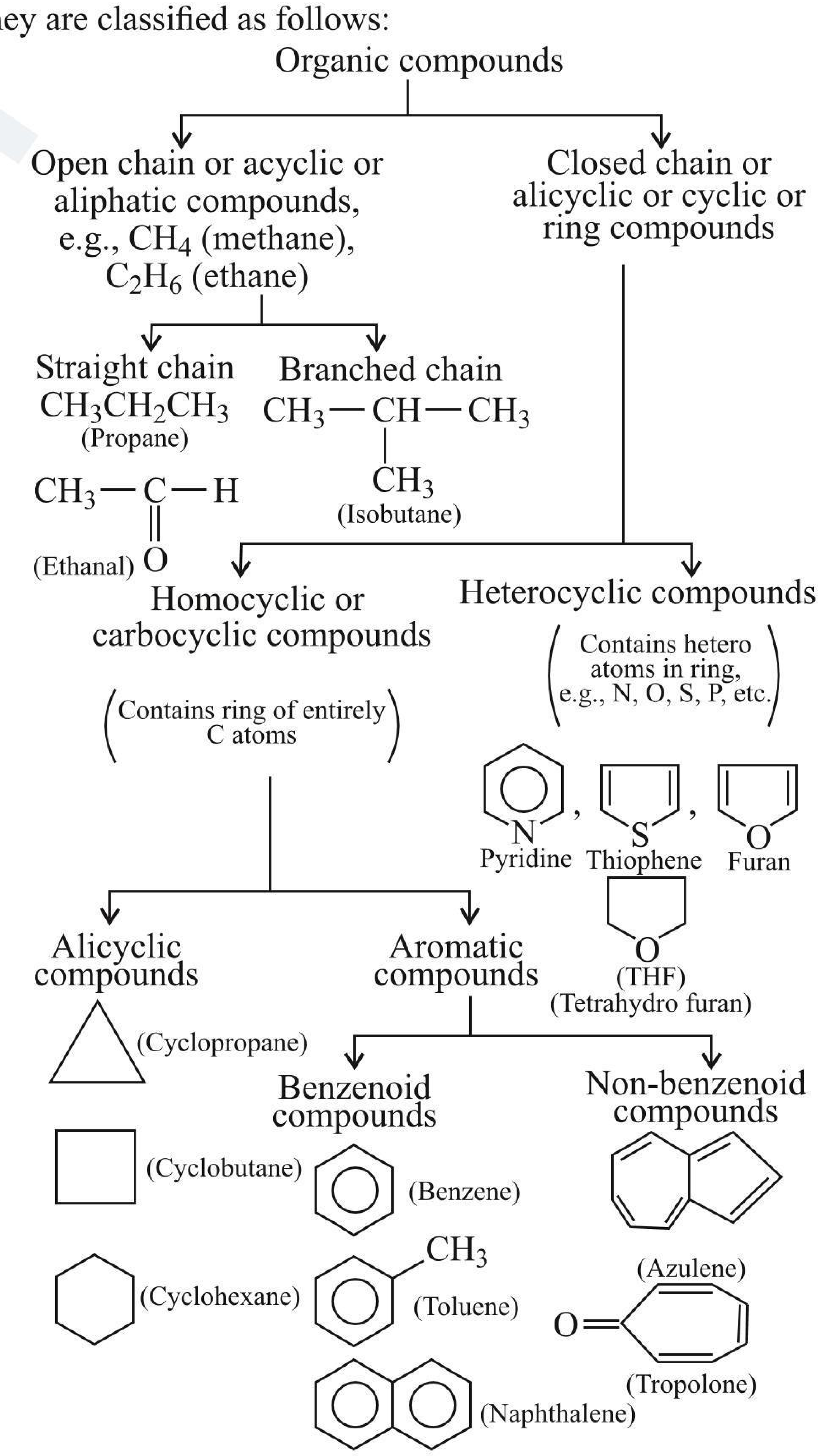
f. have generally lower melting and boiling points.

g. are classified into families of compounds such as carboxylic acids, which have similar reactive groups and chemical properties.

h. are mostly obtained from animal or vegetable kingdom in comparison to the inorganic compounds which have a mineral origin.

1.3 CLASSIFICATION OF ORGANIC COMPOUNDS

They are classified as follows:



The classification of organic compounds is basically based on the functional group. The chemical properties of a compound depends on the properties of the functional group present in it. The rest of the molecule simply affects the physical properties, e.g., melting point, boiling point, density, etc. and has very little effect on its chemical properties.

1.3.1 Homologous Series

Organic compounds containing one particular characteristic group or functional group constitute a homologous series, e.g., alkanes, alkenes, haloalkanes, alkanols, alkanals, alkanones, alkanoic acids, amines, etc.

1.3.2 Characteristics of a Homologous Series

i. Different compounds belonging to the same homologous series (containing the same functional group) are assigned by a general formula.

ii. Every member of the homologous series differs from its next higher or next lower homologue by a common difference, CH_2 (i.e., 14 amu).

iii. Different compounds belonging to the same homologous series can be prepared by a general method of preparation.

iv. Different compounds belonging to the same homologous series possess similar chemical properties (properties of the same functional group).

1.4 NOMENCLATURE OF ORGANIC COMPOUNDS

Broadly, two general ways of naming organic compounds have been used.

a. Trivial or common names: In early days whenever a new organic compound was discovered, chemists usually named it on the basis of history, e.g., wood spirit was so called as it was obtained by the destructive distillation of wood. This compound was later on named methyl alcohol (Greek: methu = wine and hule = wood). Similarly, acetic acid derives its name from vinegar (Latin: acetum = vinegar) of which it is the chief constituent. These names are called common or trivial names.

Table 1.1 Common names of organic compounds, their sources, and structures

S. No.	Common name	Source	Structure
1.	Formic acid	Formica (red ant)	HCOOH
2.	Acetic acid	Acetum (vinegar)	MeCOOH
3.	Propionic acid	Portopion (first fat)	$MeCH_2COOH$
4.	Butyric acid	Butyrum (butter)	$MeCH_2CH_2COOH$
5.	Valeric acid	Valerian (shrub)	$Me(CH_2)_3COOH$
6.	Caproic acid	Caper (goat)	$Me(CH_2)_4COOH$
7.	Urea	Urine	NH_2CONH_2
8.	Malic acid	Malum (apple)	CH_2COOH | $CH(OH)COOH$
9.	Methyl alcohol	Methu hule (Methu = wine, hule = wood)	MeOH

b. IUPAC names: The IUPAC nomenclature of organic compounds is a systematic method of naming organic compounds as recommended by the International Union of Pure and Applied Chemistry (IUPAC). This system uses substitutive nomenclature, which is based on the principal group and principal chain.

The IUPAC rules for the naming of alkanes form the basis of the substitutive nomenclature of most other compounds.

1.5 IUPAC RULES FOR NAMING ORGANIC COMPOUNDS

These rules define a few basic names that help in memorising the names of organic compounds. The base part of the name denotes the number of carbon atoms in the parent chain. The suffix indicates the type of functional group present in the parent chain. Other attached groups are called substituents.

1.5.1 IUPAC Rules for Saturated Hydrocarbons (Alkanes)

1. The longest possible chain (parent chain) is selected. The chain should be continuous.
2. C atoms which are not included in this chain are considered substituents (side chain).
3. In case of two equal chains having the same length, the one with the larger number of side chains or alkyl groups is selected.
4. Numbering of C atoms in the parent chain starts from that end where the substituent acquires the lowest position number or locant.
5. Lowest sum rule: In case of two or more substituents, numbering is done in such a way that the sum of the positional number of substituent or locant is the lowest.
6. Position and substituent name are separated with a dash (–).
7. In case of more than one substituent, they are prefixed by their respective locants in alphabetical order.

Table 1.2 Rules for IUPAC nomenclature

IUPAC name

Prefix

Primary prefix
* Distinguish from acyclic compounds

CH_2
CH_2 CH_2
CH_2—CH_2

Cyclo + Pent + ane
Primary prefix, Word root, Primary suffix

Secondary prefix
* Added before word root primary prefix (carboxylic compounds)
* Treated as substituents and not functional group
* Added in alphabetical order -F (Fluoro), -NO (Nitroso), etc.

Word root
* Basic unit of name
* Denotes the no. of C atoms in the longest chain such as C1 Meth, C2-Eth, C3-Prop C4-But, C5-Pent, etc.

Suffix

Primary suffix
* Added to word root
* Indicates whether the C chain is saturated or unsaturated

Saturated: (for single bond) **—ane**

Unsaturated: (for one double bond) **—ene**

(for two double bond) **—diene**

(for one triple bond) **—yne**

(for two triple bonds) **—diyne**

Secondary suffix
* Added to primary suffix
* Indicates the nature of the functional group, e.g., alcohol (— OH), aldehyde (— CHO),

CH_3CH_2OH
Eth + ane + ol
Word root, primary suffix, Secondary suffix

A complete IUPAC name consists of

Sec. prefix	+	Pr. prefix	+	Word root	+	Pr. suffix	+	Sec. suffix
4-Bromo		Cyclo		Hex		an(e)		1-ol

= 4-Bromo cyclo hexan-1-ol

Table 1.3 Names of some parent hydrocarbons

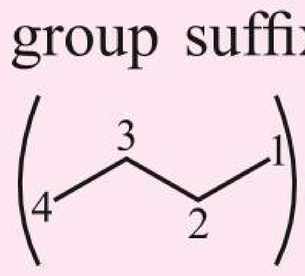
Name	Molecular formula	No. of C atoms	Name	Molecular formula	No. of C atoms
Methane	CH_4	1	Undecane	$C_{11}H_{24}$	11
Ethane	C_2H_6	2	Dodecane	$C_{12}H_{26}$	12
Propane	C_3H_8	3	Eicosane (Icosane)	$C_{20}H_{42}$	20
Butane	C_4H_{10}	4	Uncosane	$C_{21}H_{44}$	21
Pentane	C_5H_{12}	5	Docosane	$C_{22}H_{46}$	22
Hexane	C_6H_{14}	6	Triacontane	$C_{30}H_{62}$	30
Heptane	C_7H_{16}	7	Hentria contane	$C_{31}H_{64}$	31
Octane	C_8H_{18}	8	Dotriacontane	$C_{32}H_{66}$	32
Nonane	C_9H_{20}	9	Tetracontane	$C_{40}H_{82}$	40
Decane	$C_{10}H_{22}$	10	Pentacontane	$C_{50}H_{102}$	50
			Hexacontane	$C_{60}H_{122}$	60
			Heptacontane	$C_{70}H_{142}$	70
			Hectane	$C_{100}H_{202}$	100

Table 1.4 Some functional groups and classes of organic compounds

1. Alkanes

General formula: C_nH_{2n+2} IUPAC group suffix: -ane

Example $\left(H_3\overset{4}{C}\,\overset{3}{C}H_2\,\overset{2}{C}H_2\,\overset{1}{C}H_3\right)$ or $(Me(CH_2)_2Me)$



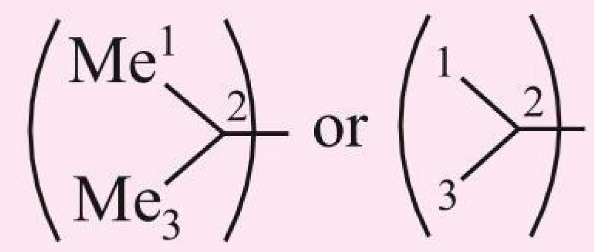
Butane (IUPAC name)

Some groups abbreviated in the book are as follows:

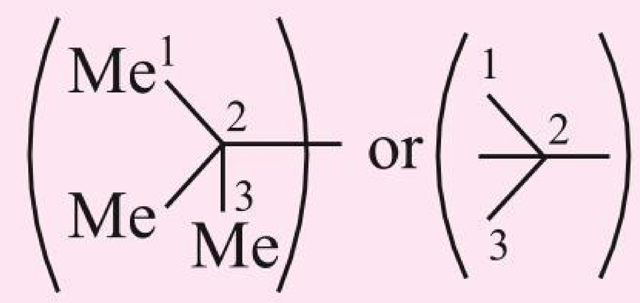
i. Methyl, CH_3 — (or) Me. **ii.** Ethyl, C_2H_5 — (or) Et.

iii. *n*-Propyl, *n*-C_3H_7 — (or) *n*-Pr. **iv.** *n*-Butyl, *n*-C_4H_9 — (or) *n*-Bu.

v. Isopropyl, *i*-Pr $\overset{3}{C}H_3-\overset{2}{C}H(\overset{1}{C}H_3)-$ or $\overset{3}{Me}-\overset{2}{C}H(\overset{1}{Me})-$ or

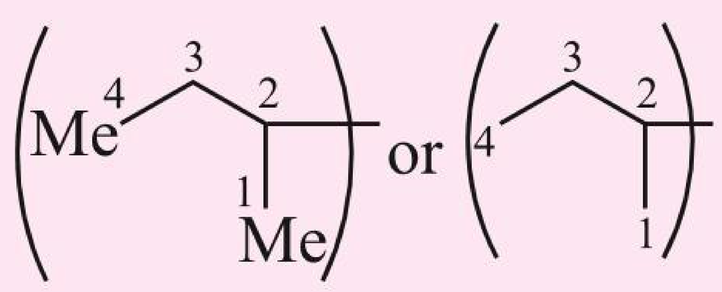


vi. *tert*-Butyl, *t*-butyl or *t*-Bu, or $CH_3-C(CH_3)_2-$ or

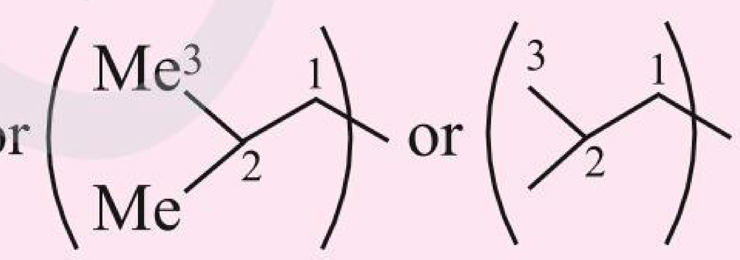


vii. Secondary butyl or *sec*-butyl or *sec*-Bu, or $\overset{4}{C}H_3-\overset{3}{C}H_2-\overset{2}{C}H(\overset{1}{C}H_3)-$ or $\overset{4}{Me}-\overset{3}{C}H_2-\overset{2}{C}H(\overset{1}{Me})-$

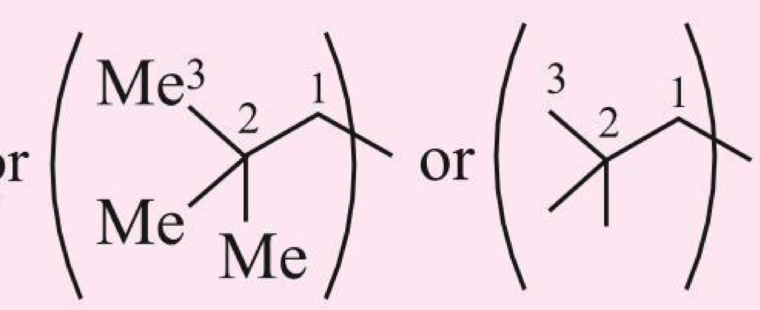
or



viii. Isobutyl, *i*-Bu, or $\overset{3}{C}H_3-\overset{2}{C}H(CH_3)-\overset{1}{C}H_2-$ or $\overset{3}{Me}-\overset{2}{C}H(Me)-\overset{1}{C}H_2-$ or



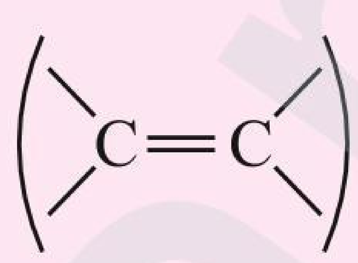
ix. Neopentyl or $\overset{3}{C}H_3-\underset{2}{C}(CH_3)_2-\overset{1}{C}H_2-$ or $\overset{3}{Me}-\underset{2}{C}(Me)_2-\overset{1}{C}H_2-$ or



2. Alkenes

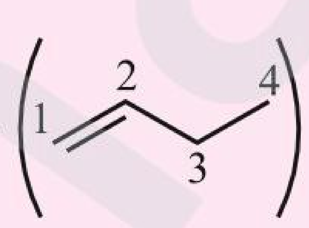
General formula: $C_nH_{2n} \Rightarrow \left(\begin{array}{l}\text{suffix} = -\text{ene} \\ \text{Alkane} \xrightarrow[+\text{ ene}]{-\text{ ane}} \text{alkene}\end{array}\right)$

Functional group structure: $\left(>C=C<\right)$

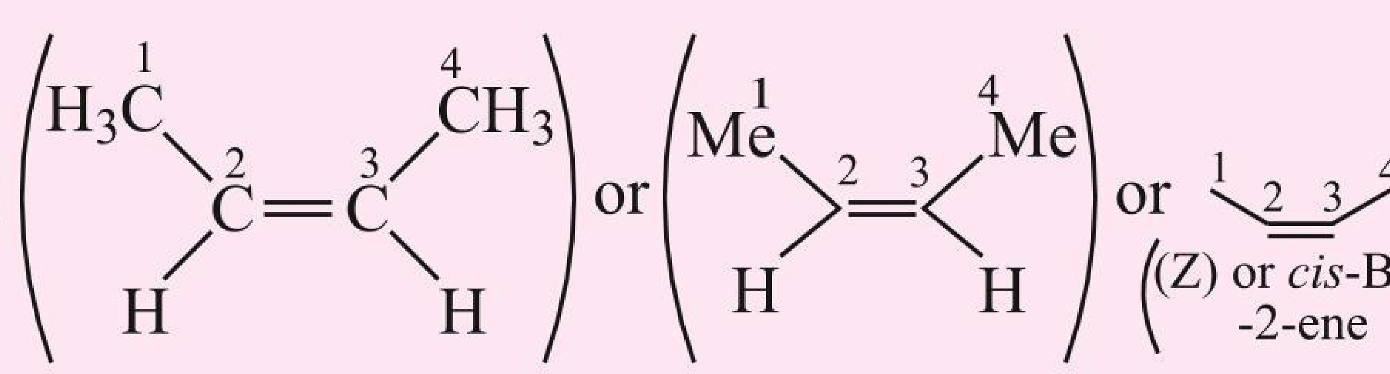


Example

i. $(\overset{1}{C}H_2=\overset{2}{C}H\overset{3}{C}H_2\overset{4}{C}H_3)$ or

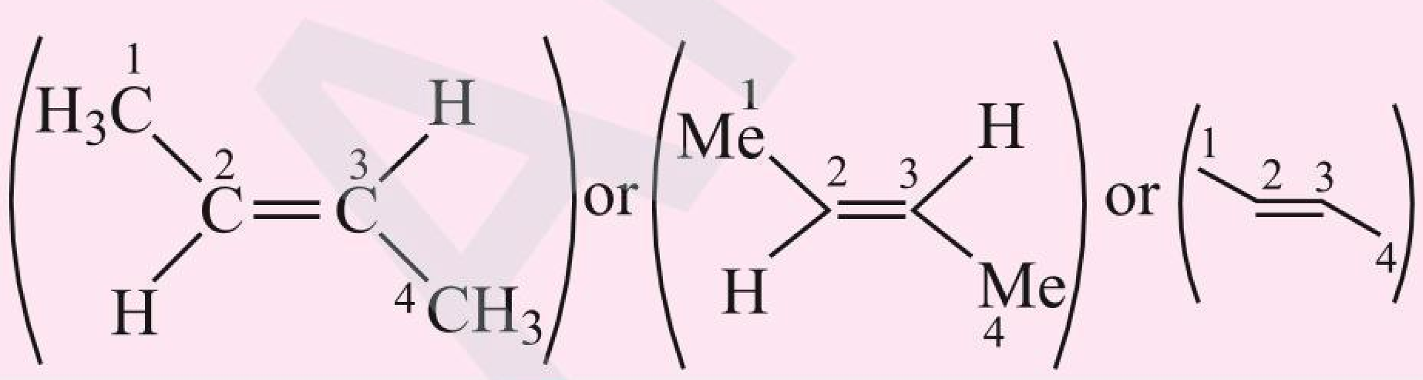


But-1-ene (IUPAC) **ii.**



(Z) or *cis*-But-2-ene

iii.



(E) or *trans*-But-2-ene

3. Alkadienes

General formula: C_nH_{2n-2} $\left(\begin{array}{l}\text{Suffix} = \text{diene} \\ \text{Alkane} \xrightarrow[+\text{ diene}]{-\text{ ne}} \text{Alkadiene}\end{array}\right)$

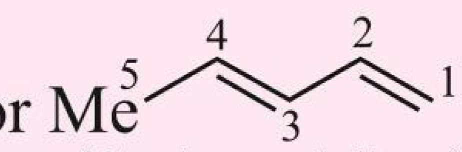
Functional group structures: These are of three types:

$>C=C=C<$ Cumulative diene

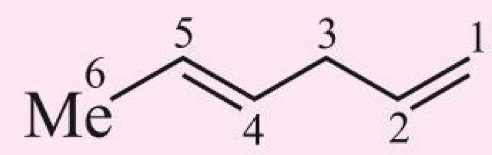
$>C=CH-HC=C<$ Conjugated dienes

$>C=CH-CH_2-CH=C<$ Isolated diene

Example

i. $\overset{5}{C}H_3-\overset{4}{C}H=\overset{3}{C}H-\overset{2}{C}H=\overset{1}{C}H_2$ or 
(Penta-1,3-diene) (Conjugated diene)

ii. $\overset{3}{C}H_2=\overset{2}{C}=\overset{1}{C}H_2$ or $\overset{3}{=}C^{2}\overset{1}{=}$ (Propa-1,2-diene) (Cumulative diene)

iii. $\overset{6}{C}H_3-\overset{5}{C}H=\overset{4}{C}H-\overset{3}{C}H_2-\overset{2}{C}H=\overset{1}{C}H_2$ or  (Penta-1,4-diene) (Isolated diene)

4. Alkyne

General formula: C_nH_{2n-2} (Suffix = -yne; Alkane $\xrightarrow[+\text{ yne}]{-\text{ ane}}$ Alkyne)

Functional group structure: ($-C\equiv C-$)

Example

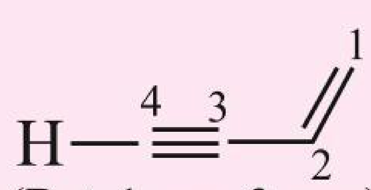
i. $HC\equiv CH$ or $H-\equiv-H$ Ethyne (IUPAC) Acetylene (Common name)

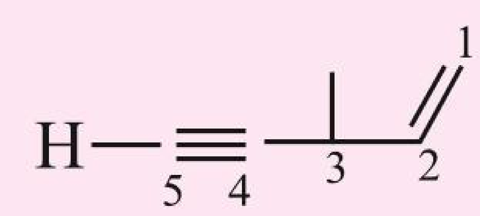
ii. $\overset{4}{C}H_3-\overset{3}{C}\equiv\overset{2}{C}-\overset{1}{C}H_3$ or $\overset{4}{Me}-\overset{3}{\equiv}\overset{2}{}-\overset{1}{Me}$ or $\overset{4}{-}\overset{3}{\equiv}\overset{2}{-}\overset{1}{}$ (But-2-yne)

5. Alkenyne

General formula: C_nH_{2n-4} Suffix = – ene-yne Functional group structure: $\left(-C\equiv C-C=C\langle\right)$

Example

i. $H\overset{4}{C}\equiv\overset{3}{C}-\overset{2}{C}H=\overset{1}{C}H_2$ or 
(But-1-ene-3-yne)

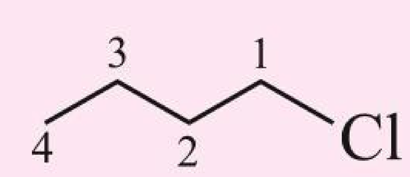
ii. $H-\underset{5}{}\equiv\underset{4}{C}-\underset{3}{\overset{CH_3}{C}}-\underset{2}{C}H=\underset{1}{C}H_2$ or 
(3-Methyl but-1-ene-4-yne)

Rule: When both (C═C) and (C≡C) are present, then the numbering of the C chain is done from the (C═C) bond side.

6. Halides

General formula: $C_n H_{2n+1}X$ (X = F, Cl, Br, I) (RX) Suffix = halo

Functional group structure: — X

Example $\overset{4}{C}H_3\overset{3}{C}H_2\overset{2}{C}H_2\overset{1}{C}H_2Cl$ or  or 
(1-Chloro butane)

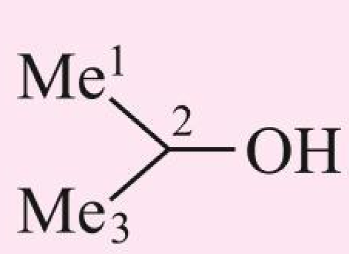
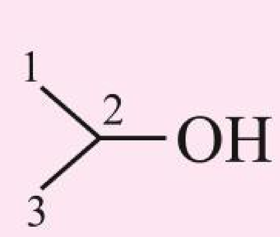
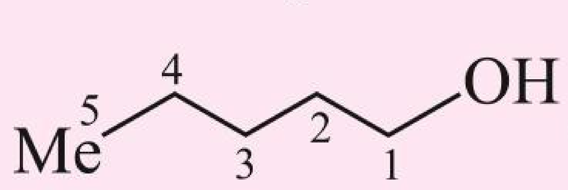
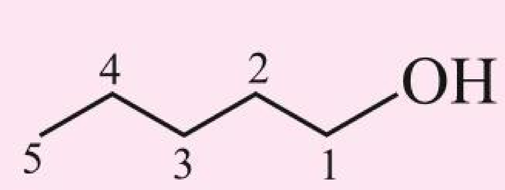
7. Alcohols

General formula: $C_nH_{2n+1}OH$ (R—OH)

IUPAC suffix: -ol, Common name = Alcohol IUPAC prefix: -hydroxy.

$$\text{Alkan}\text{e} \xrightarrow[+\text{ ol}]{-\text{ e}} \text{Alkanol}$$

Functional group structure: —OH

Example	IUPAC name	Common name
i. CH_3OH	Methanol	Methyl alcohol or Carbinol or Zerone
ii. CH_3CH_2OH or EtOH or 	Ethanol	Ethyl alcohol
iii. $\underset{3}{C}H_3-\underset{2}{\overset{\overset{1}{C}H_3}{C}}H-OH$ or  or 	Propan-2-ol	Isopropyl alcohol
iv. $CH_3(CH_2)_4OH$ or  or 	Pentan-1-ol	*n*-Amyl alcohol

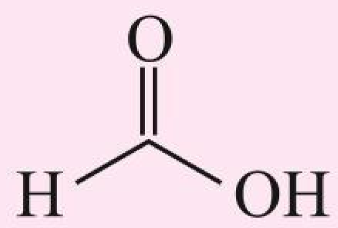
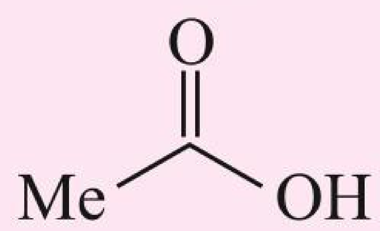
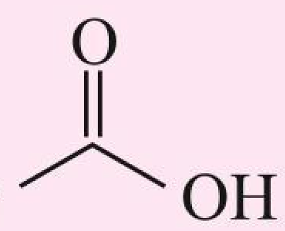
8. Carboxylic acids

General formula: $C_nH_{2n+1}COOH$ (R—COOH)

IUPAC suffix: -oic acid (IUPAC) Common name = Acid IUPAC prefix: -carboxy

$$\text{Alkane} \xrightarrow[\text{+ oic acid}]{-\text{e}} \text{Alkanoic acid}$$

Functional group structure: (—COOH)

Example	IUPAC name	Common name
i. HCOOH or 	Methanoic acid	Formic acid
ii. CH_3COOH or  or 	Ethanoic acid	Acetic acid

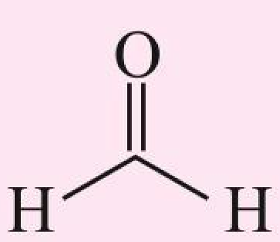
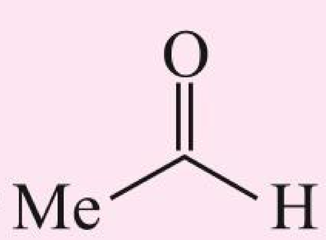
9. Aldehydes

General formula: C_nH_{2n+1}. CHO (R—CHO)

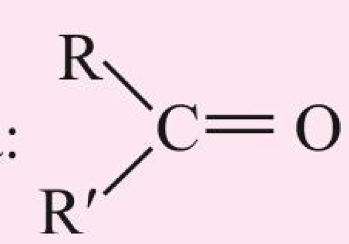
IUPAC suffix: -al, Common name = Aldehyde IUPAC prefix: formyl or oxo-

$$\text{Alkane} \xrightarrow[\text{+ al}]{-\text{e}} \text{Alkanal}$$

Functional group structure: (—CHO) or $\left(-\overset{\text{H}}{\overset{|}{\text{C}}}=\text{O}\right)$

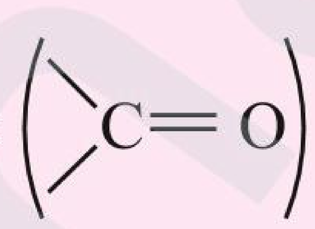
Example	**IUPAC name**	**Common name** (derived from acid)
i. HCHO or 	Methanal	Formaldehyde
ii. CH_3CHO or 	Ethanal	Acetaldehyde

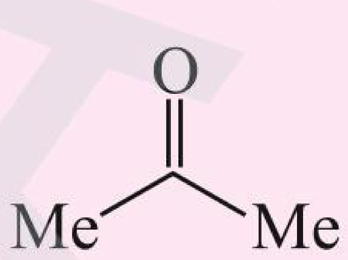
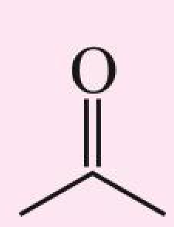
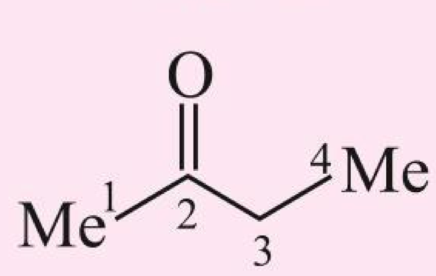
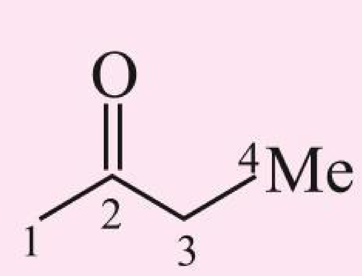
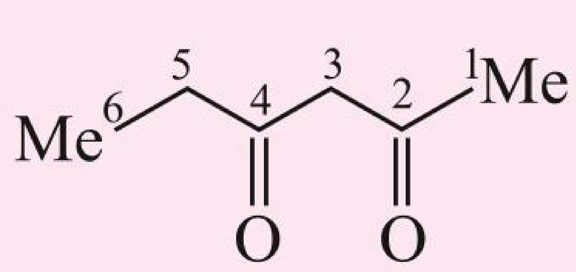
10. Ketones

General formula: 

IUPAC suffix: -one, Common name: Ketone IUPAC prefix: -oxo,

$$\text{Alkane} \xrightarrow[\text{+ one}]{-\text{e}} \text{Alkanone}$$

Functional group structure: 

Example	**IUPAC name**	**Common name**
i. CH_3COCH_3 or  or 	Propan-2-one	Acetone
ii. $CH_3COCH_2CH_3$ or  or 	Butan-2-one	Ethyl methyl ketone
iii. $\overset{6}{C}H_3\overset{5}{C}H_2\overset{4}{C}O\overset{3}{C}H_2\overset{2}{C}O\overset{1}{C}H_3$ or 	Hexane-2,4-dione (e is not dropped)	—

11. Nitriles

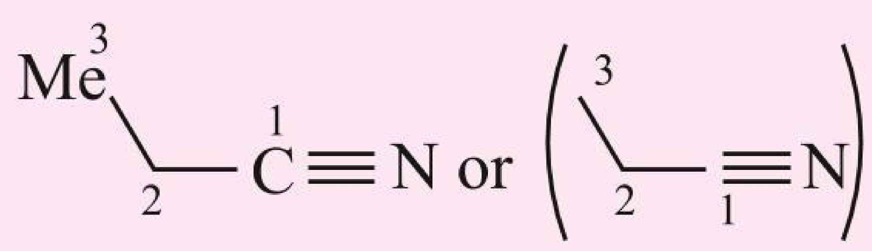
General formula: $C_nH_{2n+1}CN$ (R—C≡N)

IUPAC suffix: nitrile, Common name: Cyanide

IUPAC prefix: cyano

$$\text{Alkane} \xrightarrow[\text{+ nitrile}]{} \text{Alkanenitrile (e is not dropped)}$$

Functional group structure: (—C≡N)

Example	IUPAC name	Common name
i. $\overset{2}{C}H_3\overset{1}{C}N$ or $\overset{2}{Me}—\overset{1}{C}≡N$ or $(\overset{2}{—}\overset{1}{C}≡N)$	Ethane nitrile	Methyl cyanide or Acetonitrile
ii. $\overset{3}{C}H_3\overset{2}{C}H_2\overset{1}{C}N$ or 	Propane nitrile	Ethyl cyanide or Propionitrile

12. Ethers

General formula: (R—O—R′)

IUPAC suffix: oxy, Common name: (Ether)

IUPAC prefix: alkoxy (smaller chain) alkane (larger chain)

Functional group structure: (R—O— R′)

Example	IUPAC name	Common name
i. $CH_3—O—CH_3$ or $(CH_3)_2O$ or Me_2O	Methoxy methane	Dimethyl ether
ii. $CH_3CH_2—O—CH_3$ or Et—O—Me	Methoxy ethane	Ethylmethyl ether
iii. $CH_3CH_2—O—CH_2CH_3$ or Et—O—Et or Et_2O	Ethoxy ethane	Diethyl ether

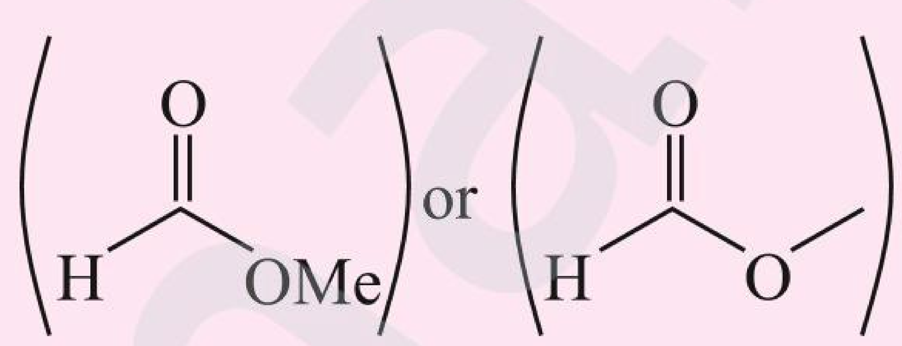
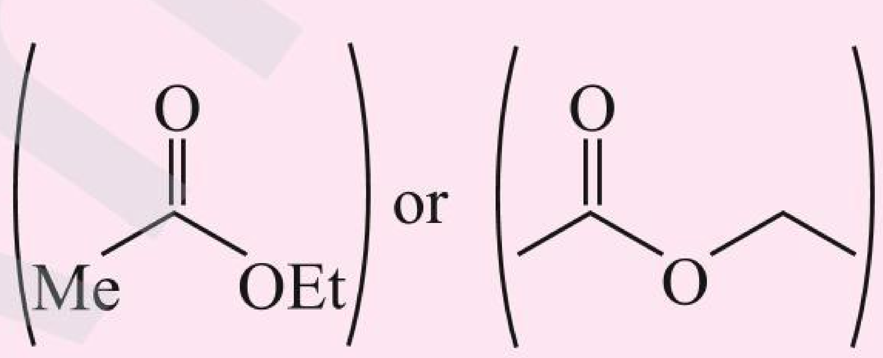
13. Esters

General formula: $\left(R—\overset{\displaystyle O}{\overset{\|}{C}}—OR'\right)$

IUPAC suffix: -oate

IUPAC prefix: alkoxy carbonyl

Functional group structure: (— COOR) or $\left(—\overset{\displaystyle O}{\overset{\|}{C}}—OR\right)$

Example	IUPAC name	Common name (derived from acid)
i. $HCOOCH_3$ or 	Methyl methanoate	Methyl formate
ii. $CH_3COOCH_2CH_3$ or 	Ethyl ethanoate	Ethyl acetate

14. Acyl halides

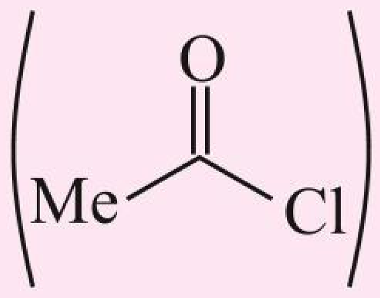
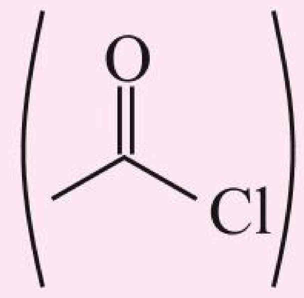
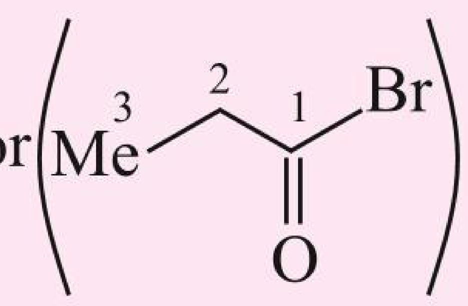
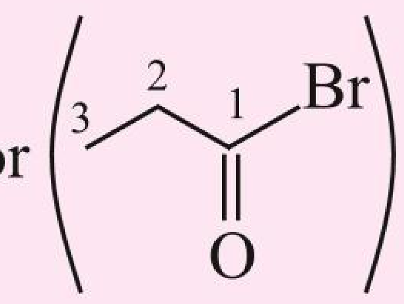
General formula: $\left(R—\overset{\displaystyle O}{\overset{\|}{C}}—X\right)$ (X = F, Cl, Br, I).

IUPAC suffix: -oyl halide

IUPAC prefix: halocarbonyl

$$\text{Alkane} \xrightarrow[\text{oyl halide}]{-\,e} \text{Alkanoyl halide}$$

Functional group structure: $(-\overset{\overset{\large O}{\|}}{C}-X)$

Example	**IUPAC name**	**Common name** (derived from acid)
i. HCOCl	Methanoyl chloride	Formyl chloride
ii. CH_3COCl or  or 	Ethanoyl chloride	Acetyl chloride
iii. $\overset{3}{C}H_3\overset{2}{C}H_2\overset{1}{C}OBr$ or (EtCOBr) or  or 	Propanoyl bromide	Propionyl bromide

15. Amides

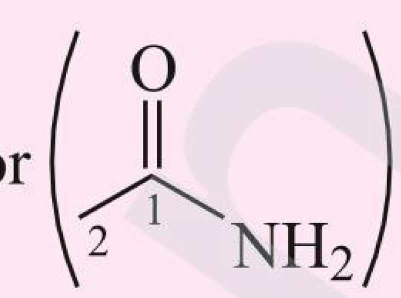
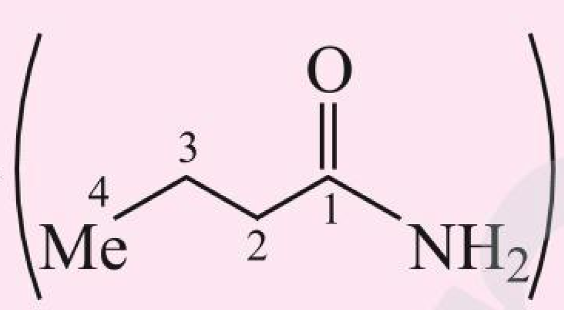
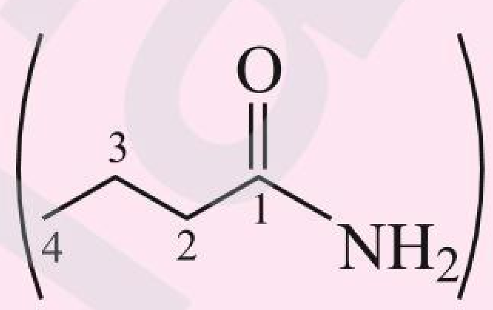
General formula: $(R-\overset{\overset{\large O}{\|}}{C}-NH_2)$

IUPAC suffix: amide

IUPAC prefix: carbamoyl

$$\text{Alkane} \xrightarrow[+\text{ amide}]{-\text{ e}} \text{Alkanamide}$$

Functional group structure: $\left(-\overset{\overset{\large O}{\|}}{C}-NH_2,\ -\overset{\overset{\large O}{\|}}{C}-NHR,\ -\overset{\overset{\large O}{\|}}{C}-N\begin{smallmatrix}R\\R\end{smallmatrix}\right)$

Example	**IUPAC name**	**Common name**
i. $HCONH_2$ or 	Methanamide	Formamide
ii. CH_3CONH_2 or $\overset{2}{Me}\overset{1}{C}ONH_2$ or 	Ethanamide	Acetamide
iii. $\overset{4}{C}H_3(\overset{3}{C}H_2)_2\overset{1}{C}ONH_2$ or $PrCONH_2$ or  or 	Butanamide	Butyramide

16. Anhydrides

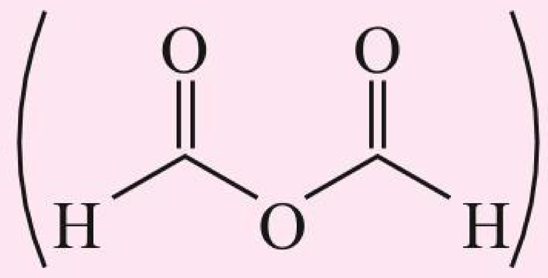
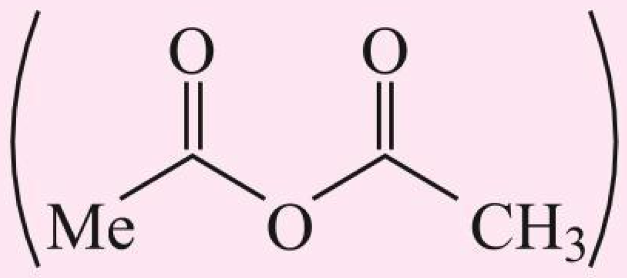
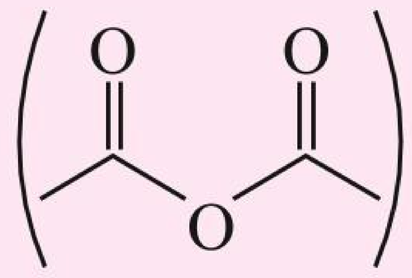
General formula: $\left(R-\overset{\overset{\large O}{\|}}{C}-O-\overset{\overset{\large O}{\|}}{C}-R\right)$

IUPAC suffix: -oic anhydride

IUPAC prefix: -acetoxy or acytyloxy or $\left(-O-\overset{\overset{\large O}{\|}}{C}-Me\right)$

$$\text{Alkanoic acid} \xrightarrow[+\text{ anhydride}]{-\text{ acid}} \text{Alkanoic anhydride}$$

Functional group structure: $(-COOCO-)$ or $\left(-\overset{\overset{\large O}{\|}}{C}-O-\overset{\overset{\large O}{\|}}{C}-\right)$

Example	IUPAC name	Common name (derived from acid)
i. $HCOOCOH$ or $(HCO)_2O$ or 	Methanoic anhydride	Formic anhydride
ii. $CH_3COOCOH_3$ or $(CH_3CO)_2O$ or Ac_2O or  or 	Ethanoic anhydride	Acetic anhydride

Note: $\left(CH_3 - \overset{\overset{\displaystyle O}{\|}}{C} -\right)$ group is abbreviated as Ac- and (CH_3COOH) group as AcOH.

17. Acid hydrazides

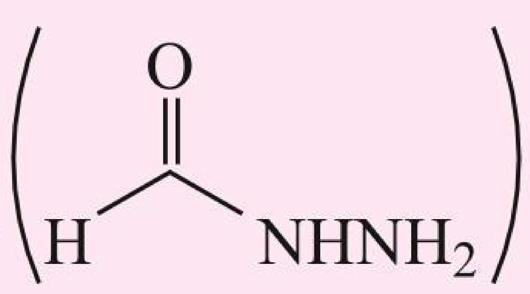
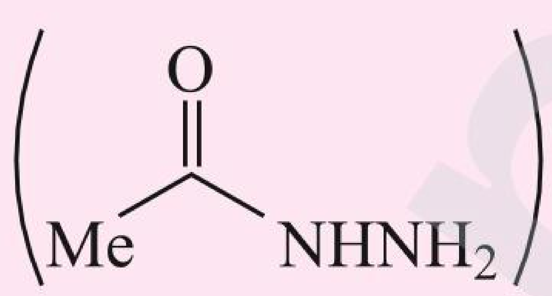
General formula: $\left(R - \overset{\overset{\displaystyle O}{\|}}{C} - NH - NH_2\right)$

IUPAC suffix: -hydrazide

IUPAC prefix: alkano

$$\text{Alkanoic acid} \xrightarrow[+\text{ hydrazide}]{-\text{ ic acid}} \text{Alkanohydrazide}$$

Functional group structure: ($—CONHNH_2$) or $\left(- \overset{\overset{\displaystyle O}{\|}}{C} - NH - NH_2\right)$

Example	IUPAC name	Common name (derived from acid)
i. $HCONHNH_2$ or 	Methanohydrazide	Formyl hydrazide
ii. $CH_3CONHNH_2$ or 	Ethanohydrazide	Acetyl hydrazide

18. Acid azides

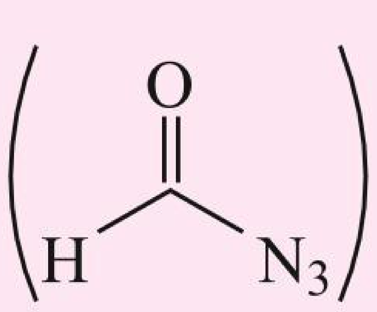
General formula: $\left(R - \overset{\overset{\displaystyle O}{\|}}{C} - N_3\right)$ or $\left[R - \overset{\overset{\displaystyle O}{\|}}{C} - N = \overset{\oplus}{N} = \overset{\ominus}{N}\right]$

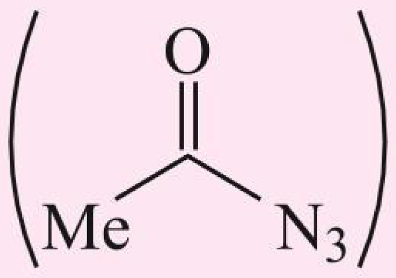
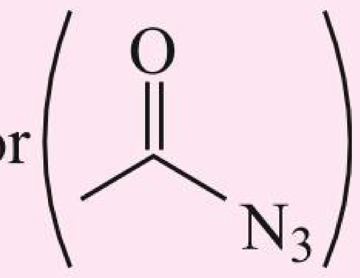
IUPAC suffix: azide

IUPAC prefix: alkano

$$\text{Alkanoic acid} \xrightarrow[+\text{ azide}]{-\text{ ic acid}} \text{Alkanoazide}$$

Functional group structure: ($—CON_3$) or $\left[- \overset{\overset{\displaystyle O}{\|}}{C} - N = \overset{\oplus}{N} = \overset{\ominus}{N}\right]$

Example	IUPAC name	Common name (derived from acid)
i. $HCON_3$ or 	Methano azide	Formyl azide

Example	IUPAC name	Common name
ii. CH_3CON_3 or  or 	Ethanoazide	Acetyl azide

19. Hydroxamic acids

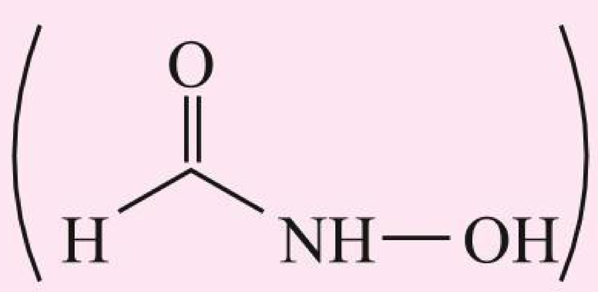
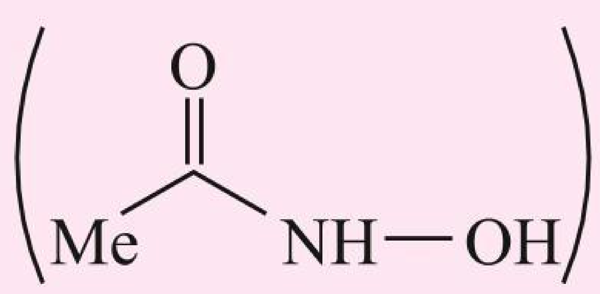
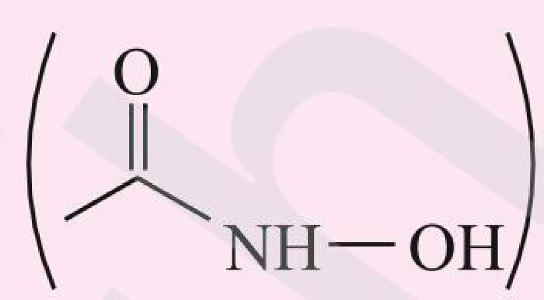
General formula: $\left(R—\overset{\overset{\displaystyle O}{||}}{C}—NH—OH\right)$

IUPAC suffix: -hydroxamic acid

IUPAC prefix: alkano

$$\text{Alkanoic acid} \xrightarrow[\text{+ hydroxamic acid}]{\text{– ic acid}} \text{Alkanohydroxyamic acid}$$

Functional group structure: (— CONHOH) or $\left(—\overset{\overset{\displaystyle O}{||}}{C}—NH—OH\right)$

Example	**IUPAC name**	**Common name** (derived from acid)
i. HCONH.OH or 	Methano hydroxamic acid	Formyl hydroxamic acid
ii. $CH_3CONHOH$ or  or 	Ethanohydrox amic acid	Acetyl hydroxamic acid

20. Thio acids

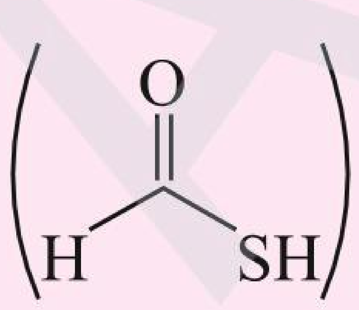
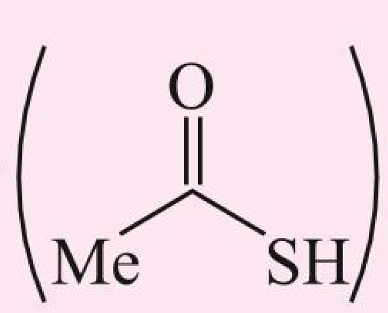
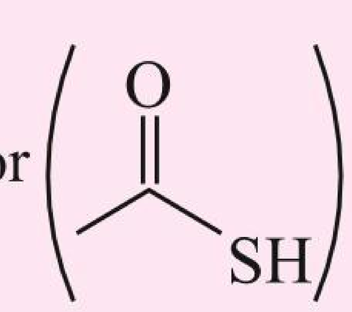
General formula: $\left(R—\overset{\overset{\displaystyle O}{||}}{C}—SH\right)$

IUPAC suffix: thioic acid

IUPAC prefix: carbothioic acid

$$\text{Alkane} \xrightarrow[\text{+ thioic acid}]{} \text{Alkanethioic acid}$$

Functional group structure: (— COSH) or $\left(—\overset{\overset{\displaystyle O}{||}}{C}—SH\right)$

Example	**IUPAC name**	**Common name** (derived from acid)
i. HCOSH or 	Methane thioic acid or Carbothioic acid	Thio formic acid
ii. CH_3COSH or  or 	Ethane thioic acid or Methane carbothioic acid	Thio acetic acid

21. Thio esters

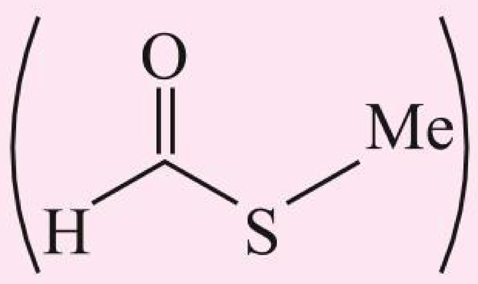
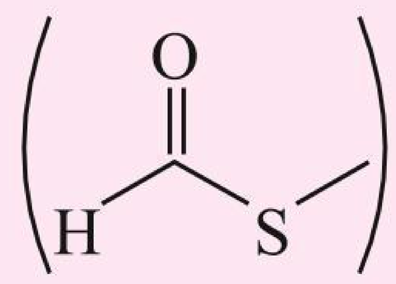
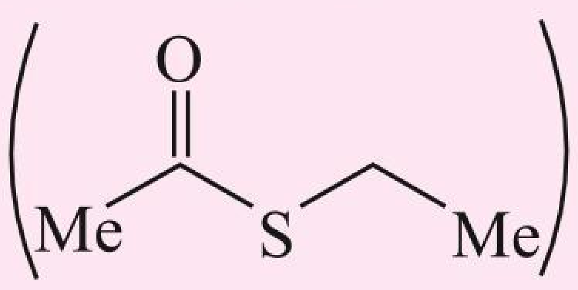
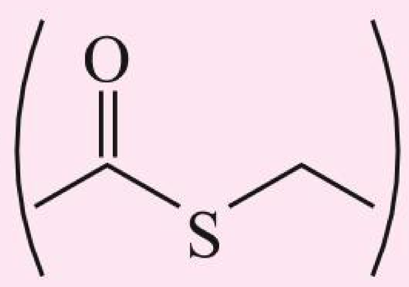
General formula: $\left(R-\overset{\overset{\Large O}{||}}{C}-S-R\right)$

IUPAC suffix: thioate

IUPAC prefix: —

oate of ester $\longrightarrow$ thioate

Functional group structure: (— COSR) or $\left(-\overset{\overset{\Large O}{||}}{C}-S-R\right)$

Example	IUPAC name	Common name (derived from acid)
i. $HCOSCH_3$ or  or 	Methyl methane thioate	Methy thio formate
ii. $CH_3COSCH_2CH_3$ or MeCOSEt or  or 	Ethyl ethane thioate	Ethyl thioacetate

22. Thioalcohols or Thiols or Mercaptans

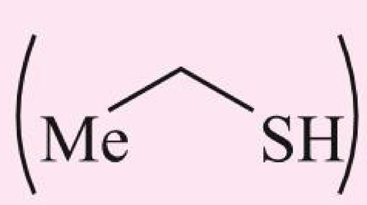
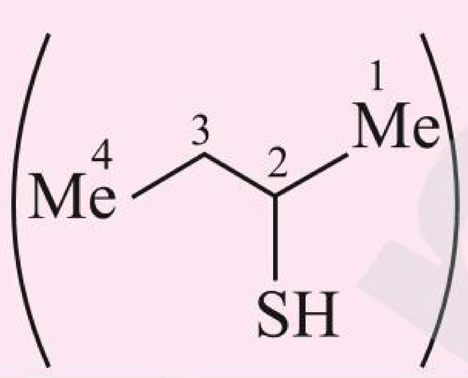
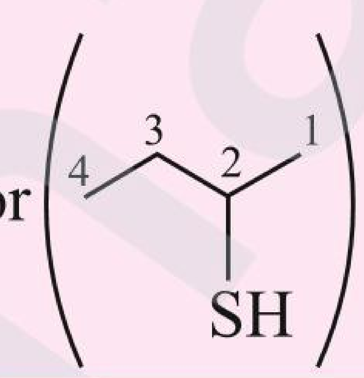
General formula: (RSH)

IUPAC suffix: thiol

IUPAC prefix: mercapto

Alkan(e) $\xrightarrow[+\text{ thiol}]{-\text{ e}}$ Alkan thiol

Functional group structure: (— SH)

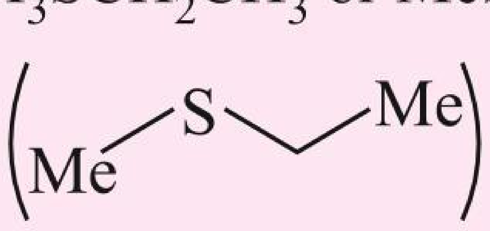
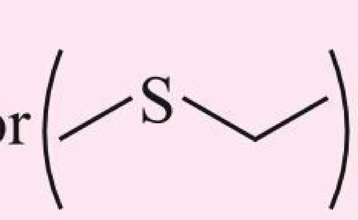
Example	IUPAC name	Common name (derived from acid)
i. CH_3SH or MeSH	Methanthiol	Methyl thioalcohol or Methylmercaptan
ii. CH_3CH_2SH or EtSH  or 	Ethan-1-thiol	Ethyl thioalcohol or Ethyl mercaptan
iii. $\overset{4}{C}H_3\overset{3}{C}H_2\overset{2}{C}H(SH)-\overset{1}{C}H_3$ or $\left(\overset{4}{Et}-\overset{2}{C}H(SH)-\overset{1}{Me}\right)$ or  or 	Butan-2-thiol	

23. Thioethers

General formula: (R — S — R)

IUPAC suffix: thioether

IUPAC prefix: alkyl

Example	IUPAC name	Common name (derived from acid)
i. CH_3SCH_3 or MeSMe or Me_2S	Methyl thio ether or (Methyl thio) Methane	Dimethyl sulphide
ii. $CH_3SCH_2CH_3$ or MeSEt or  or 	Ethyl methyl thio ether or (Methyl thio) ethane	Ethyl methyl sulphide

24. Amines

General formula: RNH_2 (1°) $RNHR$ (2°) $R-\underset{R}{\underset{|}{N}}-R$ (3°) $[R_4N^{\oplus}]X^{\ominus}$ (4° salt)

IUPAC suffix: -amine IUPAC prefix: -amino-

$$\text{Alkan(e)} \xrightarrow[+\text{ amine}]{-\text{ e}} \text{Alkanamine}$$

Functional group structure: $-NH_2$, (1°), $>NH$ (2°), $>N-$ (3°)

Example	IUPAC name	Common name
i. $CH_3CH_2NH_2$ or $EtNH_2$ or (Me⌃NH₂) or (⌃NH₂)	Ethan amine	Ethyl amine
ii. $\overset{3}{C}H_3-\underset{NH_2}{\underset{\mid}{\overset{2}{C}H}}-\overset{1}{C}H_3$ or ((Me)(Me)>—NH_2)	Propan-2-amine	Isopropylamine
iii. $CH_3-NH-CH_2CH_3$ or ((Me)(Et)>NH)	*N*-Methyl ethan-amine	Ethyl methyl amine

(If more than one alkyl group is attached to N atom, smaller alkyl group is written as N-alkyl.)

Example	IUPAC name	Common name
iv. $C_2H_5-\underset{C_2H_5}{\underset{\mid}{N}}-\overset{1}{C}H_2\overset{2}{C}H_2\overset{3}{C}H_2\overset{4}{C}H_3$	*N,N*-Diethyl butan-1-amine	*N,N*-Diethyl butylamine
v. $H_2N-\overset{1}{C}H_2-\overset{2}{C}H_2-NH_2$	Ethane-1,2-diamine (e is retained.)	—

25. Nitro compounds

General formula: (RNO_2) or $\left(R-\underset{O^{\ominus}}{\underset{|}{\overset{\oplus}{N}}}=O\right)$ or $\left(R-\underset{O}{\underset{\downarrow}{N}}=O\right)$

IUPAC suffix: alkane IUPAC prefix: nitro

They are named as nitro derivatives of the corresponding alkanes.

Functional group structure: $(-NO_2)$

Example	IUPAC name
i. CH_3NO_2	Nitro methane
ii. $CH_3(CH_2)_3NO_2$	1-Nitro butane
iii. $CH_3-CH(NO_2)CH_3$	2-Nitro propane

26. Alkyl nitrites

General formula: $(R-O-N=O)$

IUPAC suffix: nitrite IUPAC prefix: alkyl

Functional group structure: $(-O-N=O)$

Example	IUPAC name
i. CH_3-ONO or $Me-O-N=O$	Methyl nitrite
ii. $CH_3CH_2CH_2ONO$ or $Pr-O-N=O$	Propyl nitrite

27. Alkyl isocyanides or Isonitriles

General formula: $\left(R - \overset{}{\underset{\oplus}{N}} \equiv \underset{\ominus:}{C}\right)$ or $(R - N \rightleftarrows C)$

According to IUPAC recommendation, as a substituent — NC is termed carbylamino. Thus, CH_3NC is carbylamino methane and so on. But this name is not in use.

For naming isocyanides, *iso* is prefixed to the name of the corresponding cyano/nitrile compound. In another mode, the suffix carbylamine is added to the name of the alkyl group.

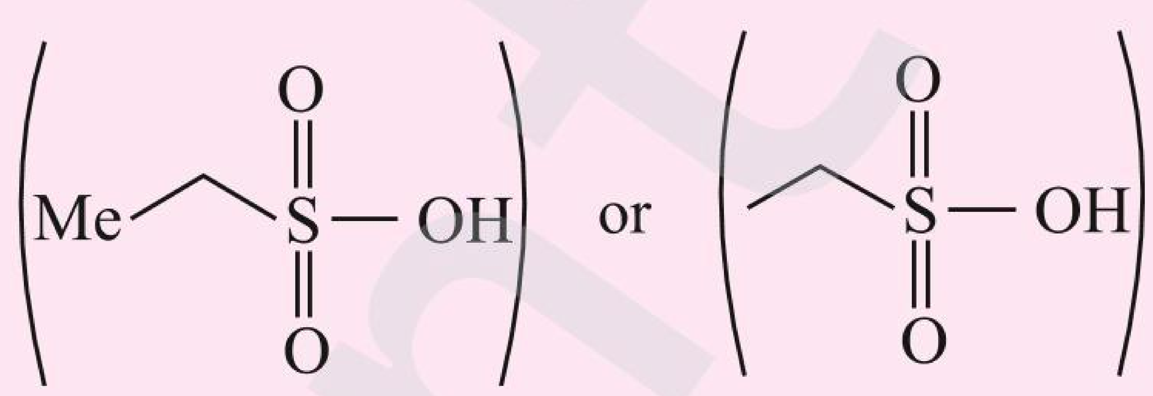
Example	Common name
i. CH_3NC	Methyl isocyanide or Acetoisonitrile or Methyl carbylamine
ii. C_2H_5NC	Ethyl isocyanide or Propionoiso nitrile or Ethyl carbylamine
iii. C_3H_7NC	Propyl isocyanide or Butyroisonitrile or Propyl carbylamine

28. Sulphonic acids

General formula: $(R - SO_3H)$

IUPAC suffix: sulphonic acid   IUPAC prefix: sulpho

Functional group structure: $(R - SO_3H)$ or $\left(-\overset{\overset{\displaystyle O}{\|}}{\underset{\underset{\displaystyle O}{\|}}{S}} - OH\right)$

Example	IUPAC name
i. CH_3SO_3H or $Me - \overset{\overset{\displaystyle O}{\|}}{\underset{\underset{\displaystyle O}{\|}}{S}} - OH$	Methyl sulphonic acid
ii. $CH_3CH_2SO_3H$ or 	Ethyl sulphonic acid

29. Imines

General formula: $RCH{=}NH$

IUPAC suffix: imine   IUPAC prefix: alkanal

$$\text{Alkane} \xrightarrow[+\text{ al imine}]{-\text{ e}} \text{Alkanal imine}$$

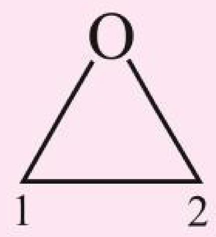
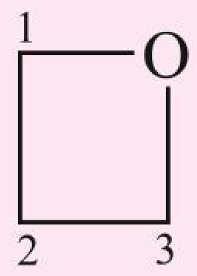
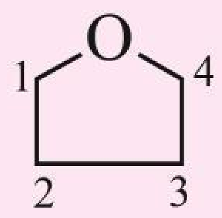
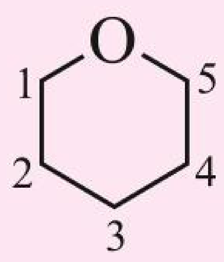
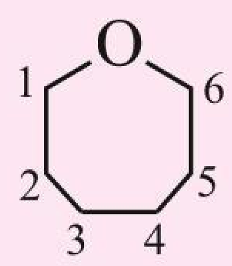
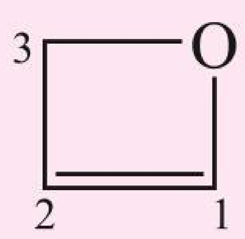
Functional group structure: $(-CH{=}NH)$

Example	IUPAC name	Common name
i. $HCH{=}NH$	Methanalimine	Formaldimine
ii. $CH_3CH{=}NH$	Ethanalimine	Acetaldimine
iii. $CH_3CH_2CH{=}NH$	Propanalimine	Propionaldimine

30. Cyclic ethers

General formula: O atom in ring

IUPAC suffix: alkane   IUPAC prefix: epoxy

Example	IUPAC name	Common name
i. 	Oxirane or 1,2-epoxy ethane	Ethylene oxide
ii. 	Oxetane or 1,3-epoxy propane	Propylene oxide
iii. 	Oxolane or 1,4-epoxy butane	Butylene oxide
iv. 	Oxane or 1,5-epoxy pentane	Pentylene oxide
v.  Hexylene oxide	Oxepane or 1,6-epoxyhexane	Hexylene oxide
vi. 	Oxetene or 1,3-epoxyprop-1-ene	

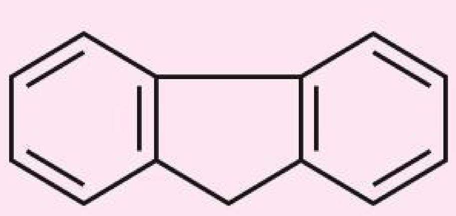
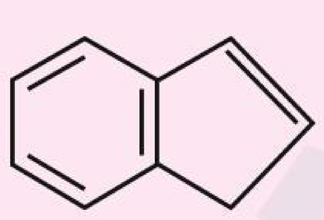
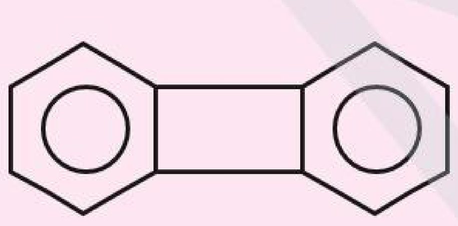
31. **Alicyclic compounds:** See Section 6.7, 'Cycloalkanes'.

32. **Bicyclo, tricyclo, spiro, and polycyclic compounds:** See Section 6.7.2.

33. **Aromatic compounds:** See Section 1.11.

34. **Fused benzene rings**

Fused benzene ring is usually indicated by the prefix benzo, without writing *ortho* before benzo (*ortho* is understood because fusion can occur with these two positions).

Example	IUPAC name	Common name
i. 	Benzocyclohexene (Double bond is common to both the rings.)	Tetralin or 1,2,3,4-Tetrahydro napthalene
ii. 	Dibenzocyclopentadiene (Double bond is common in cyclopentane ring from both sides of the benzene ring.)	Fluorene
iii. 	Benzocyclopentadiene	Indene
iv. 	Dibenzocyclobutadiene	Biphenylene

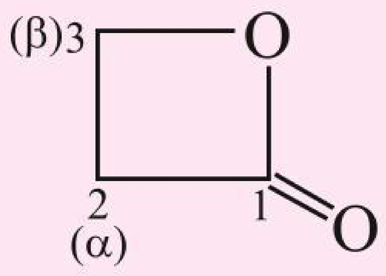
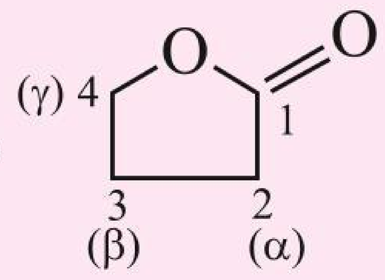
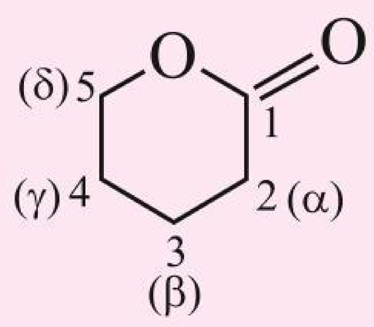
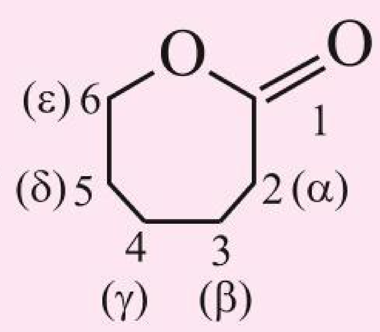
35. **Lactones or cyclic esters**

General formula: $\left(-\overset{\displaystyle O}{\underset{}{\overset{\|}{C}}}-O- \right)$ group in ring

IUPAC suffix: olide

IUPAC prefix: alkane

$$\text{Alkan(e oic acid)} \xrightarrow[+\text{ olide}]{-\text{ e oic acid}} \text{Alkanolide}$$

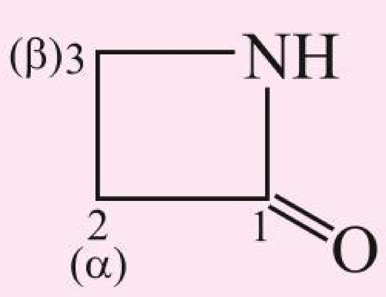
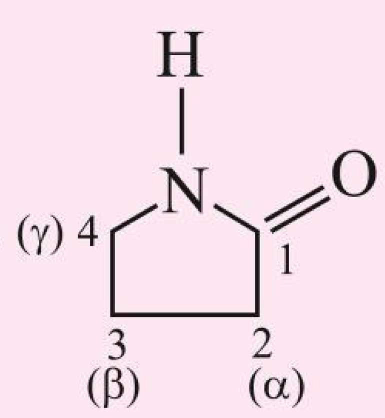
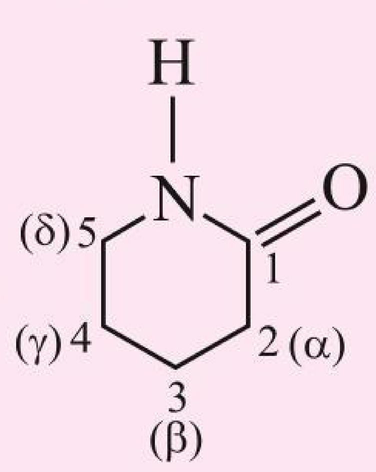
Example	IUPAC name	Common name
i. 	3-Propanolide	β-Propiolactone
ii. 	4-Butanolide	γ-Butyrolactone
iii. 	5-Pentanolide	δ-Valerolactone
iv. 	6-Hexanolide	ε-Caprolactone

36. Lactam or cyclic imides

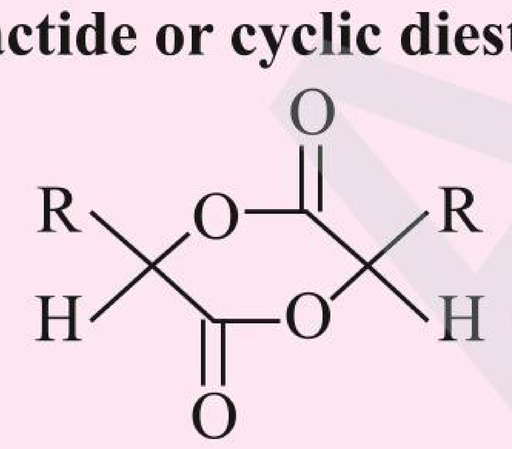
General formula: $\left(-\overset{\overset{\large O}{\|}}{C}-NH- \right)$ group in ring

IUPAC suffix: oimide IUPAC prefix: —

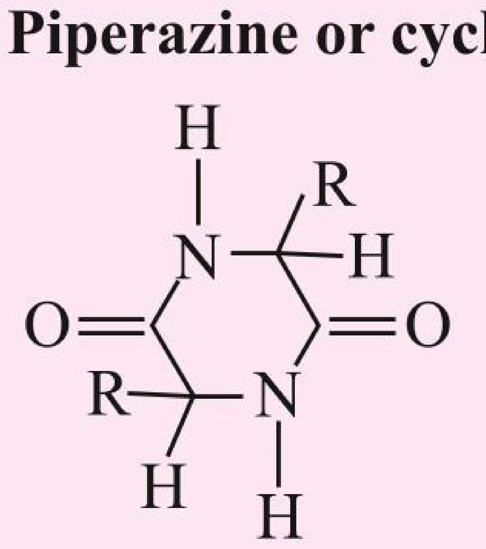
Alkan e oic acid $\xrightarrow[+\text{ oimide}]{-\text{ e oic acid}}$ Alkanoimide

Example	IUPAC name	Common name
i. 	3-Propanoimide	β-Propialactam
ii. 	4-Butanoimide	γ-Butyrolactam
iii. 	5-Pentanoimide	δ-Valerolactam

37. Lactide or cyclic diester



38. Piperazine or cyclic diimide



39. Derivatives of carbonic acid (H_2CO_3) or

$\left(HO-\overset{\overset{\large O}{\|}}{C}-OH \right)$

i. Acid chloride of carbonic acid:

$\left(Cl-\overset{\overset{\large O}{\|}}{C}-Cl \right)$ (Phosgene)

ii. Amide of carbonic acid:

a. $H_2N-\overset{\overset{\large O}{\|}}{C}-NH_2$ (urea) or (carbamide)

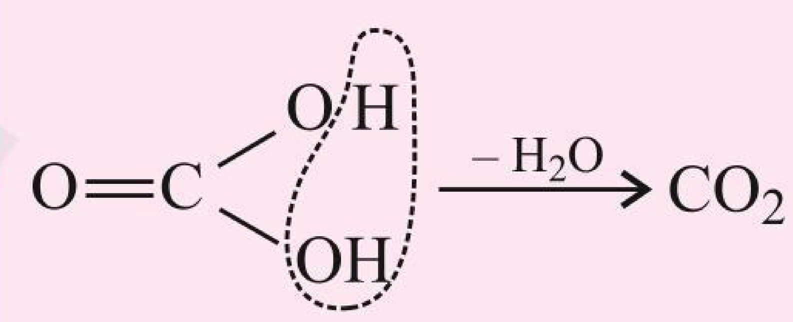
b. $H_2N-\overset{\overset{\large O}{\|}}{C}-OH$ (carbamic acid)

c. $H_2N-\overset{\overset{\large O}{\|}}{C}-NH-\overset{\overset{\large O}{\|}}{C}-NH_2$ (carbamoyl urea)

iii. Ester of carbonic acid:

$EtO-\overset{\overset{\large O}{\|}}{C}-OEt$ (Diethyl carbonate)

iv. Anhydride of carbonic acid:



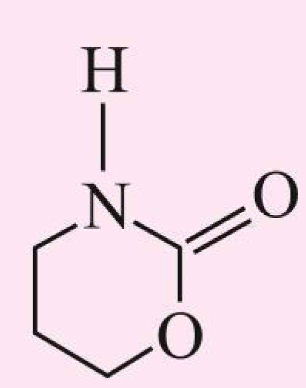
v. Ethyl carbamate: $H_2N-\overset{\overset{\large O}{\|}}{C}-OEt$

vi. Ethyl chlorocarbonate: $Cl-\overset{\overset{\large O}{\|}}{C}-OEt$

vii. Cyanamide: $H_2N-C\equiv N$

40. *N*-Alkyl carbamate or Urethane:

$\left(R-NH-\overset{\overset{\large O}{\|}}{C}-OR' \right)$

Example	IUPAC name	Common name (lactam)
i. $CH_3-NH-\overset{\overset{\large O}{\|}}{C}-OEt$	Ethyl-*N*-methyl carbamate	Urethane
 (Cyclic urethane)		

41. Some common names

i. Carbinol or Zerone (CH_3OH)

ii. Carbolic acid (PhOH, phenol)

iii. Prestone ($HOCH_2—CH_2OH$, Glycol)

iv. Lewisite gas ($ClCH{=}CHAsCl_2$)

v. BAL gas (British Anti-Lewisite):

$$\left(HSCH_2—\underset{\displaystyle SH}{\underset{|}{CH}}—CH_2OH\right)$$

vi. Mustard gas $\left(\underset{\displaystyle H_2C}{\overset{\displaystyle CH_2Cl}{|}}—S—\underset{\displaystyle CH_2}{\overset{\displaystyle CH_2Cl}{|}}\right)$

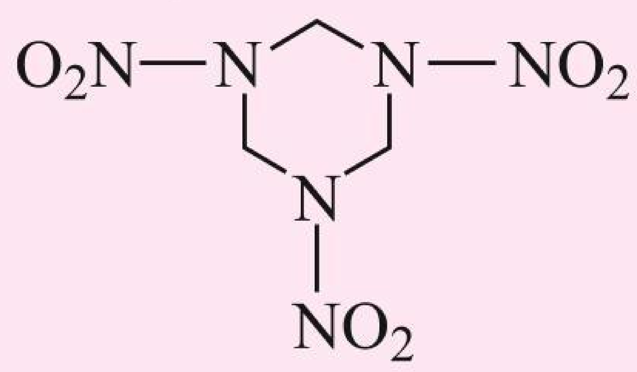
vii. Phosgene gas ($COCl_2$)

viii. Tear gas or chloropicrin ($Cl_3C—NO_2$)

ix. Weeping gas (Phenacyl chloride, $PhCOCH_2Cl$)

x. Laughing gas (N_2O)

xi. RDX (Research and developed explosive) or cyclonite.



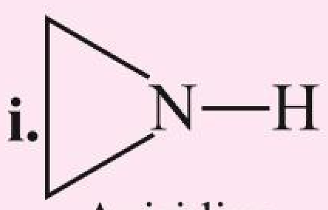
xii. Nerve gas: It is an organic derivative of H_3PO_4, i.e., alkaline phosphate, fluorophosphates, thiophosphates. It inhibits the enzyme choline stearase and causes acetylcholine poisoning and cessation of nerve transmission.

Antidote: Atropine sulphate and paralidoxine iodide

xiii. Sarin gas (MPI): Methyl fluoro phosphonium iodide

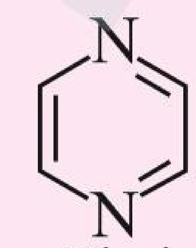
$$\left(Me—\overset{\oplus}{P}H_2 \atop \quad / \atop F\right)I^{\ominus}$$

42. Heterocyclic compounds of nitrogen

i. 
Aziridine

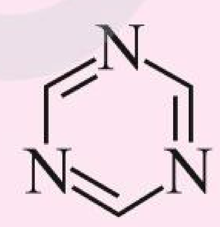
ii. 
o-Diazine
(pyridazine)

iii. 
m-Diazine
pyrimidine

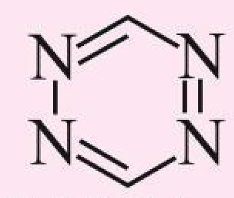
iv. 
p-Diazine
(pyrazine)

v. 
1,2,3-Triazine

vi. 
1,2,4-Triazine

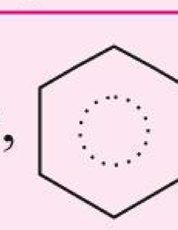
vii. 
1,3,5-Triazine

viii. 
1,2,3,4-Tetrazine

ix. 
1,2,4,5-Tetrazine

x. 
Purine

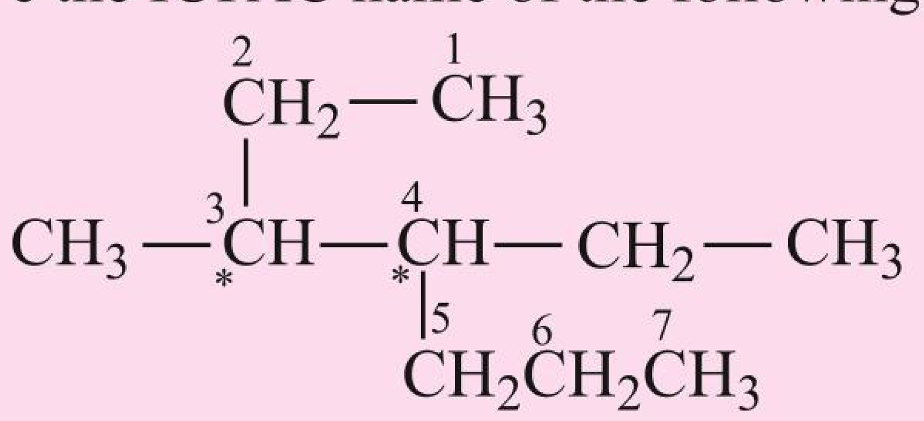
Table 1.5 Summary of some functional groups and classes of organic compounds

Class of compounds	Functional group structure	IUPAC group prefix	IUPAC group suffix	Example
Alkanes	—	—	-ane	Butane, $CH_3(CH_2)_2CH_3$
Alkenes	>C=C<	—	-ene	But-l-ene, $CH_2 = CHCH_2CH_3$
Alkynes	–C≡C–	—	-yne	But-l-yne, $CH{\equiv}CCH_2CH_3$
Arenes	—	—	—	Benzene, 
Halides	–X (X = F, Cl, Br, I)	halo-	—	1-Bromobutane, $CH_3(CH_2)_2CH_2Br$
Alcohols	–OH	hydroxy-	-ol	Butan-2-ol, $CH_3CH_2CHOHCH_3$
Aldehydes	–CHO	formyl, or oxo	-al	Butanal, $CH_3(CH_2)_2CHO$
Ketones	>C=O	oxo-	-one	Butan-2-one, $CH_3CH_2COCH_3$
Nitriles	–C≡N	cyano	nitrile	Pentanenitrile, $CH_3CH_2CH_2CH_2CN$
Ethers	–R–O–R–	alkoxy-	—	Ethoxyethane, $CH_3CH_2OCH_2CH_3$
Carboxylic acids	–COOH	carboxy	-oic acid	Butanoic acid, $CH_3(CH_2)_2CO_2H$
Carboxylate ions	$–COO^-–$	—	-oate	Sodium butanoate, $CH_3(CH_2)_2\,CO_2^{\ominus}Na^{\oplus}$
Esters	–COOR	alkoxycarbonyl	-oate	Methyl propanoate, $CH_3CH_2COOCH_3$
Acyl halides	–COX (X = F, Cl, Br, I)	halocarbonyl	-oyl halide	Butanoyl chloride, $CH_3(CH_2)_2COCl$
Amines	$–NH_2$, > NH, > N–	amino-	-amine	Butan-2-amine, $CH_3CHNH_2CH_2CH_3$

Amides	$-CONH_2$, $-CONHR$, $-CONR_2$	-carbamoyl	-amide	Butanamide, $CH_3(CH_2)_2CONH_2$
Nitro compounds	$-NO_2$	nitro	—	1-Nitrobutane, $CH_3(CH_2)_3NO_2$
Sulphonic acids	$-SO_3H$	sulpho	sulphonic acid	Methylsulphonic acid CH_3SO_3H

ILLUSTRATION 1.1

Give the IUPAC name of the following compound:



Sol. Sum of locants (*) = 3 + 4 = 7

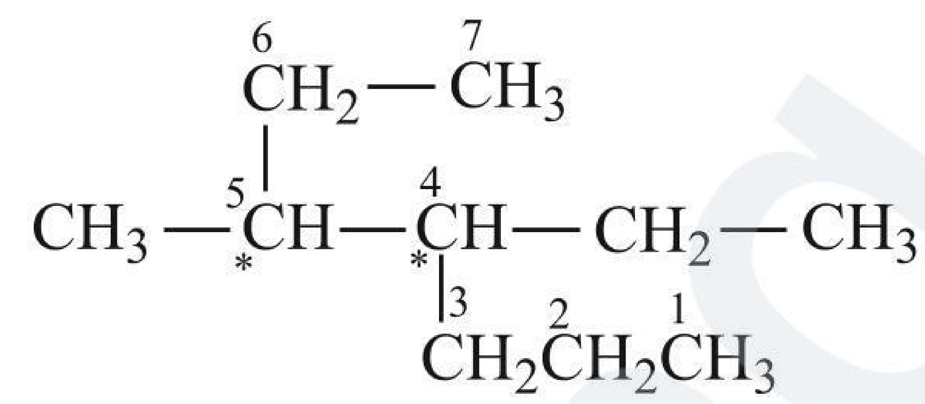
Therefore, the numbering is done according to this rule:

—CH_2—CH_3 is another substituent (Ethyl).

—CH_3 is the substituent (Methyl).

So the IUPAC name of this compound is 4-Ethyl-3 methyl-heptane, i.e., locant-substituent name parent alkane (word root).

('e' of ethyl comes before 'm' of methyl)



Sum of locants (*) = 4 + 5 = 9

(The sum of the locant is larger than in the previous case.)

Therefore, the numbering is not done according to this rule:

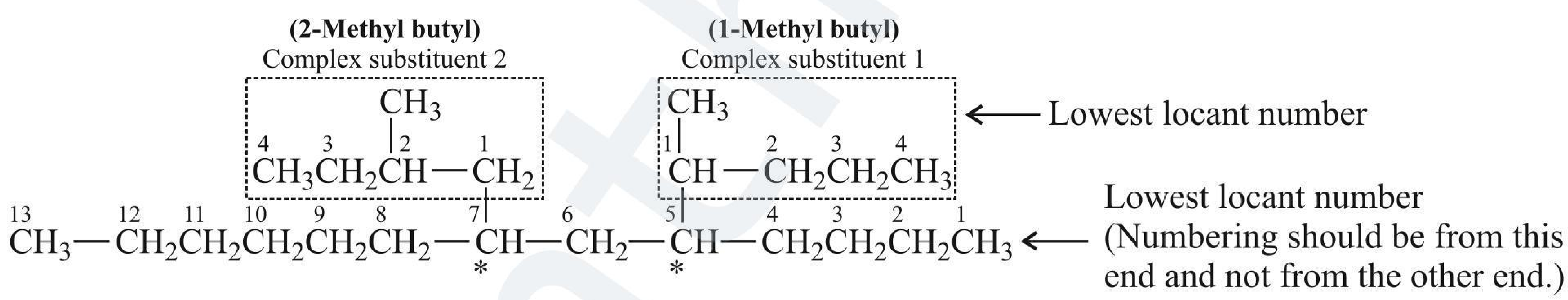
—CH_2—CH_3 is another substituent (Ethyl).

—CH_3 is the substituent (Methyl).

ILLUSTRATION 1.2

Give the IUPAC name of the compounds if it contains complex substituents and other substituents.

Sol. In case of complex substituent and other substituents, the complex substituent begins with the first letter of its complete name. In case of two same complex substituents, one with the lowest positional number or locant is named first. This is called priority of citation.



The IUPAC name of this compound is

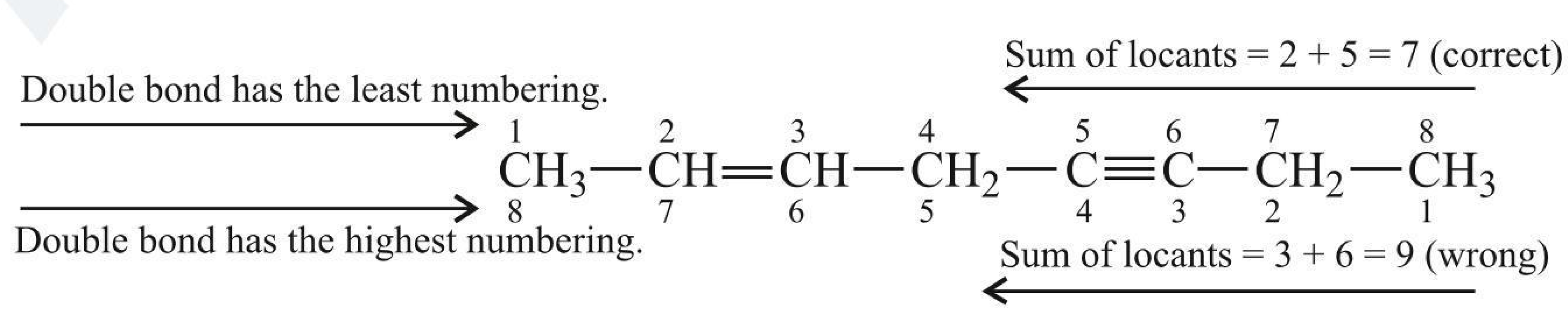
5-(1-Methyl butyl)-7-(2-methyl butyl) tridecane

Priority of citation (5 < 7, 1 < 2); Locant 1 comes before 2.

1.5.2 IUPAC Rules for Unsaturated Hydrocarbons (Alkenes and Alkynes)

1. All the rules of alkanes are applicable here also.
2. The parent or the longest chain is selected irrespective of the ═ or ≡ bonds.
3. The numbering is done from the end which is nearer to the ═ bond followed by the ≡ bond, and according to the lowest sum of locant rule.
4. The numbering or sum rule will follow the alphabetical order of the substituent.

For example

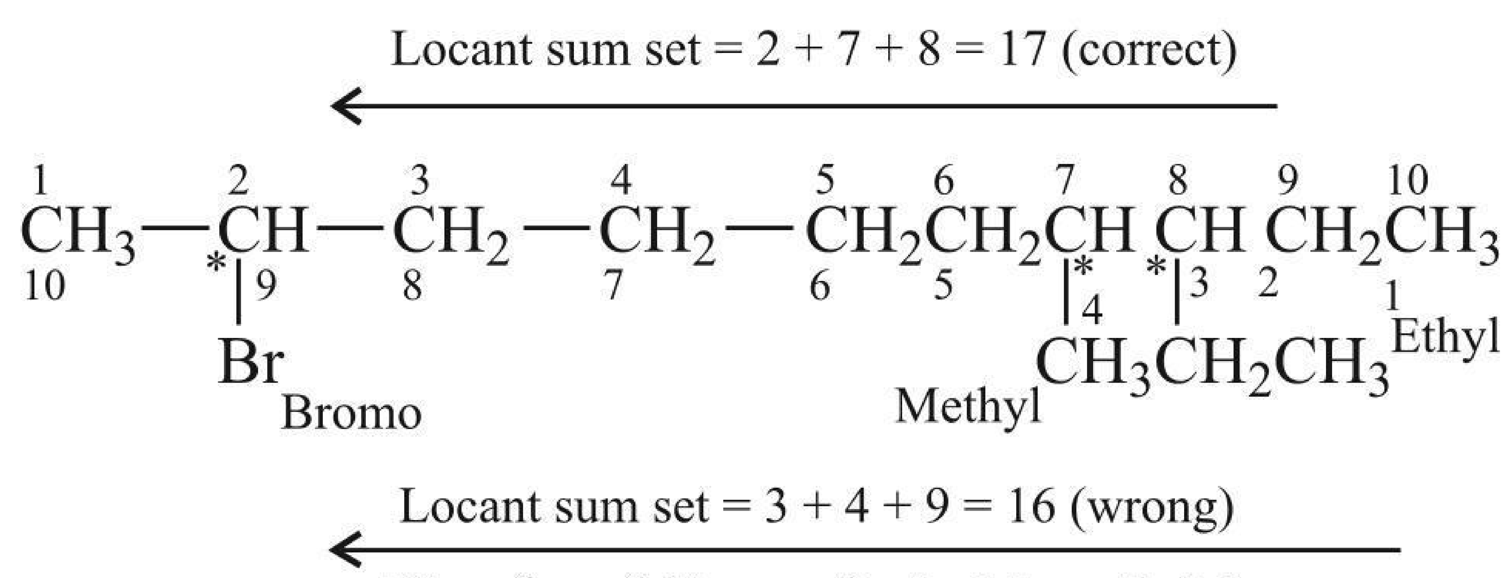


Correct name: Oct-2-en-5-yne

1.5.3 IUPAC Rules for Haloalkanes

1. Substituents are named in alphabetical order.
2. In case of two or more locants, the locant set with the greater number is preferred over the smaller number locant.

For example

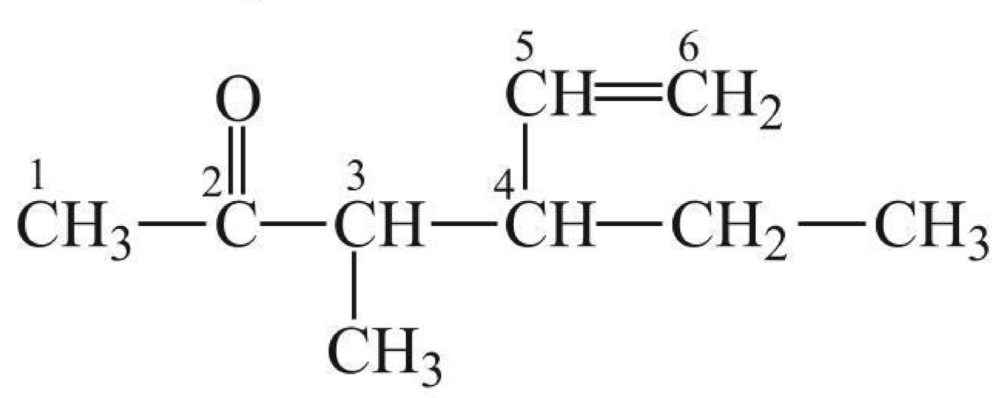


Therefore, 2-Bromo-8-ethyl-7-methyl decane (since *b* of bromo comes before *e* of ethyl and *m* of methyl)

1.5.4 IUPAC Rules for Functional Compounds

While numbering the longest chain, the functional group should acquire the lowest number followed by other substituents and the family of multiple bonds even if it violates the lowest sum rule.

For example

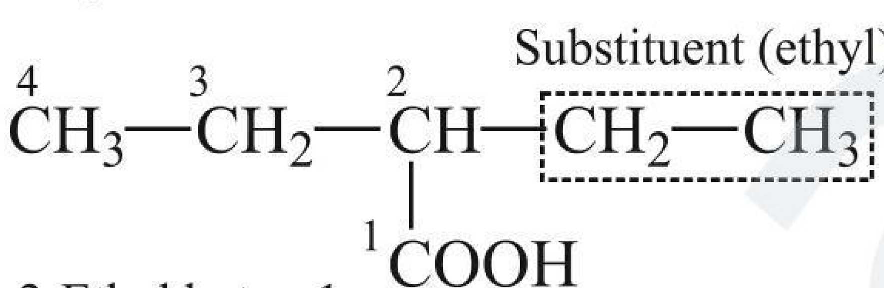


C=O (Functional group) acquires the lowest number
CH_3 (Substituent) methyl
$CH{=}CH_2$ (= bond)

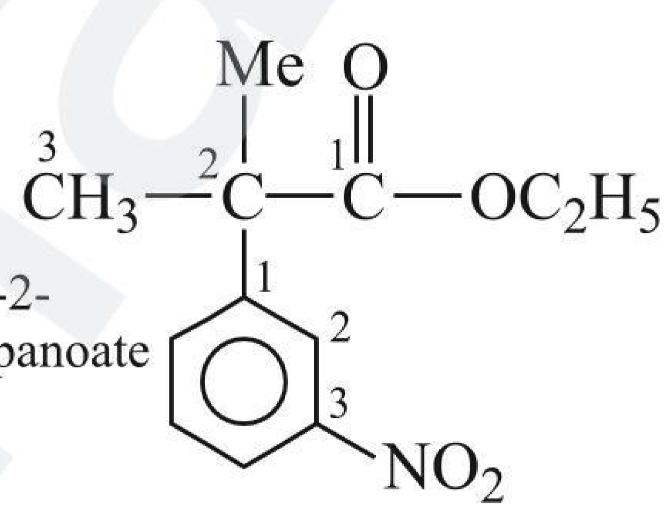
1.5.5 IUPAC Rules For Chain Terminating Functional groups (—CHO, —COOH, —$CONH_2$, —COCl, —C≡N)

These chain terminating groups are included in the numbering, starting from the end where it acquires the lowest number followed by other substituents in alphabetical order.

For example



2-Ethyl butan-1-oic acid



Ehtyl 2-methyl-2-(3-nitro phenyl) propanoate

1.5.6 IUPAC Rules for Polyfunctional Compounds

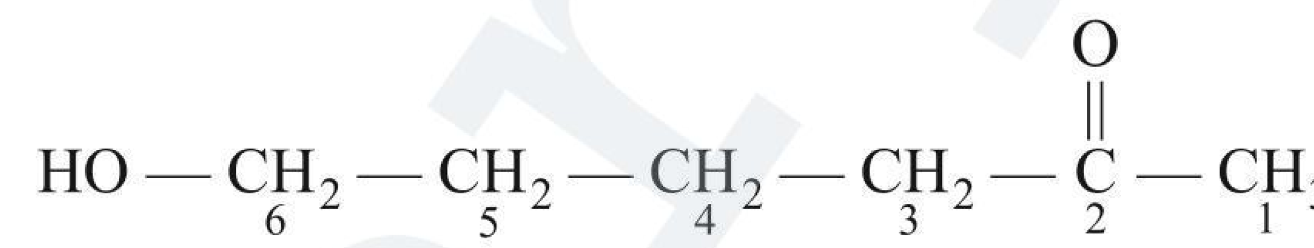
In case of polyfunctional compounds, one of the functional groups is chosen as the principal functional group and the compound is named on that basis. The remaining functional groups, which are subordinate functional groups, are named as substituents using the appropriate prefixes.

The decreasing order of priority of some functional groups is

— COOH > — SO_3H > — COOR (ester) > — COCl (acylhalide) > — $CONH_2$ (amide) > — C ≡ N (nitriles) > — CH=O (aldehyde) > >C=O (keto) > — OH (alcohol) > —NH_2 (amine) > >C=C< (alkene) > —C ≡ C—(alkyne)

The — R (alkyl group), Ph or C_6H_5— (phenyl), halogens (F, Cl, Br, I), —NO_2, alkoxy (—OR), etc., are always prefix substituents. Thus, a compound containing both an alcohol and a keto group is named hydroxy alkanone since the keto group is preferred to the hydroxyl group.

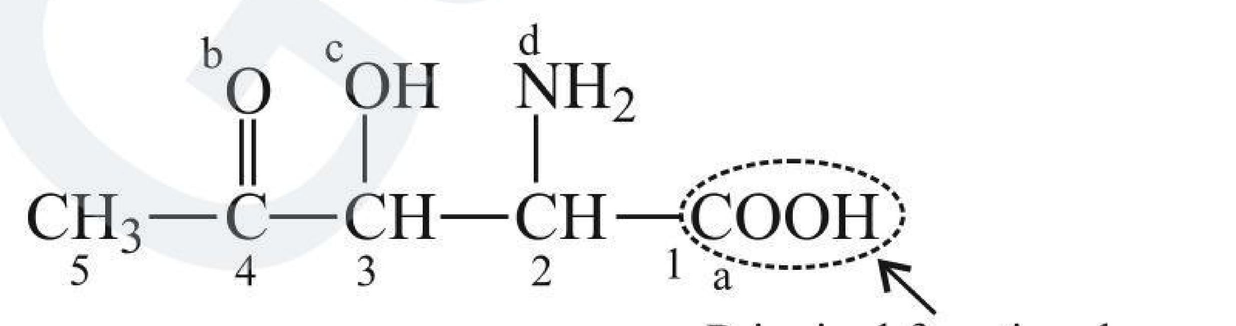
For example

i. 

6-Hydroxyhexan-2-one

ii. 

4-Chlorobut-1-ene

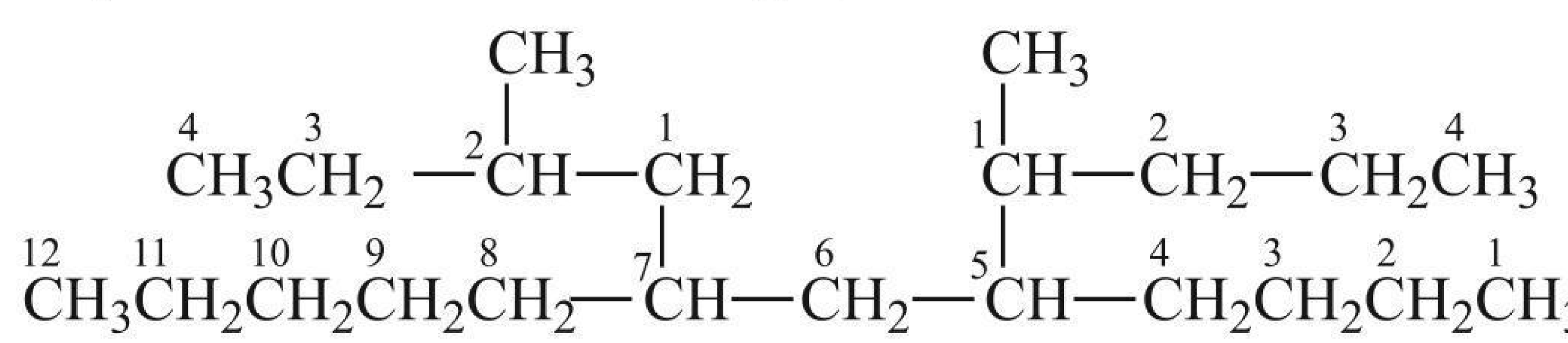
iii. 

(a, b, c, and d are in the decreasing preference order but written in IUPAC name in alphabetical order.)

IUPAC name: 2-Amino-3-hydroxy-4-oxopentan-1-oic acid.

1.5.7 When the Names of Two or More Substituents are Composed of Identical Words

The priority of citation is given to the substituent which has the lowest locant at the first cited point of difference within the complex substituent. For example,



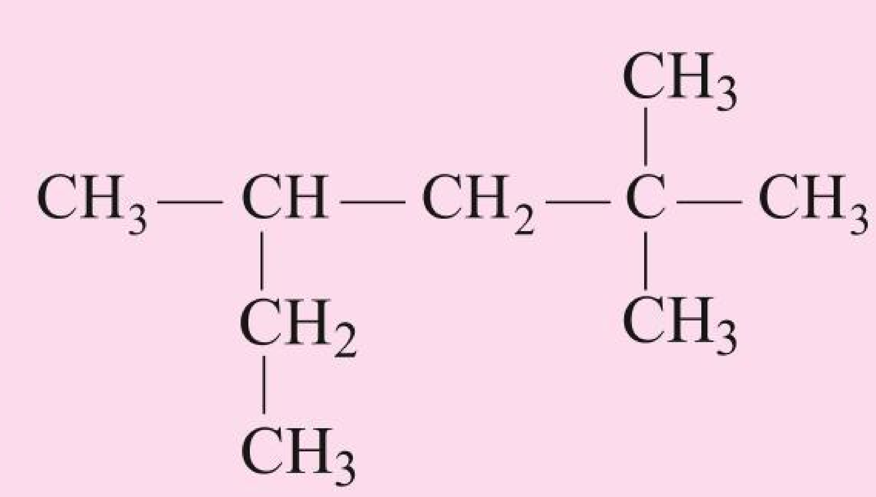
5-(1-Methylbutyl)-7-(2-methylbutyl) dodecane

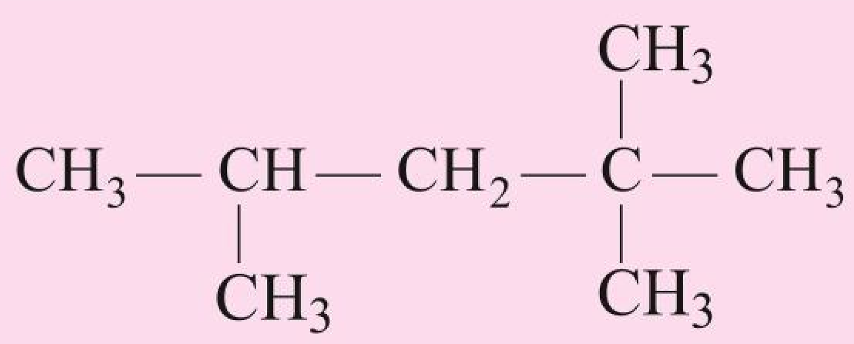
Note Note

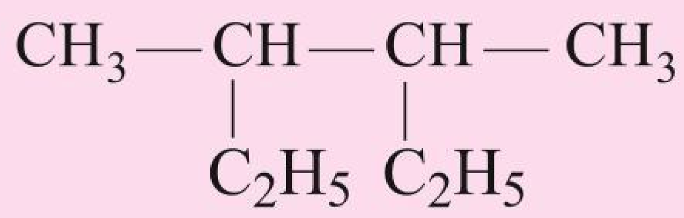
It may be noted here that '1-methylbutyl' group is given the priority of citation over '2-methylbutyl' group because locant 1 comes before 2.

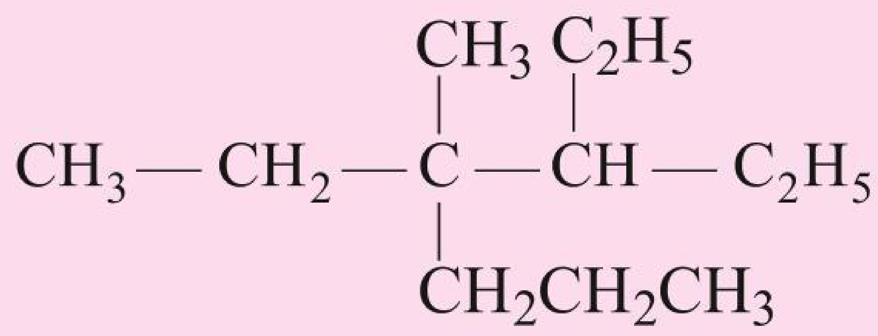
Illustration 1.3

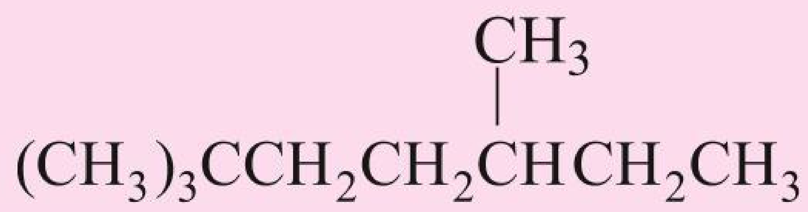
a. Give the IUPAC names of the following alkanes:

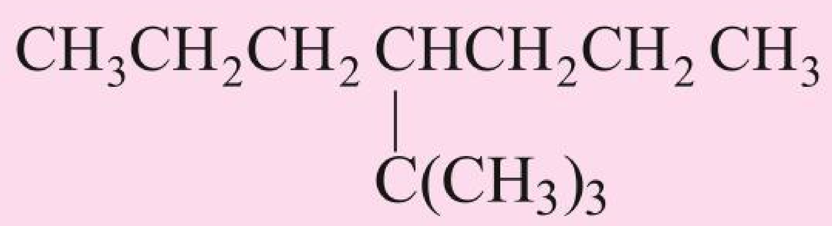
i. 

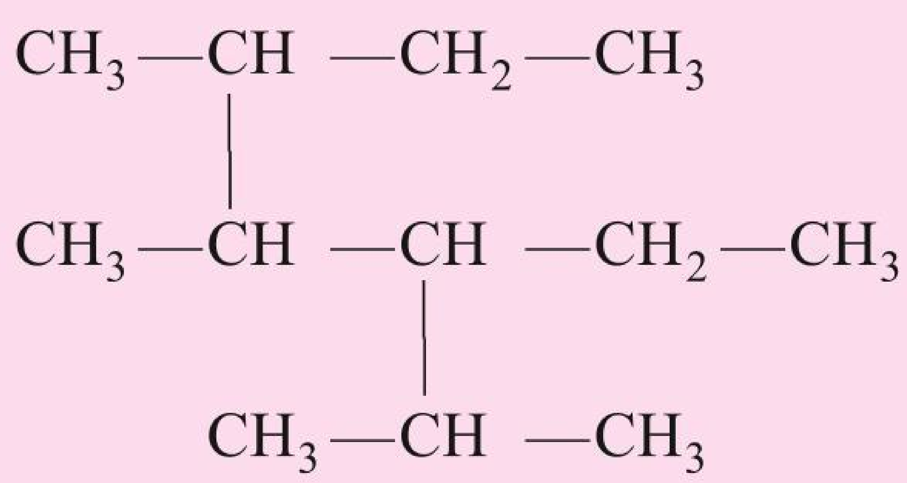
ii. 

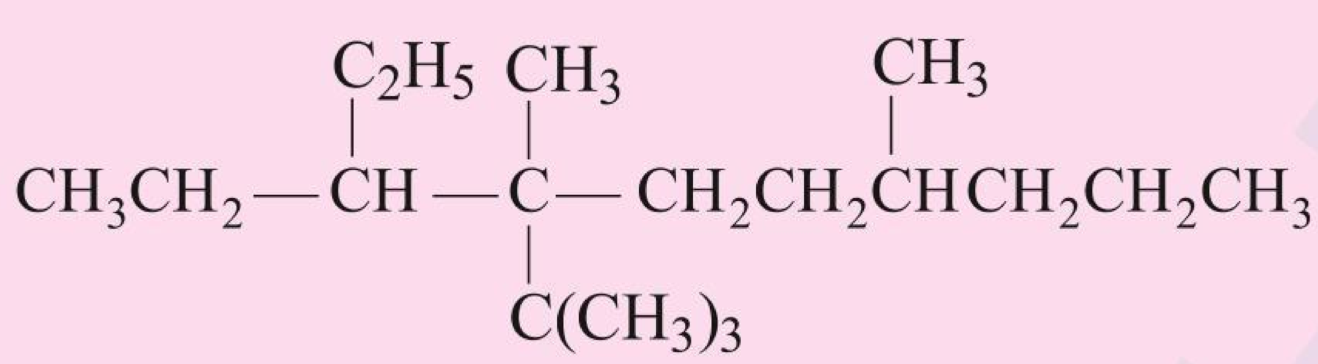
iii. 

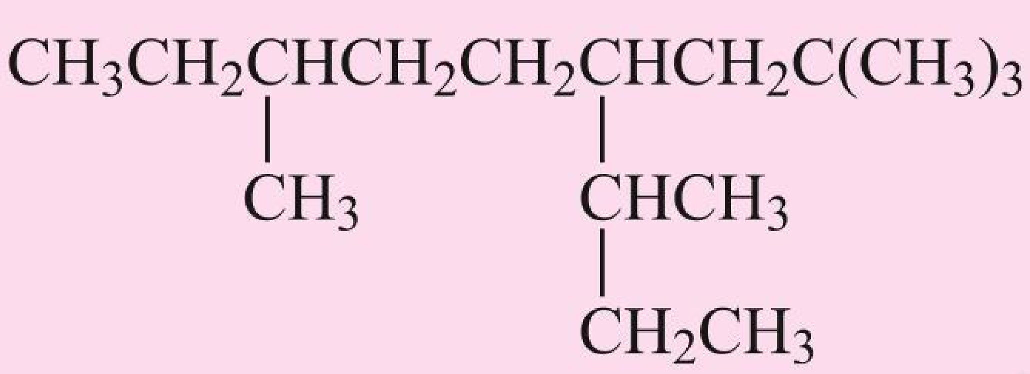
iv. 

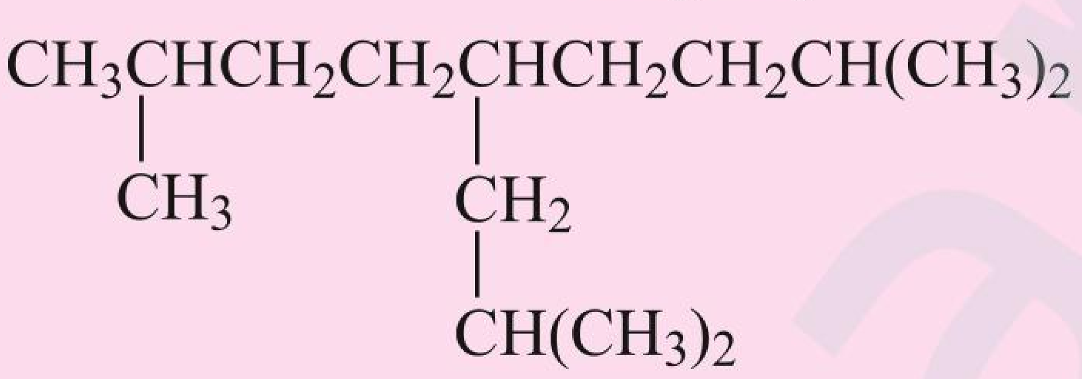
v. 

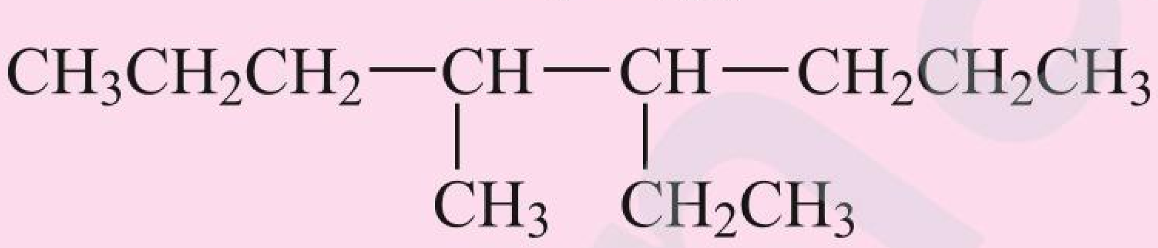
vi. 

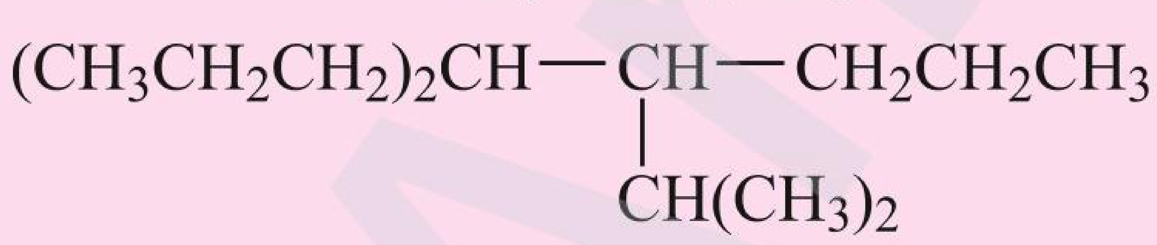
vii. 

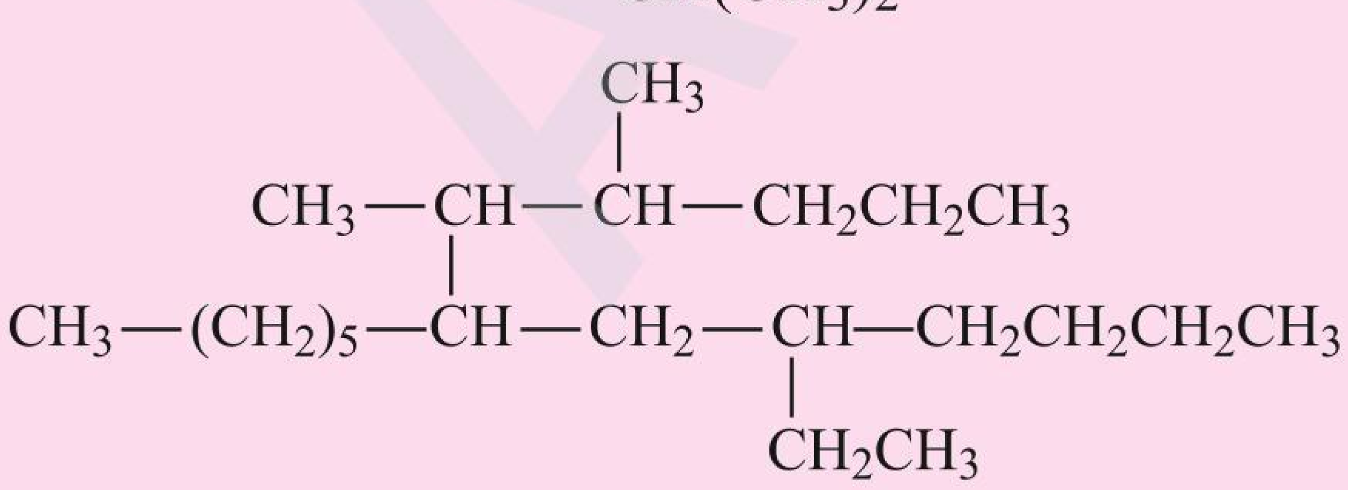
viii. 

ix. 

x. 

xi. 

xii. 

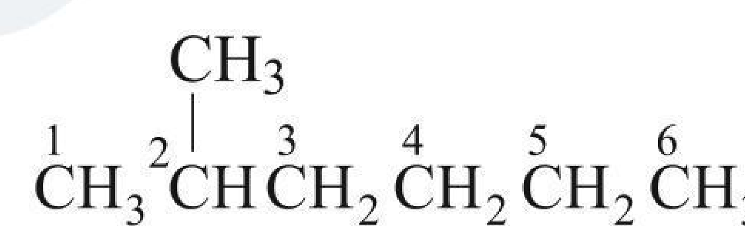
xiii. 

b. What is wrong with the following names? Draw the structures they represent and give their correct names.

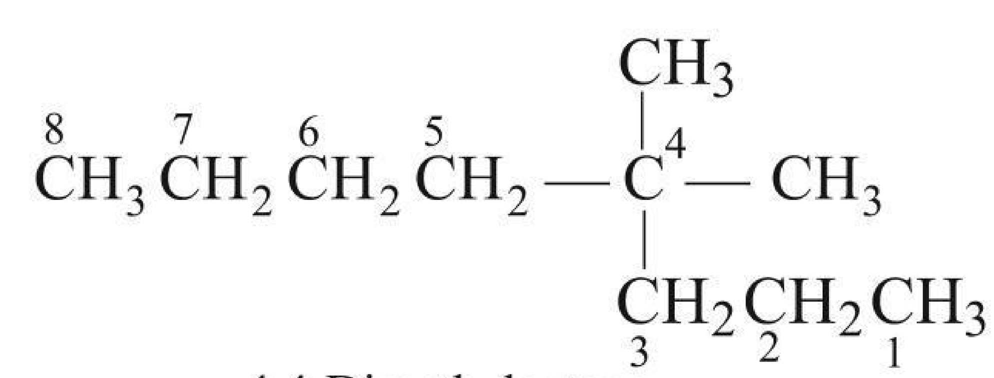
i. 1,1-Dimethylpentane
ii. 2-Methyl-2-propylhexane
iii. 3-Dimethylpentane
iv. 4,4-Dimethyl-3-ethylpentane
v. 4-(2-Methylethyl) heptane

Sol.

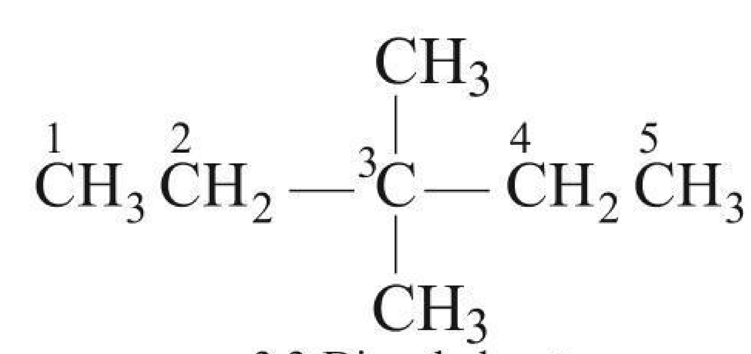
a. **i.** 2,2,4-Trimethylhexane
ii. 2,2,4-Trimethylpentane
iii. 3,4-Dimethylhexane
iv. 3,4-Diethyl-4-methylheptane
v. 2,2,5-Trimethylheptane
vi. 4-(1,1-Dimethylethyl)heptane
vii. 3-Ethyl-2,4,5-trimethylheptane
viii. 4-(1,1-Dimethylethyl)-3-ethyl-4,7-dimethyl decane
ix. 2,2,7-Trimethyl-4-(1-methylpropyl) nonane
x. 2,8-Dimethyl-5-(2-methylpropyl) nonane
xi. 5-Ethyl-4-methyloctane
xii. 4-(1-Methylethyl)-5-propyloctane
xiii. 7-(1,2-Dimethylpentyl)-5-ethyl tridecane

b. **i.** 

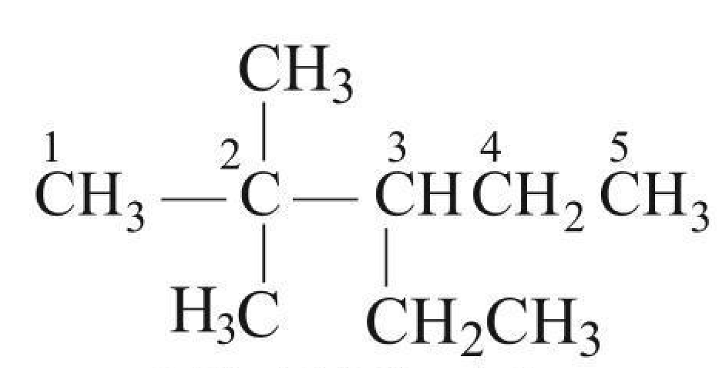
2-Methylhexane

ii. 

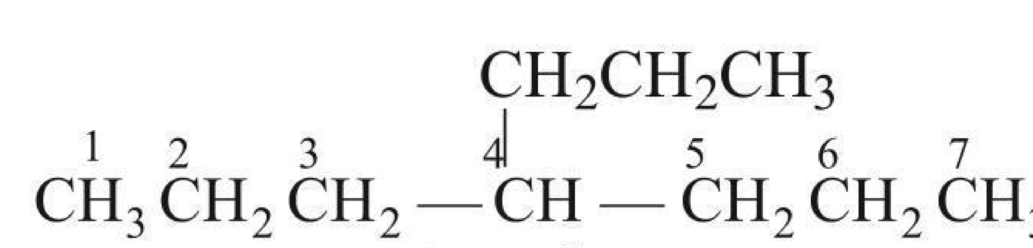
4,4-Dimethyloctane

iii. 

3,3-Dimethylpentane

iv. 

3-Ethyl-2,2-dimethylpentane

v. 

4-Propylheptane

1.5.8 When Both Double and Triple Bonds are Present in the Compound

In such cases, their locants are written immediately before their respective suffixes and the terminal 'e' from the suffix 'ene' is dropped while writing their complete names. It may be emphasised here that such unsaturated compounds are always named as derivatives of alkyne rather than alkene.

For example:

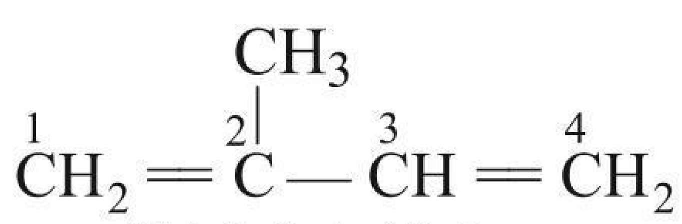
i. $\overset{5}{C}H_3-\overset{4}{C}H=\overset{3}{C}H-\overset{2}{C}\equiv\overset{1}{C}H$

Pent + 3-en (e) + 1-yne
= Pent-3-en-1-yne
(Formerly 3-Penten-1-yne)

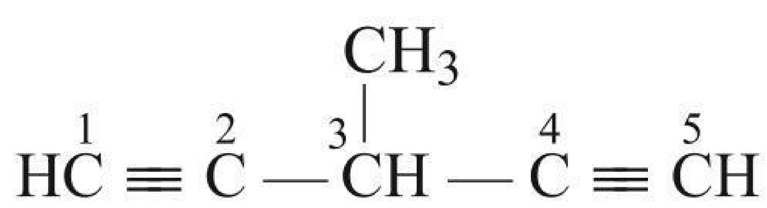
ii. $H\overset{5}{C}\equiv\overset{4}{C}-\overset{3}{C}H_2-\overset{2}{C}H=\overset{1}{C}H_2$

Pent-1-en (e) + 4-yne
= Pent-1-en-4-yne
(Formerly 1-Penten-4-yne)

These rules are further illustrated by the following additional examples:

iii. 

2-Methylbuta-1,3-diene
(Formerly 2-Methyl-1,3-butadiene)

iv. 

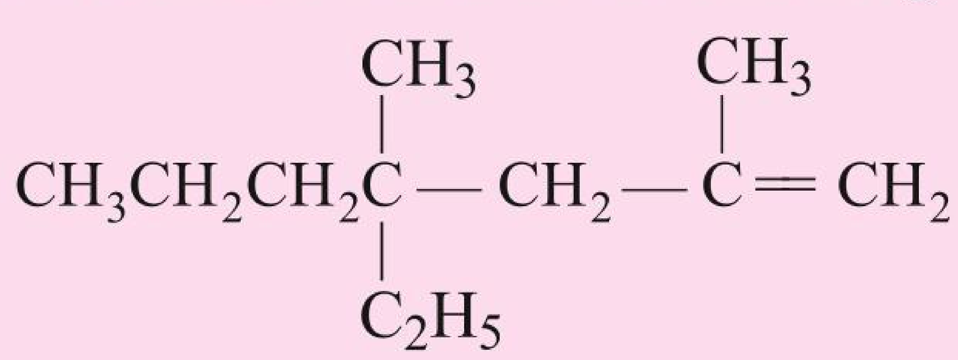
3-Methylpenta-1,4-diyne
(Formerly 3-Methyl-1,4-Pentadiyne)

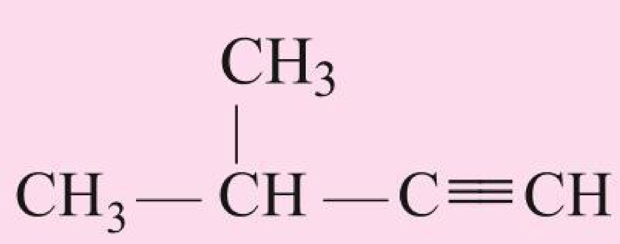
v. $H\overset{6}{C}\equiv\overset{5}{C}-\overset{4}{C}H=\overset{3}{C}H-\overset{2}{C}H=\overset{1}{C}H_2$

Hexa-1,3-diene-5-yne
(Formerly 1,3-Hexadiene-5-yne)

ILLUSTRATION 1.4

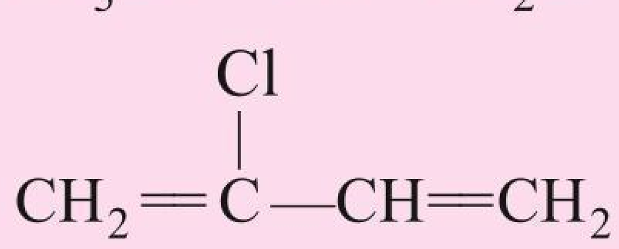
Give the IUPAC names of the following compounds:

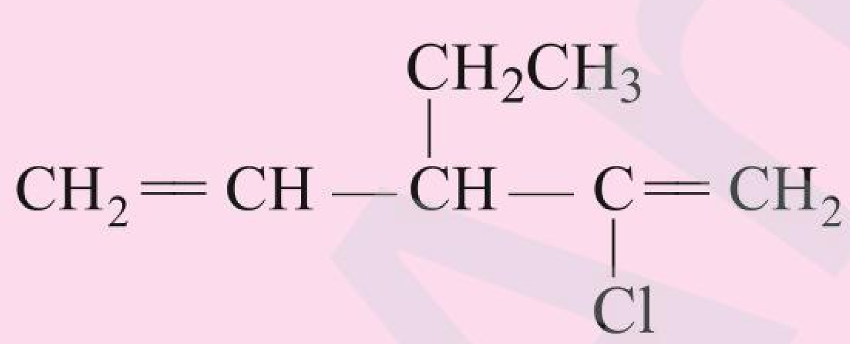
i. 

ii. 

iii. $C_6H_5-CH=CH-CH_2Cl$

iv. $CH_3CH=CH-CH_2Br$

v. 

vi. 

vii. $(CH_3)_3C-CH=CH_2$

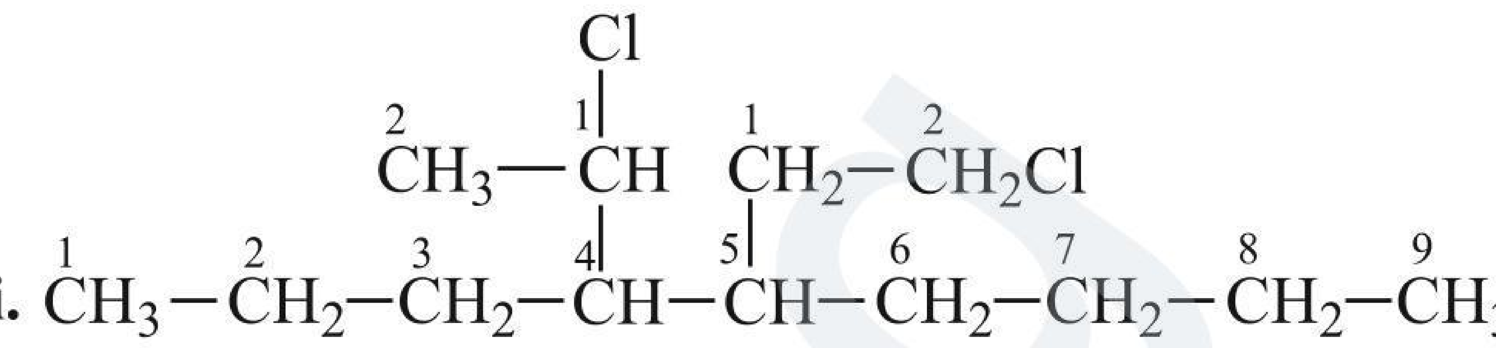
Sol.

i. 4-Ethyl-2,4-dimethyl hept-1-ene
ii. 3-Methylbut-1-yne
iii. 3-Chloro-1-phenylprop-1-ene
iv. 1-Bromo but-2-ene
v. 2-Chloro buta-1,3-diene
vi. 2-Chloro-3-ethyl penta-1,4-diene
vii. 3,3-Dimethyl-1-butene

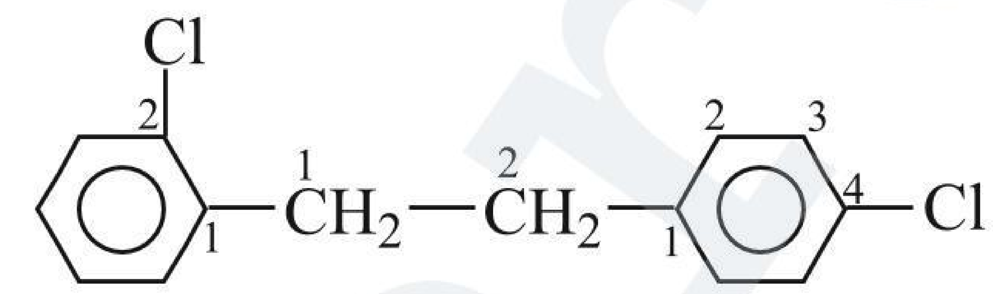
1.5.9 When Two or More Prefixes Consist of Identical Roman Letters (Words)

The priority for citation is given to that group which contains the lowest locant at the first point of difference.

For example

i. 

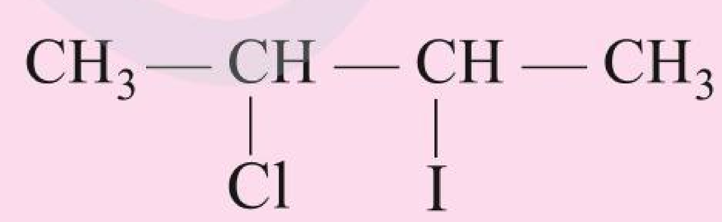
4-(1-Chloroethyl)-5-(2-chloroethyl) nonane

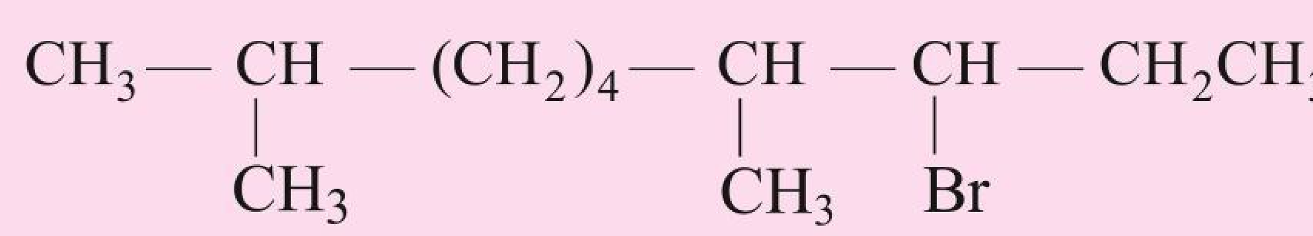
ii. 

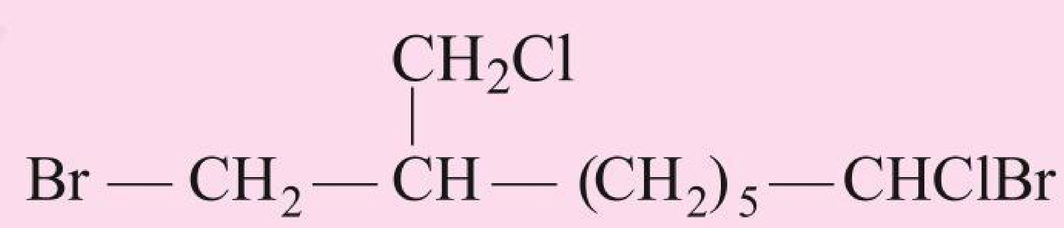
1-(2-Chlorophenyl)-2-(4-chlorophenyl) ethane
(OR) (1-(o-chlorophenyl)-2-(p-chlorophenyl) ethane

ILLUSTRATION 1.5

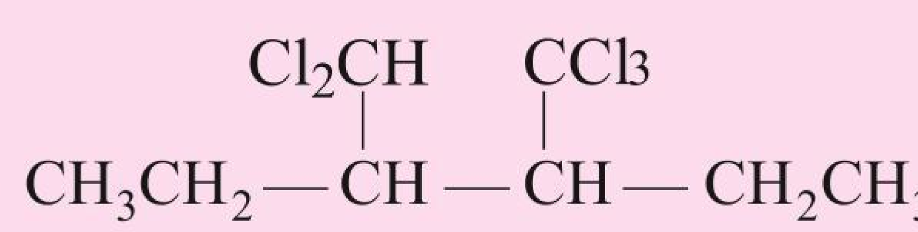
Give the IUPAC name of the following compounds:

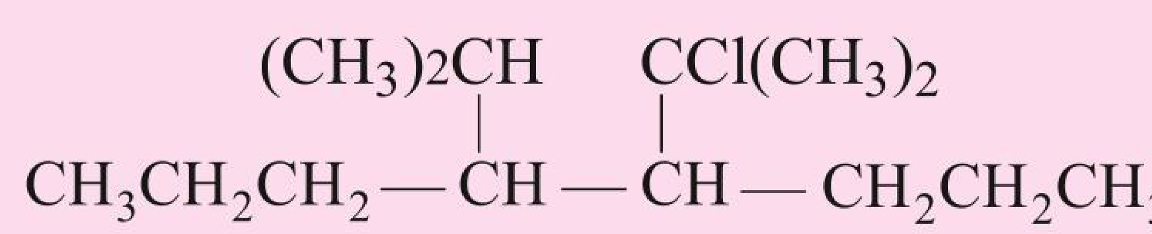
i. 

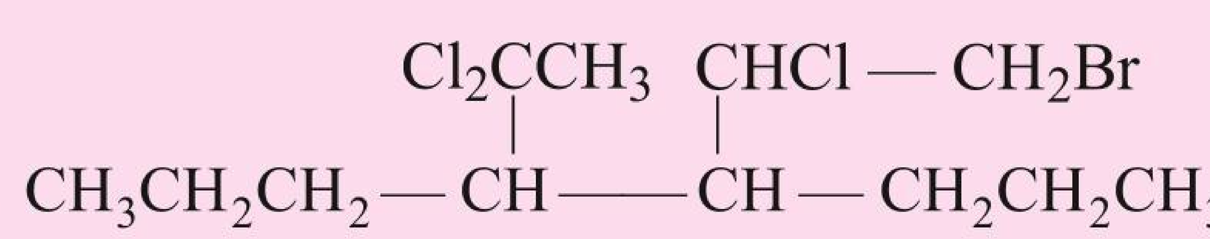
ii. 

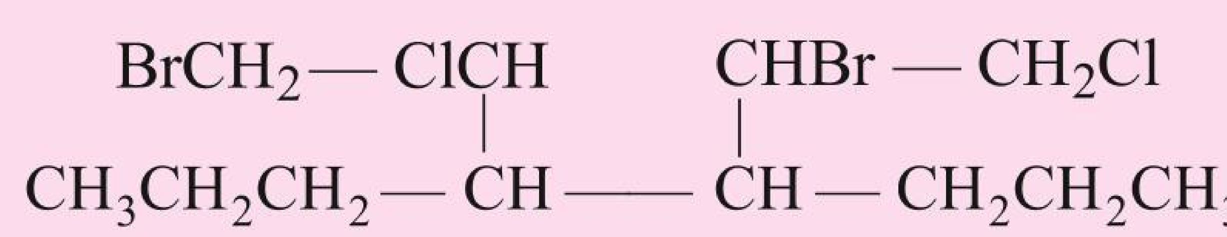
iii. 

iv. $(CCl_3)_4C$

v. 

vi. 

vii. 

viii. 

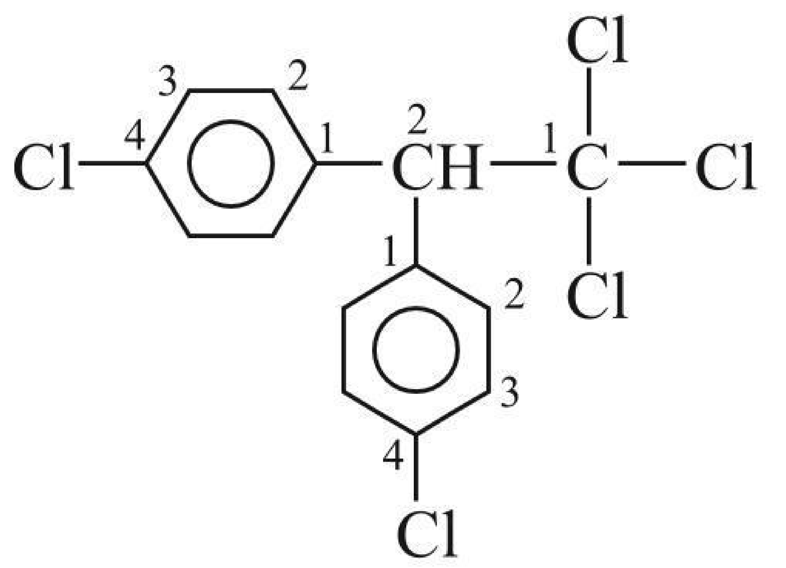
Sol.

i. 2-Chloro-3-iodobutane
ii. 8-Bromo-2,7-dimethyl decane
iii. 1,8-Dibromo-1-chloro-7-(chloromethyl) octane
iv. 1,1,1,3,3,3,-Hexachloro-2,2-bis(trichloromethyl) propane
v. 3-(Dichloromethyl)-4-(trichloromethyl) hexane
vi. 4-(1-Chloro-1-methylethyl)-5-(1-methylethyl) octane
vii. 4-(2-Bromo-1-chloro)-5-(1,1-dichloroethyl) octane
viii. 4-(1-Bromo-2-chloroethyl)-5-(2-bromo-1-chloro-ethyl) octane

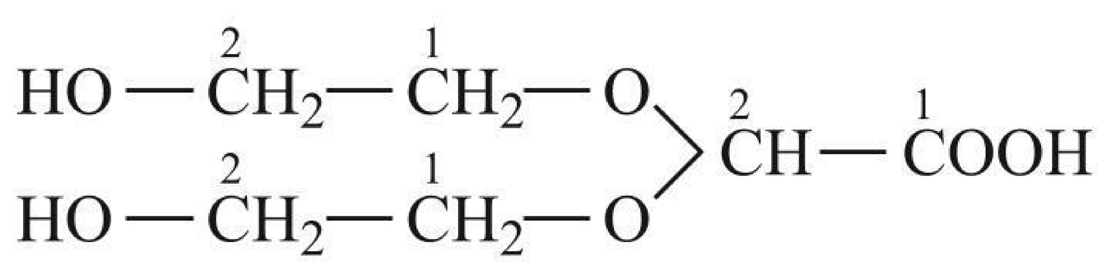
1.5.10 When the Organic Molecule Contains More Than One Similar Complex Substituents

In such cases, the numerical prefixes, such as di, tri, tetra, etc., are replaced by bis, tris, tetrakis, etc., respectively.

For example

i. 

1,1,1-Trichloro-2,2-bis(4-chlorophenyl) ethane
(OR)
(DDT) or Dichlorodiphenyl trichloroethane

ii. 

2,2-bis(2-Hydroxyethoxy) ethanoic acid

Illustration 1.6

Give the IUPAC names of the following compounds:

i. $C_2H_5-\underset{\underset{CH_2}{\|}}{C}-CH_2OH$

ii. $CH_3-CH=CH-CHO$

iii. $NH_2-CH_2-CH_2-CH_2-NH_2$

iv. $CH_3-CH=CH-COOH$

v. $(CH_3)_2C=CHCOCH_3$

vi. $CH_2=CH-CN$

vii. $OCH-CHO$

viii. $HOOC-C\equiv C-COOH$

ix. $CH_2=\overset{\overset{CH_3}{|}}{C}-COOCH_3$

x. $CH_3CH_2-\overset{\overset{CH_3}{|}}{C}(Cl)-CH_2-CONHCH_2CH_3$

xi. $CH_3-\overset{\overset{OCH_3}{|}}{CH}-CHO$

xii. $[(CH_3)_2CH]_3COH$

xiii. $C_6H_5-CH=CH-COOH$

Sol.

i. 2-Ethylprop-2-en-1-ol
ii. But-2-en-1-al
iii. Propane-1,3-diamine
iv. But-2-en-1-oic acid
v. 4-Methylpent-3-en-2-one
vi. Prop-2-ene-1-nitrile
vii. Ethane-1,2-dial
viii. But-2-yne-1,4-dioic acid
ix. Methyl 2-methylprop-2-en-1-oate
x. 3-Chloro-N-ethyl-3-methylpentan-1-amide
xi. 2-Methoxypropan-1-al
xii. 2,4-Dimethyl-3-(1-methylethyl)pentan-3-ol
xiii. 3-Phenylprop-2-en-1-oic acid

1.5.11 Poly-Functional Compounds Containing More than Two Like-Functional Groups

According to the latest convention (1993 recommendation for IUPAC nomenclature), if an unbranched carbon chain is directly linked to more than two like-functional groups, then the organic compound is named as a derivative of the parent alkane which does not include the carbon atoms of the functional groups.

For example

i. $N\equiv C-\overset{1}{C}H_2-\overset{\overset{C\equiv N}{|}}{\overset{2}{C}H}-\overset{3}{C}H_2-C\equiv N$

Propane-1,2,3-tricarbonitrile
(Not 3-cyanopentane-1,5-dinitrile)

ii. $HOOC-\overset{1}{C}H_2-\overset{2}{C}H_2-\overset{\overset{COOH}{|}}{\overset{3}{C}H}-\overset{4}{C}H_2-\overset{5}{C}H_2-COOH$

Pentane-1,3,5-tricarboxylic acid
(Not 4-Carboxyheptane-1,7-dioic acid)

iii. $HOOC-\overset{1}{C}H_2-\overset{\overset{COOH}{|}}{\underset{\underset{OH}{|}}{\overset{2}{C}}}-\overset{3}{C}H_2-COOH$

2-Hydroxypropane-1,2,3-tricarboxylic acid
(Not 3-Carboxy-3-hydroxypentane 1,5-dioic acid)

1.5.12 When All the Three Like Groups are not Directly Linked to the Unbranched Carbon Chain

The two like groups are included in the parent chain while the third group which forms the side chain is considered a substituent group.

For example

i. $N\equiv C-CH_2-CH_2-\overset{\overset{CH_2C\equiv N\text{ (Substituent group)}}{|}}{CH}-CH_2-C\equiv N$

3-(Cyanomethyl) hexane-1,6-dinitrile

ii. $HOOC-CH_2-CH_2-CH_2-\overset{\overset{CH_2COOH\text{(Substituent group)}}{|}}{CH}-CH_2-COOH$

3-(Carboxymethyl) heptane-1,7-dioic acid

Illustration 1.7

Give the IUPAC names of the following polyfunctional compounds:

i. $CH_3CH_2O-CH_2-CHOH-CH_3$

ii. $CH_3-\underset{NO_2}{\underset{|}{CH}}-CH_2-\overset{CH_3}{\overset{|}{\underset{OH}{\underset{|}{C}}}}-CH_3$

iii. $CH_2=\overset{H_3C}{\overset{|}{C}}-\overset{OH}{\overset{|}{CH}}-CH_2CN$

iv. $CH_3-\underset{CONH_2}{\underset{|}{CH}}-COOH$

v. $HOOCCH_2-\underset{COOH}{\underset{|}{CH}}-CH_2COOH$

vi. $CH_3-\underset{COCH_3}{\underset{|}{CH}}-CH_2-\underset{NH_2}{\underset{|}{CH_2}}$

vii. $HOOCCH_2-\overset{CHO}{\overset{|}{\underset{COOH}{\underset{|}{C}}}}-CH_2COOH$

viii. $CH_3NH-CH_2CH_2-\overset{CH_3}{\overset{|}{\underset{CHO}{\underset{|}{C}}}}-CH_2CH_3$

xi. $HC\equiv C-\overset{CH_3}{\overset{|}{CH}}-CHOH-CH_2COCl$

x. $CH_3\underset{Br}{\underset{|}{CH}}-\overset{CH_3}{\overset{|}{\underset{OH}{\underset{|}{C}}}}-CO_2CH_3$

xi. $CH_2=\overset{CH_3}{\overset{|}{C}}-CHOH-CH_2-CHO$

Sol.

- **i.** 1-Ethoxypropan-2-ol
- **ii.** 2-Methyl-4-nitropentan-2-ol
- **iii.** 3-Hydroxy-4-methylpent-4-ene-1-nitrile
- **iv.** 2-Carbamoylpropan-1-oic acid or 2-carboxamidopropan-1-oic acid
- **v.** Propane-1,2,3-tricarboxylic acid
- **vi.** 5-Amino-3-methylpentan-2-one
- **vii.** 2-Formylpropane-1,2,3-tricarboxylic acid
- **viii.** 4-Aminomethyl-2-ethyl-2-methylbutan-1-al
- **ix.** 3-Hydroxy-4-methyl hex-5-yn-1-oylchloride
- **x.** Methyl 3-bromo-2-hydroxy-2-methylbutan-1-oate
- **xi.** 3-Hydroxy-4-methylpent-4-en-1-al

Illustration 1.8

a. Rewrite the following structural formulae in bond line notation.

i. $CH_3CH_2CH_2CH_2CH_3$

ii. $(CH_3)_2CHCH_2CH(CH_3)NH_2$

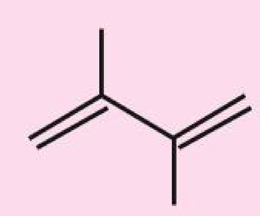
iii. $(CH_3CH_2)_2NCH_3$

iv. $CH_3CHBrCH_2CH_2CHO$

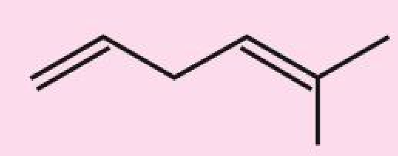
v. $CH_3-\overset{O}{\overset{||}{C}}-\underset{CH_3}{\underset{|}{CH}}-O-CH_2-\overset{O}{\overset{||}{C}}-OH$

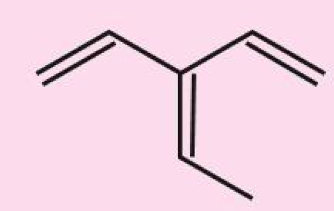
vi. $(CH_3)_2CH\overset{OH}{\overset{|}{C}}(CH_3)CH=CH_2$

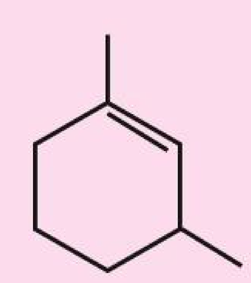
b. Give the IUPAC names for the following compounds:

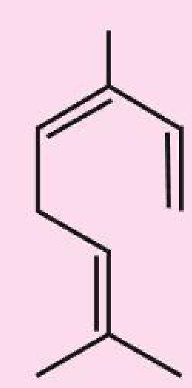
i. 

ii. 

iii. 

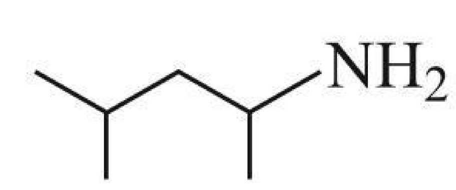
iv. 

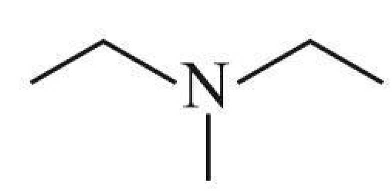
v. 

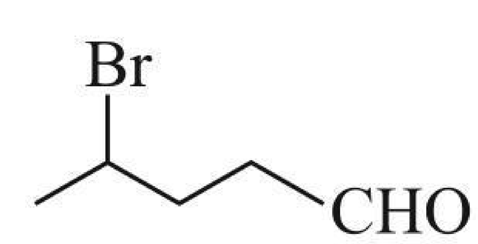
vi. 

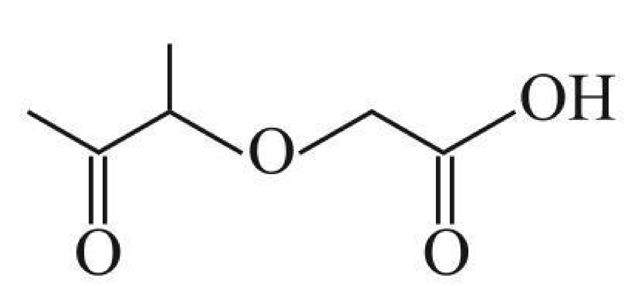
Sol.

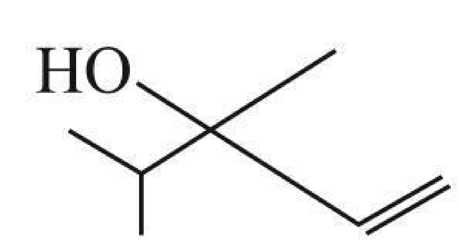
a. i. 

ii. 

iii. 

iv. 

v. 

vi. 

b. i. 2,3-Dimethylbuta-1,3-diene

ii. But-1-en-3-yne

iii. 5-Methylhexa-1,4-diene

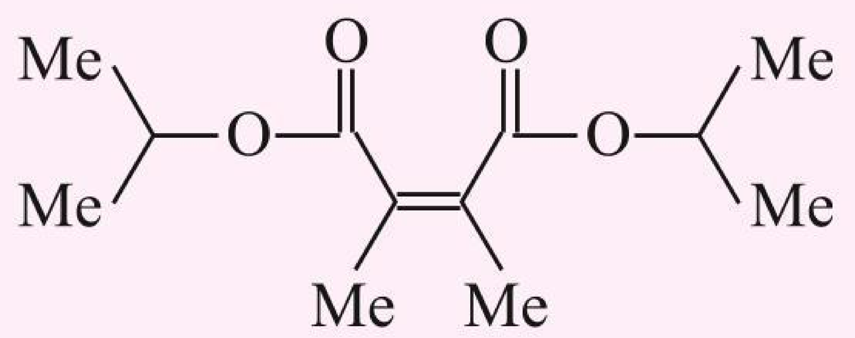
iv. 3-Ethenyl penta-1,3-diene or 3-Vinylpenta-1,3-diene

v. 1,3-Dimethylcyclohex-1-ene

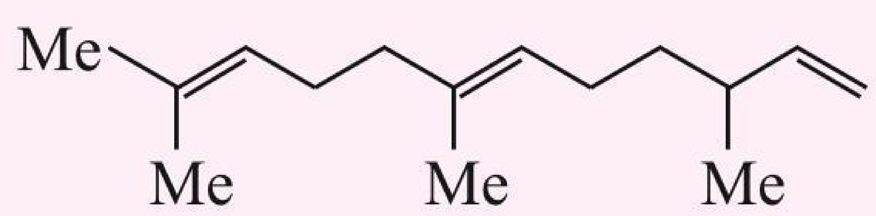
vi. 3,7-Dimethylocta-1,3,6-triene

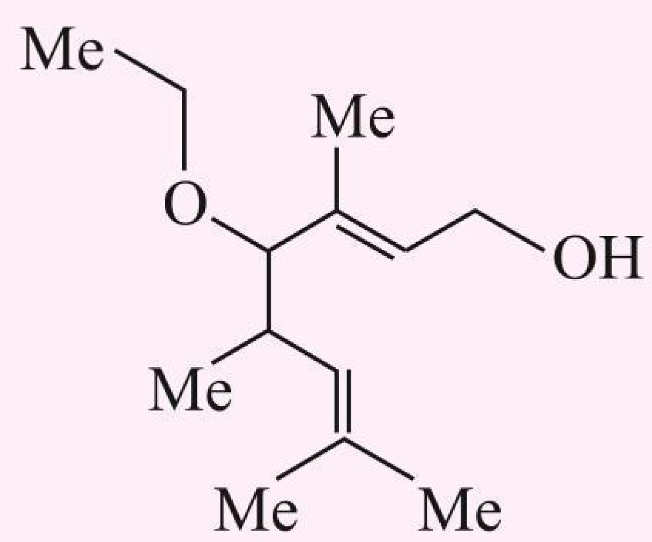
Concept Application Exercise 1.1

Write the IUPAC name of the following compounds:

1. 

2. 

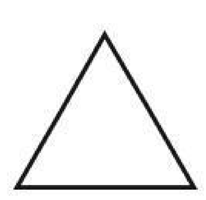
3. 

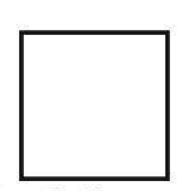
4. 

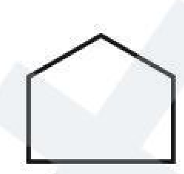
1.6 NAMING OF ALICYCLIC COMPOUNDS

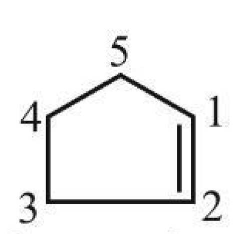
1. IUPAC suffix: ane, ene, yne

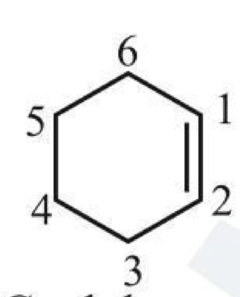
IUPAC prefix: cyclo

I. 
Cyclopropane

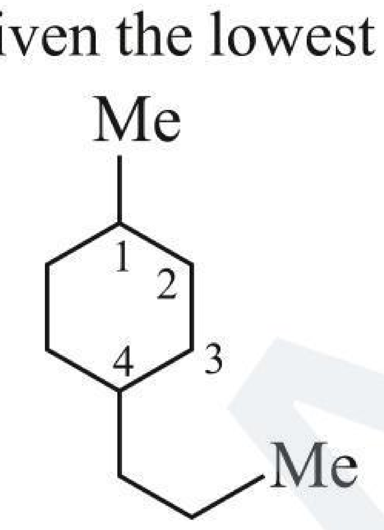
II. 
Cyclobutane

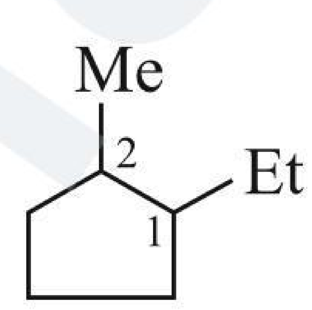
III. 
Cyclopentane

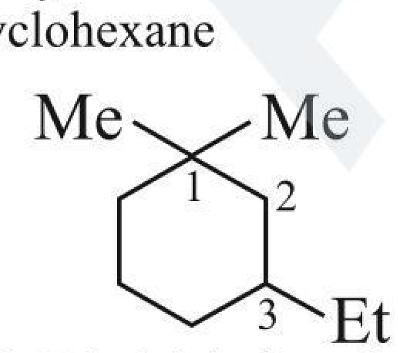
IV. 
Cyclopent-1-ene

V. 
Cyclohexene

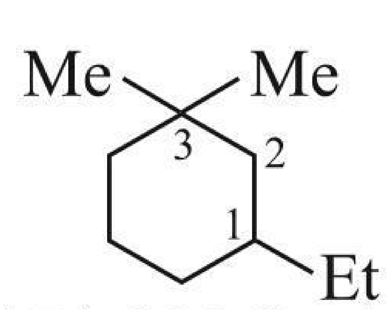
2. If two or more alkyl groups or other substituents are present in the ring, their positions are indicated, e.g., 1, 2, 3,..., etc. The substituent which comes first in the alphabetical order is given the lowest number, as per the lowest sum rule, e.g.,

I. 
1-Methyl-4-
-propylcyclohexane

II. 
1-Ethyl-2-methyl
cyclopentane

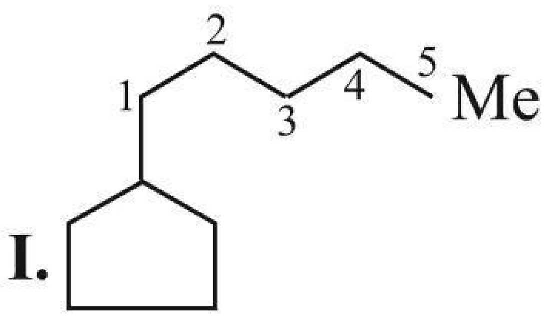
III. 
3-Ethyl-1,1-dimethyl
cyclohexane
(Lowest sum = 3 + 1 + 1 = 5)
(More branched gets lower number)

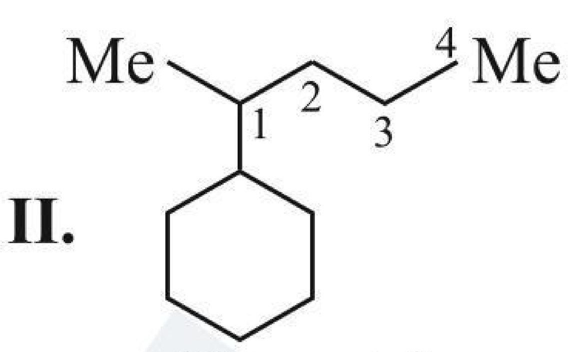
(not)

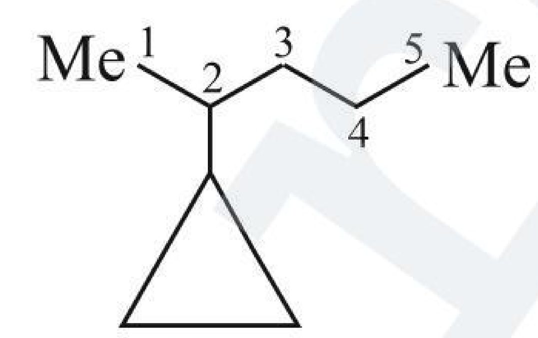

1-Ethyl-3,3-dimethyl
cyclohexane
(Violates lowest sum rule
= 1 + 3 + 3 = 7)

3. a. If the ring contains equal or more number of C atoms than the alkyl groups attached to it, it is named as alkyl cycloalkane.

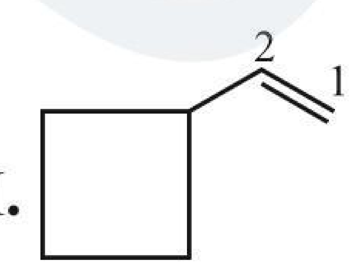
b. If the ring contains lesser number of C atoms than the alkyl groups attached to it, it is named cycloalkyl alkane, e.g.,

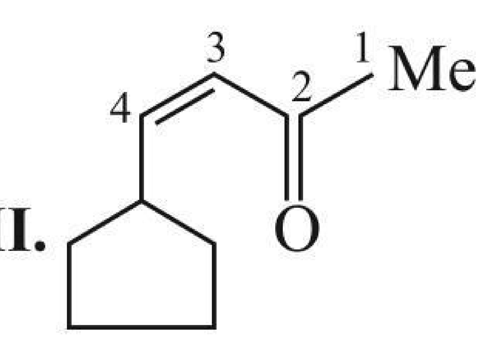
I. 
(Same number of C atoms
in ring and side chain)
Pentyl cylopentane
(Alkyl cycloalkane)

II. 
(Ring contains
more C atoms than
side chain)
1-Methyl butyl
cyclohexane

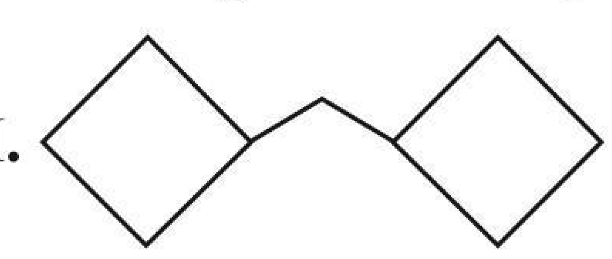
III. 
(Ring contains less C atoms than side chain)
2-Cyclopropyl pentane
(Cycloalkyl alkane)

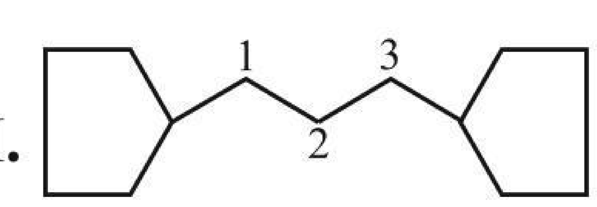
c. If the side chain contains a functional group or a multiple bond, then alicyclic ring is considered substituent irrespective of the size of the ring, e.g.,

I. 
2-Cyclobutyl
ethene

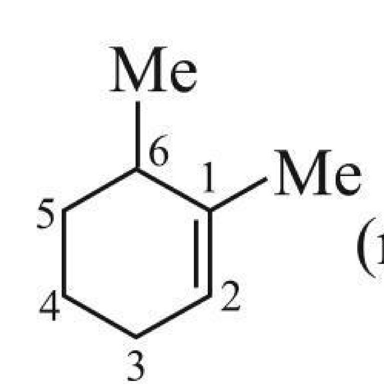
II. 
4-Cyclopentyl but
-3-en-2-one

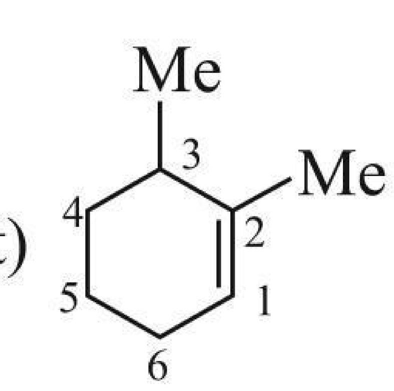
d. If more than one alicyclic ring is attached to a single chain, then the compound is named as cycloalkyl alkane (i.e., derivative of alkane) irrespective of the number of C atoms in the ring or the chain, e.g.,

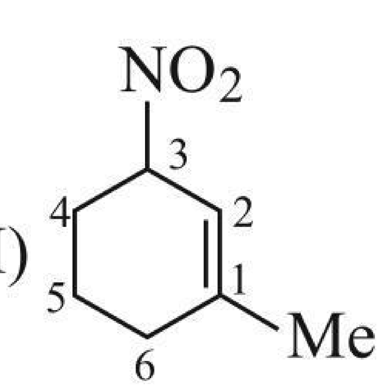
I. 
Dicyclobutyl
methane

II. 
1,3-Dicyclopentyl propane

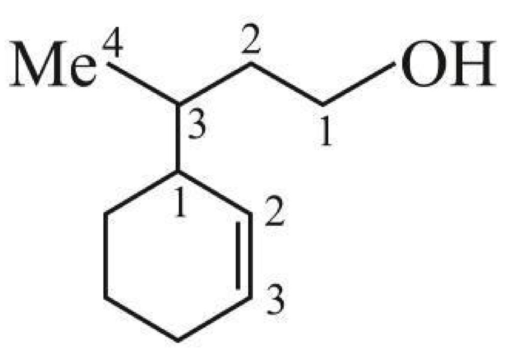
e. If double or triple (multiple) bonds and some other substituents are present in the ring the numbering is done in such a way that the multiple bond gets the lowest number, e.g.,

(I) 
1,6-Dimethyl cyclohex-
1-ene (correct)
[Lowest number for
Me substituent and
also for double bond,
although it violates
lowest sum rule]

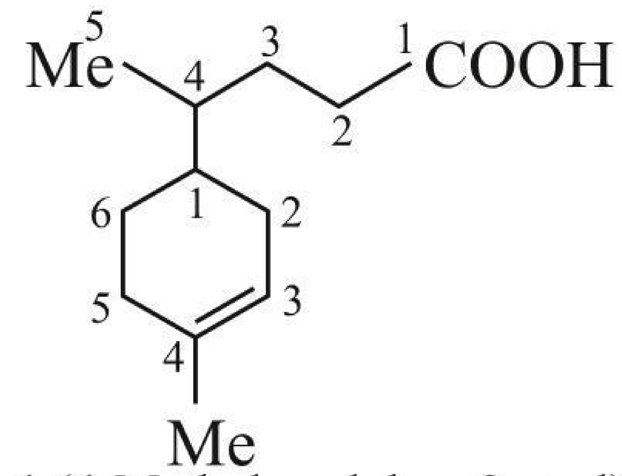
(not) 
2,3-Dimethyl
cyclohex-1-ene
(incorrect)

(II) 
1-Methyl-3-nitro
cyclohex-1-ene

f. If the ring contains a multiple bond and the side chain contains a functional group, then the ring is considered the substituent and the compound is named a derivative of the side chain, e.g.,

I. 

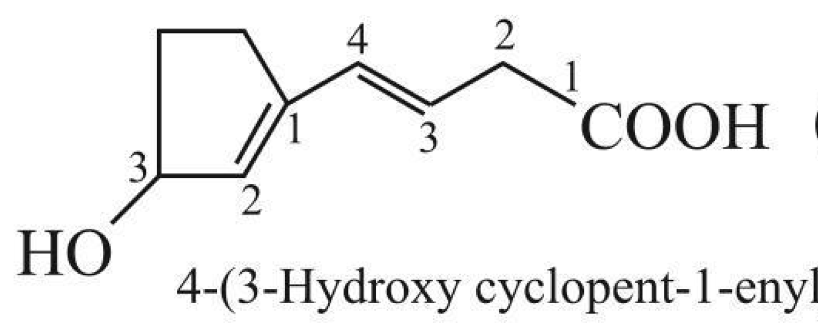
3-(Cyclohex-2-enyl) butan-1-ol

II. 

4-(4-Methyl cyclohex-3-enyl) pentanoic acid

g. If the ring and side chain both contain functional groups, then

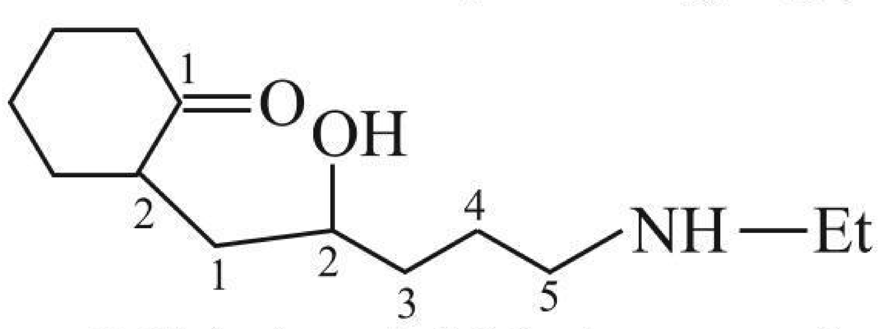
i. If the side chain contains higher priority of functional group (i.e., —COOH > $-\overset{\text{O}}{\overset{\|}{\text{C}}}-$), then the compound is named the derivative of the side chain, e.g.,



(Priority of – COOH > – OH)

4-(3-Hydroxy cyclopent-1-enyl) but-3-en-1-oic acid

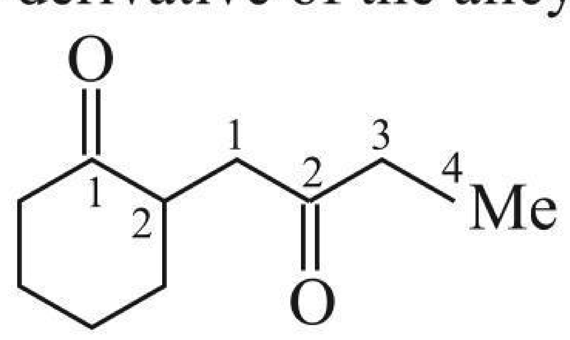
ii. If the ring contains higher priority of functional group, then the compound is named the derivative of the alicyclic ring, e.g.,



2-(5-Amino ethyl-2-hydroxy pentyl) cyclohexan-1-one

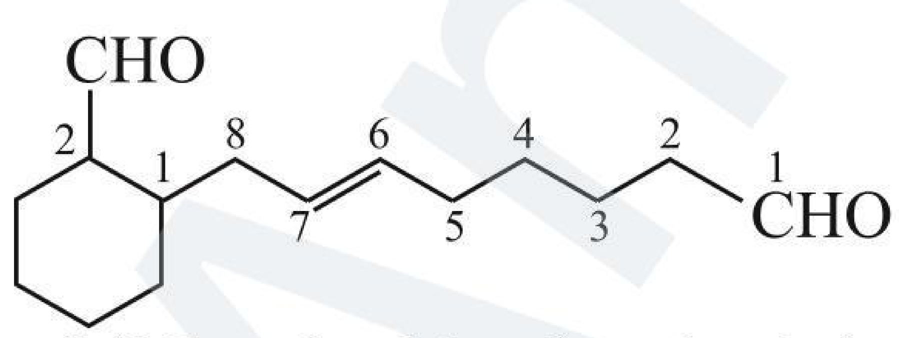
h. If both the side chain and the alicyclic ring contain the same functional group, then it is of two types.

i. If the number of C atoms of the alicyclic ring is equal or greater than that of the side chain, then it is named the derivative of the alicyclic ring, e.g.,



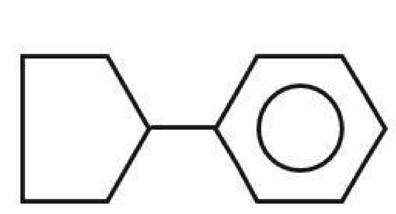
2-(2-Oxobutyl) cyclohexan-1-one

ii. If the number of C atoms of the side chain is greater than that of the alicyclic ring, then it is named the derivative of the side chain, e.g.,

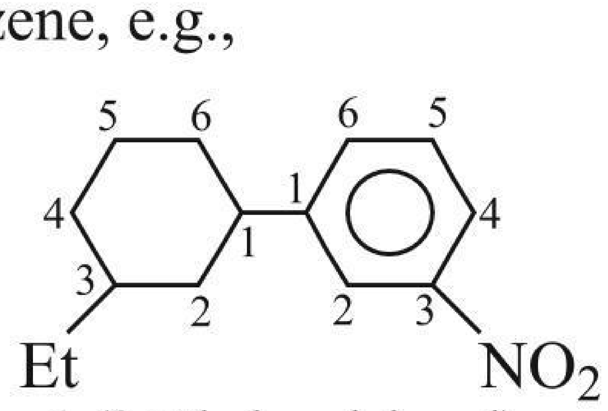


8-(2-Formyl cyclohexyl) oct-6-en-1-al

i. If an alicyclic ring is directly attached to the benzene ring, it is named the derivative of benzene, e.g.,

I. 

Cyclopentyl benzene

II. 

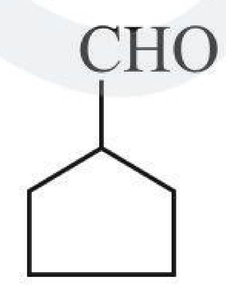
1-(3-Ethyl cyclohexyl)-3-nitro benzene

j. If an alicyclic ring is directly attached to the carbon containing the functional group, the C atom of the functional group is not included in the parent name of the alicyclic ring.

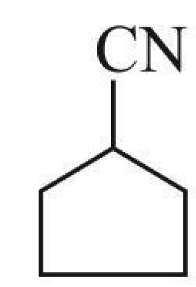
The following prefixes and suffixes are used for functional groups.

	Functional group	Prefix	Suffix
1.	—COOH	Carboxy	Carboxylic acid
2.	—CH=O	Formyl	Carbaldehyde
3.	$-\underset{\text{O}}{\underset{\|}{\text{C}}}-\text{OR}$	Carbalkoxy or Alkyl carbonyl	Alkyl carboxylate
4.	$-\underset{\text{O}}{\underset{\|}{\text{C}}}-\text{NH}_2$	Carbamoyl	Carboxamide
5.	—C≡N	Cyano	Carbonitrile
6.	$-\overset{\text{O}}{\overset{\|}{\text{C}}}-\text{X}$ (X = F, Cl, Br, I)	Halocarbonyl	Carbonyl halide

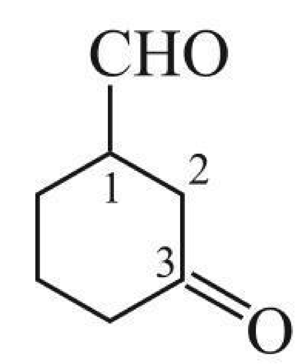
For example:

I. 

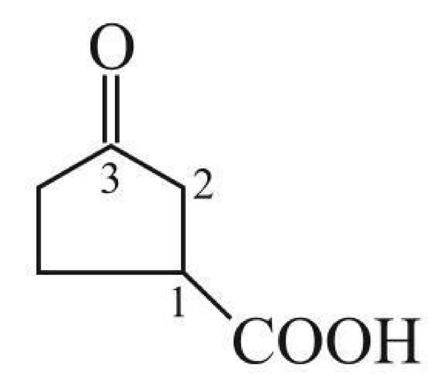
Cyclopentane carbaldehyde

II. 

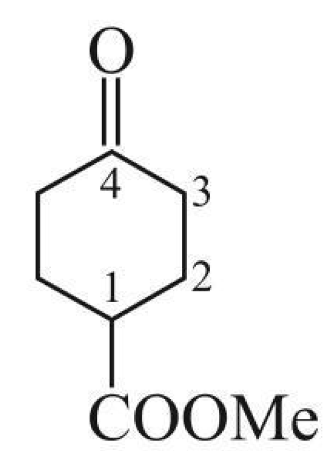
Cyclopentane carbonitrile

III. 

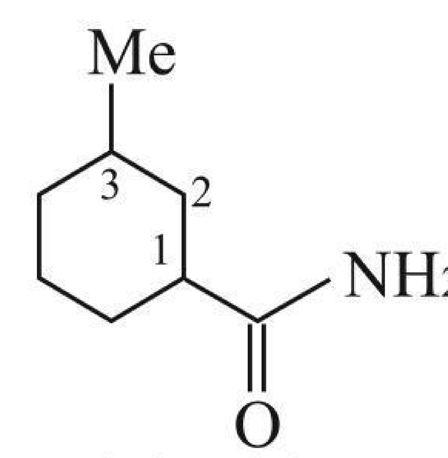
3-Oxo cyclohexane carbaldehyde

IV. 

3-Oxo cyclopentane -1-carboxylic acid

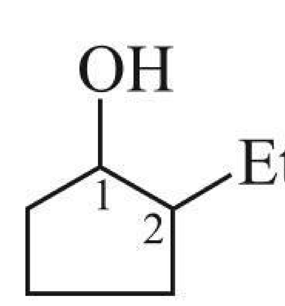
V. 

Methyl(4-oxo cyclohexane) -1- carboxylate

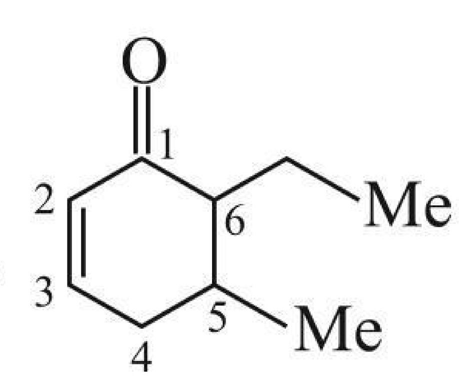
VI. 

3-Methyl cyclohexane -1-carboxamide

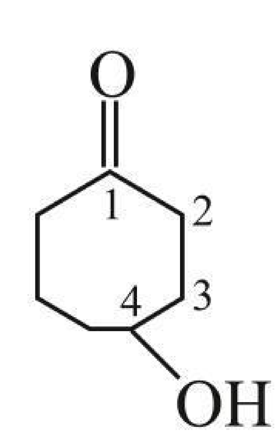
k. If some functional group along with other substituent is present in the ring, the lowest number is given to the principal functional group by using the appropriate prefix or suffix, e.g.,

I. 

2-Ethyl cyclo pentan-1-ol

II. 

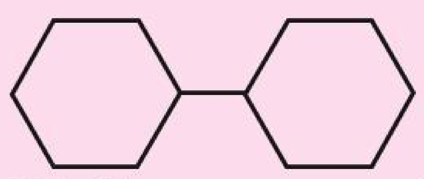
6-Ethyl-5-methyl cyclohex-2-en-1-one

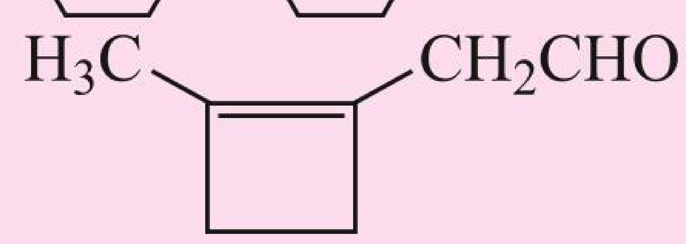
III. 

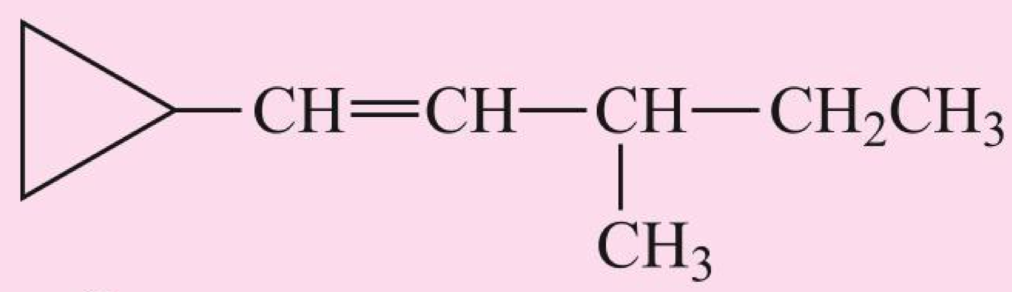
4-Hydroxycyclo heptan-1-one

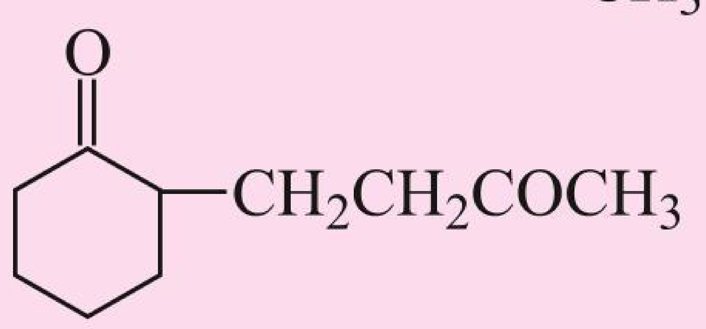
ILLUSTRATION 1.9

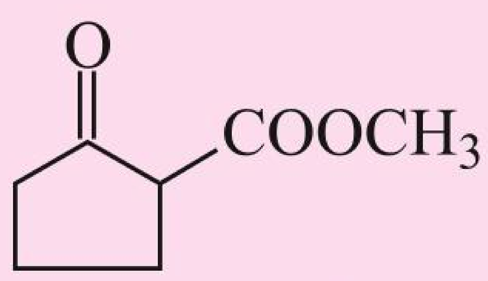
Give the IUPAC names of the following compounds:

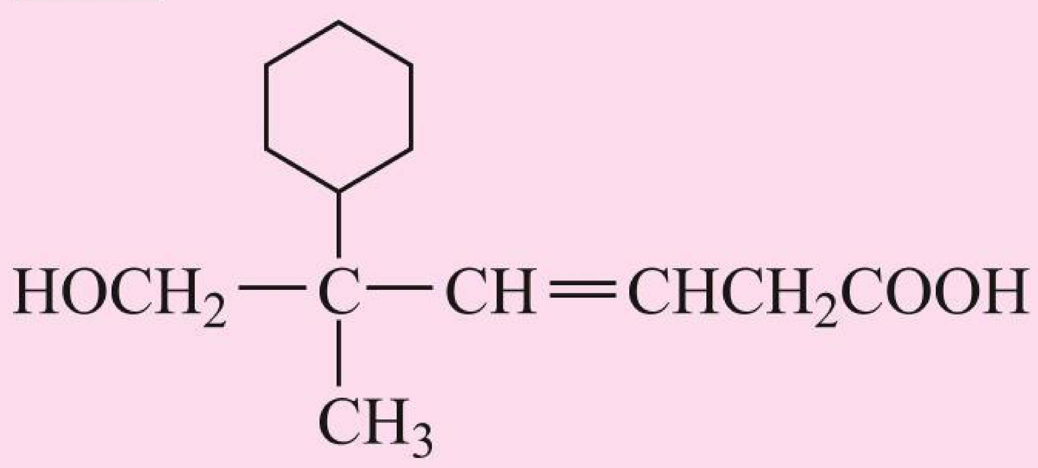
i. 

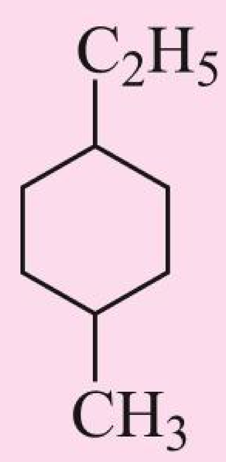
ii. 

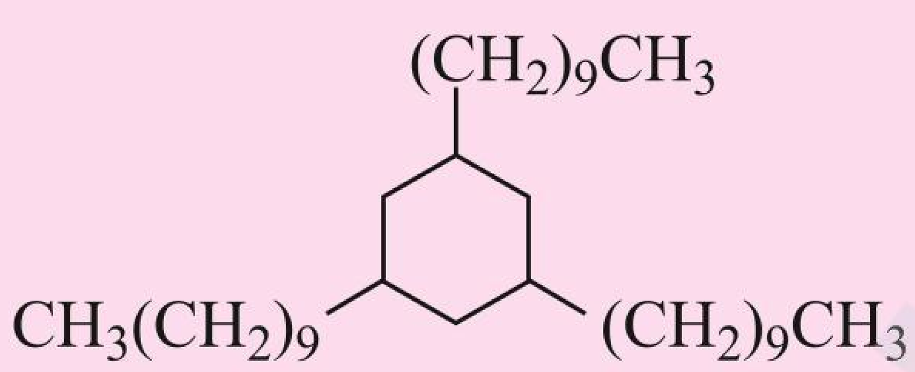
iii. 

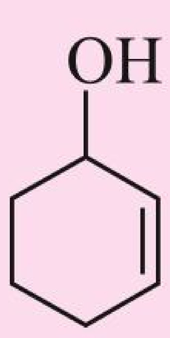
iv. 

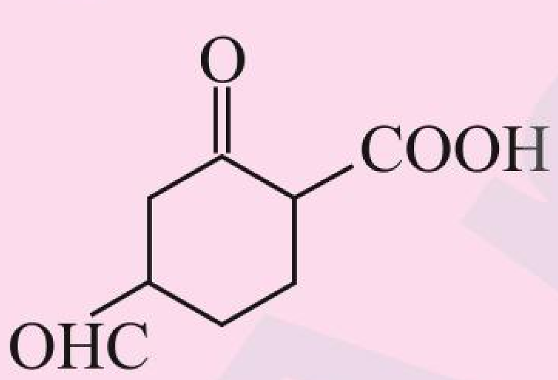
v. 

vi. 

vii. 

viii. 

ix. 

x. 

Sol.

- **i.** Cyclohexylcyclohexane
- **ii.** 2-(2-Methylcylobut-1-enyl)ethanal
- **iii.** 1-Cyclopropyl-3-methylpent-1-ene
- **iv.** 2-(3-Oxobutyl)cyclohexan-1-one
- **v.** Methyl(2-oxocylopentyl)-1-carboxylate or methanoate
- **vi.** 5-Cyclohexyl-6-hydroxy-5-methylhex-3-en-1-oic acid
- **vii.** 1-Ethyl-4-methylcyclohexane
- **viii.** 1,3,5-Tridecylcyclohexane
- **ix.** Cyclohex-2-en-1-ol
- **x.** 4-Formyl-2-oxocyclohexane-1-carboxylic acid (OR) 4-Carbaldehyde-2-oxocyclohexane-methanoic acid.

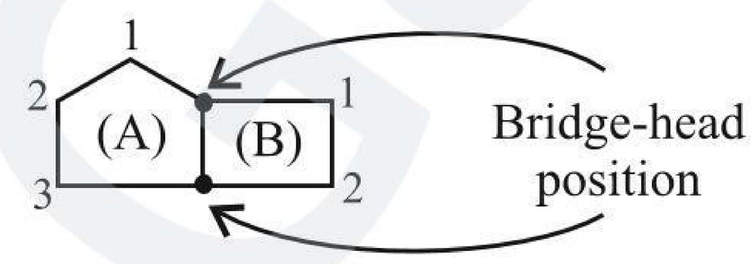
1.7 NAMING OF BICYCLO COMPOUNDS

Compounds with two fused cycloalkane rings are called bicyclo compounds. They are cyclo alkanes having two or more atoms in common.

The prefix bicyclo is followed by the name of the alkane whose number of C atoms is equal to the number of C atoms in the two rings.

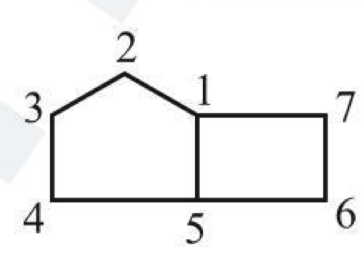
The bracketed numbers show the number of C atoms (except bridge-head position C atoms) in each bridge and they are cited in decreasing order.

For example

a. 

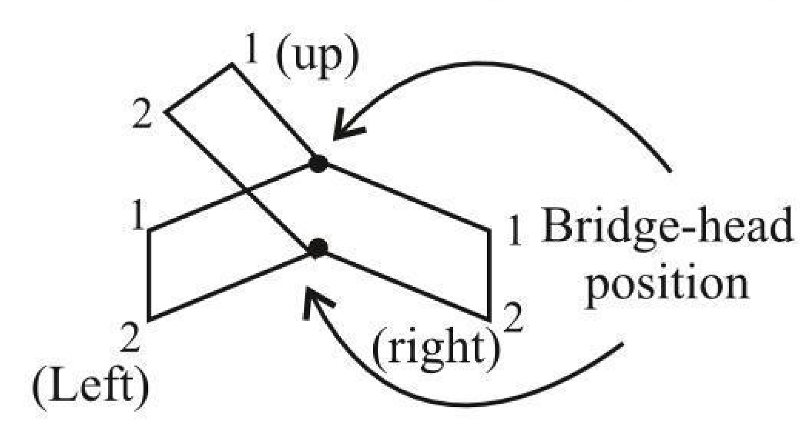
- **i.** Number of C atoms in ring A = 3
- **ii.** Number of C atoms in ring B = 2
- **iii.** Number of C atoms between bridge-head position = 0
- **iv.** Total C atoms = 3 (in ring A) + 2 (in ring B) + 2 (bridge-head position) = 7

Numbering in accordance with naming the compound.



(Numbering starts from the bridge-head to the larger ring and then back to the smaller ring.)

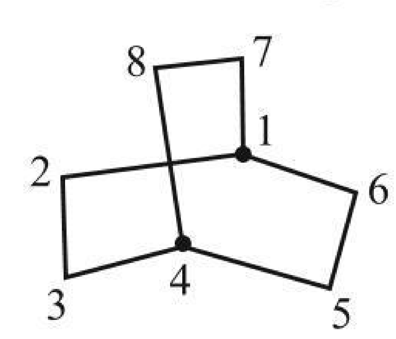
IUPAC name: Bicyclo [3.2.0] heptane

b. 

- **i.** Number of C atoms to the left of the bridge-head = 2
- **ii.** Number of C atoms to the right of the bridge- head = 2
- **iii.** Number of C atoms to the up position of bridge- head = 2

Total C atoms = 2 + 2 + 2 + 2 (bridge-head position) = 8

Numbering in accordance with naming of compound:

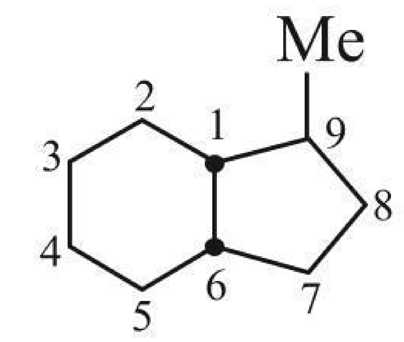


IUPAC name: Bicyclo [2.2.2.] octane.

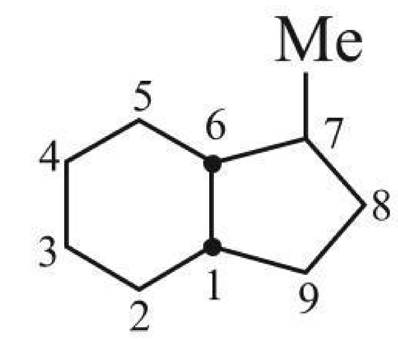
c. If substituents are present, number the bridge-head proceeding first along the longest bridge-head (i.e., the larger ring), then along the next longest bridge-head, and back to the first bridge-head. The shortest bridge is numbered last.

Out of the two bridge-head C atoms, start numbering from that bridge-head C atom from where the position of the substituent is lowest.

For example



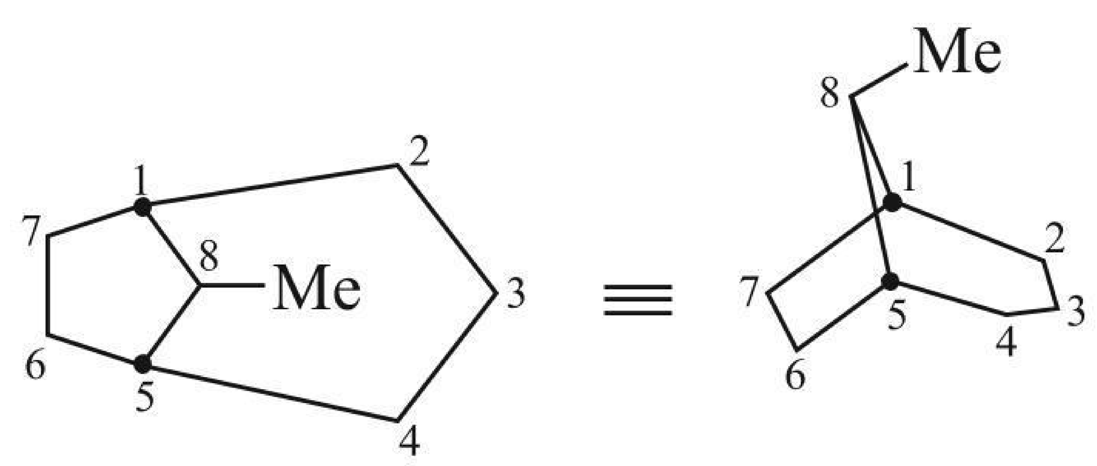
(**Wrong numbering** since the position of the substituent is at C-9)



(**Correct numbering** since the position of the substituent is at the lowest number, i.e., at C-7)

IUPAC name: 7-Methyl bicyclo [4.3.0] nonane

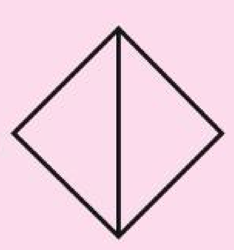
Numbering from the longest bridge-head (i.e., from the larger ring) to the next longest bridge-head (i.e., to the smaller ring).

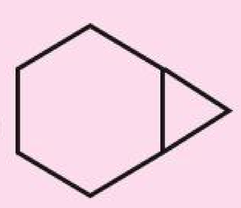


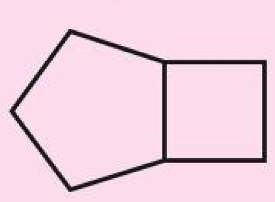
8-Methyl bicyclo [3.2.1] octane.

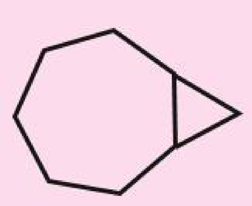
Illustration 1.10

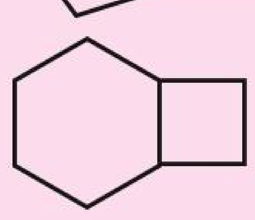
Give the IUPAC names of the following compounds:

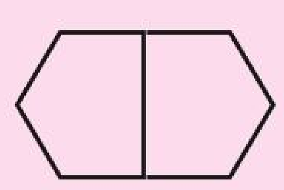
a. 

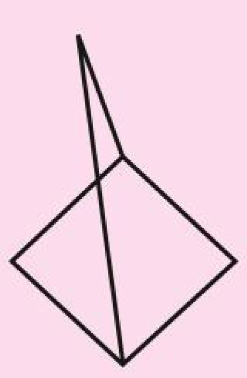
b. 

c. 

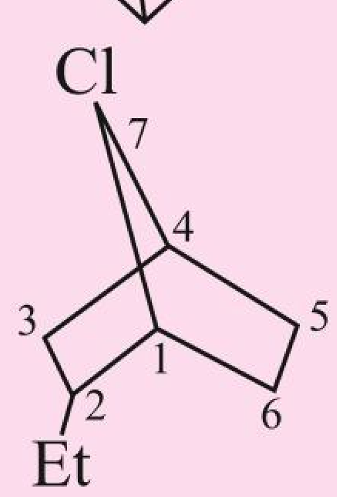
d. 

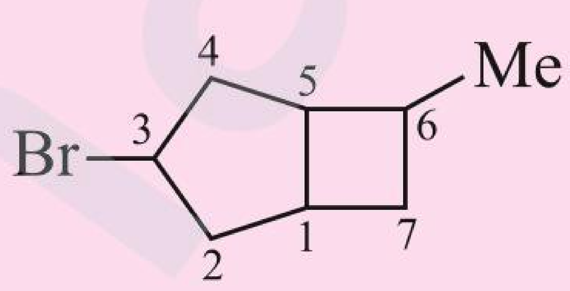
e. 

f. 

g. 

h. 

i. 

j. 

Sol.

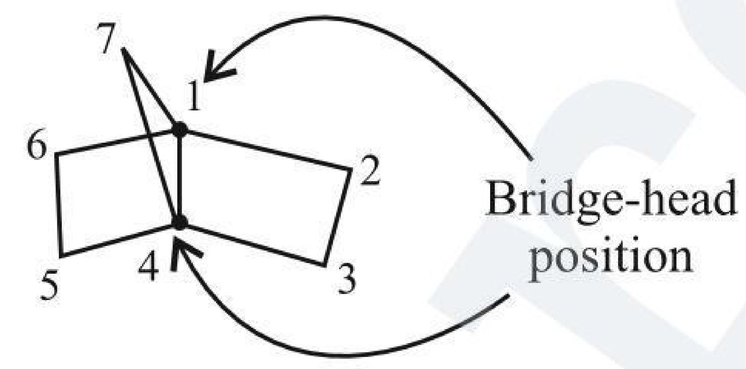
a. Bicyclo [1.1.0] butane
b. Bicyclo [4.1.0] heptane
c. Bicyclo [3.2.0] heptane
d. Bicyclo [5.1.0] octane
e. Bicyclo [4.2.0] octane
f. Bicyclo [3.3.0] octane
g. Bicyclo [1.1.1] pentane
h. Bicyclo [2.1.1] hexane
i. 7-Chloro-2-ethyl bicyclo [2.2.1] heptane
j. 3-Bromo-6-methyl bicyclo [3.2.0] heptane

1.8 TRICYCLO COMPOUNDS

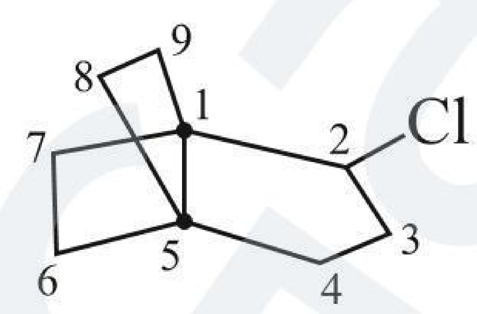
Compounds with three fused rings are called tricyclo compounds. The prefix tricyclo is followed by the name of alkane whose number of C atoms is equal to the number of C atoms in the rings.

The bracketed numbers show the number of C atoms (except the bridge-head position) in each bridge and they are cited in decreasing order.

For example:

i. 

Tricyclo [2.2.1.0] heptane

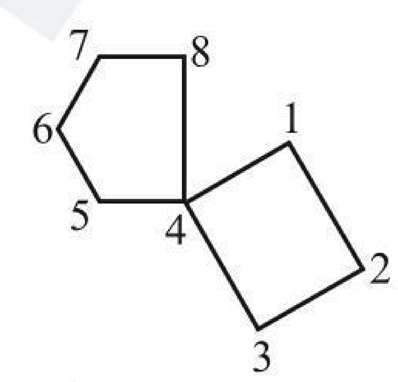
ii. 

2-Chlorotricyclo [3.2.2.0] nonane

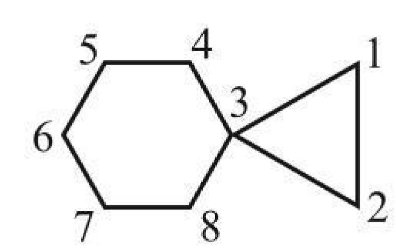
1.9 NAMING OF SPIRANES

Spiranes are polycyclics that share only one C atom. In substituted spiranes, the numbering is started next to the fused C atom in the lower-membered ring.

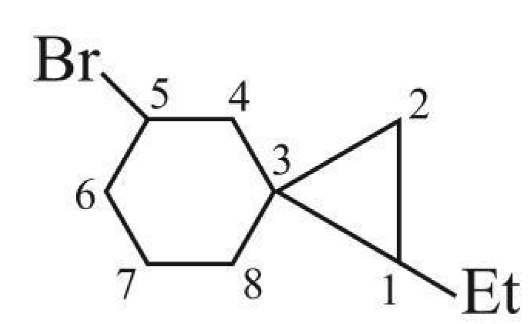
For example:



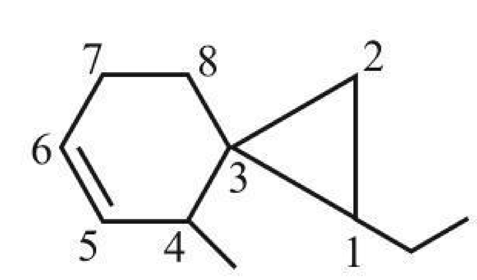
Spiro [3.4] octane



Spiro [2.5] octane



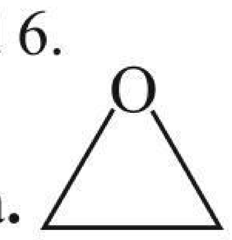
5-Bromo-1-ethyl spiro [2.5] octane



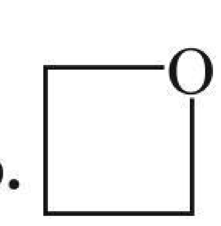
1-Ethyl-4-methyl spiro [2.5]-oct-5-ene

1.10 NAMING OF CYCLIC ETHERS

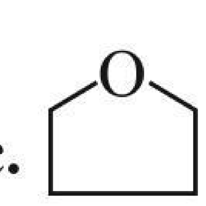
The IUPAC names for cyclic ethers $(CH_2)_nO$, where n = 2, 3, 4, 5, and 6.

a. 

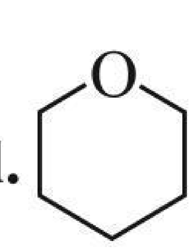
Oxirane

b. 

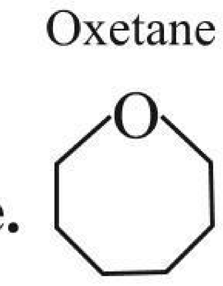
Oxetane

c. 

Oxolane

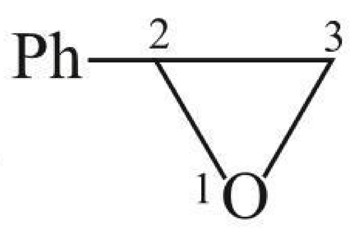
d. 

Oxane

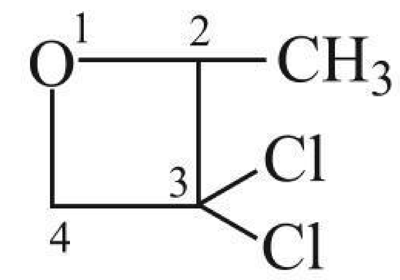
e. 

Oxepane

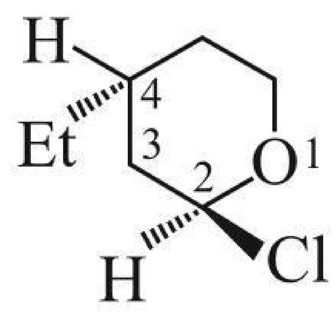
For example:

a. 

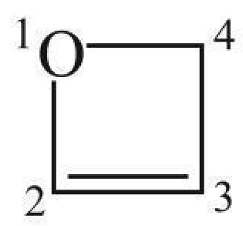
2-Phenyloxirane

b. 

3,3-Dichloro-2-methyl oxetane

c. 

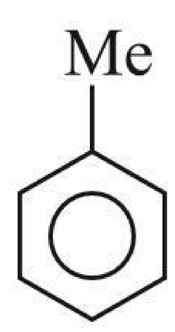
trans-2-Chloro-4-ethyloxane

d. 

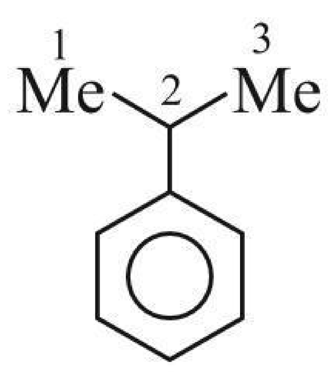
Oxetene

1.11 AROMATIC COMPOUNDS

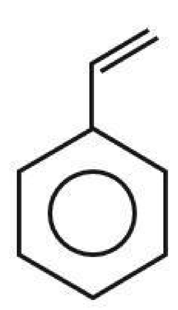
a. No specific rules are required to name aromatic compounds. However, they are named substituted benzene, e.g.,

i. 

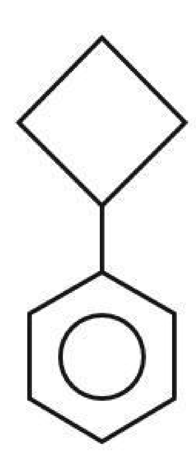
Methyl benzene (Toluene)

ii. 

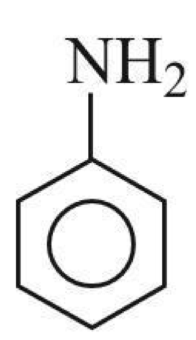
Isopropylbenzene or 2-Phenyl propane or Cumene

iii. 

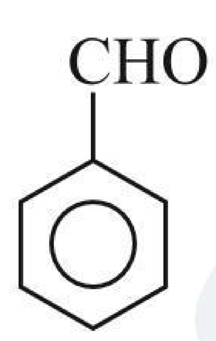
Vinyl benzene (Styrene)

iv. 

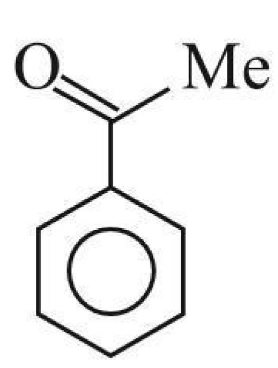
Cyclobutyl benzene

v. 

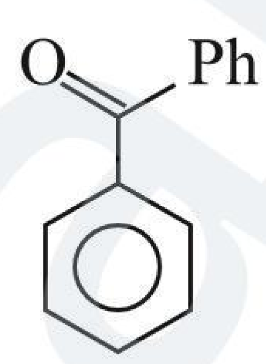
Aniline

vi. 

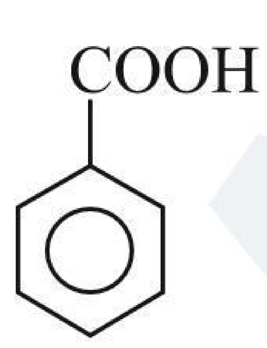
Benzaldehyde

vii. 

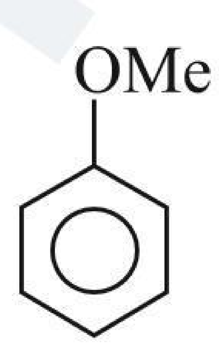
Acetophenone

viii. 

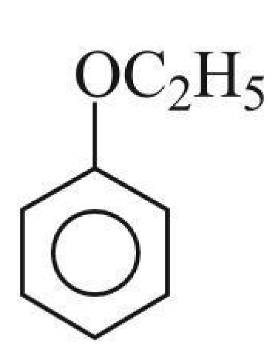
Benzophenone

ix. 

Benzoic acid

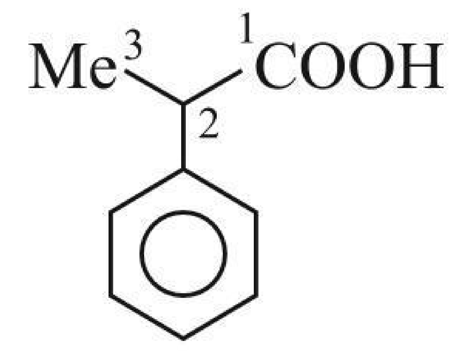
x. 

Anisole (Methoxy benzene)

xi. 

Phenetole (Ethoxy benzene)

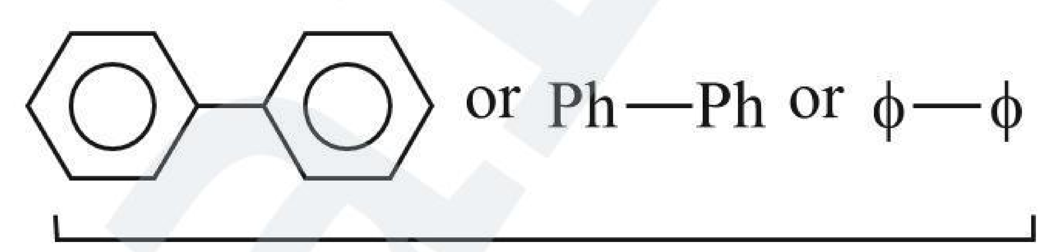
b. When large and complex groups are attached to the benzene ring, the molecule is named as an alkane, alkene, etc., and benzene as side chain derivatives, abbreviated as, Ph—, or C_6H_5— or ϕ. When the benzene ring contains some substituents, it is abbreviated as Ar —.

i. 

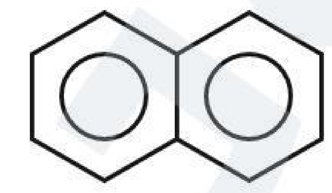
2-Phenyl propanoic acid

ii. 

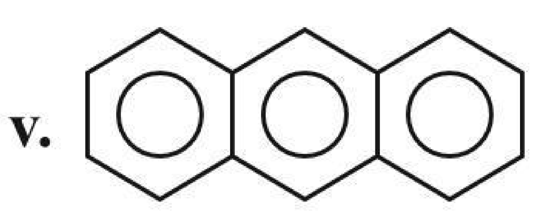
Diphenyl methane

iii. 

Biphenyl or phenyl benzene

iv. 

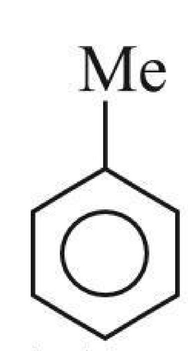
Naphthalene

v. 

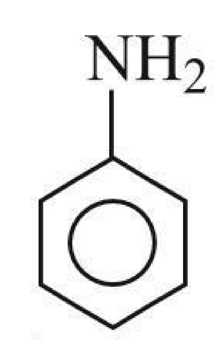
Anthracene

1.12 NOMENCLATURE OF SUBSTITUTED BENZENE COMPOUNDS

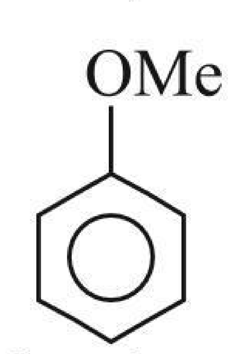
a. Mono-substituted benzene compounds:
According to IUPAC nomenclature, the substituent is placed as prefix and benzene as suffix.
However, common names (written in bracket) of many substituted compounds are commonly used, e.g.,

I. 

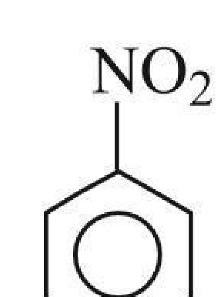
Methyl benzene (Toluene)

II. 

Amino benzene (Aniline)

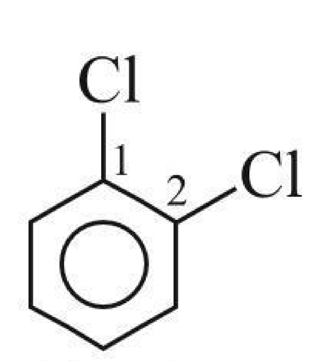
III. 

Methoxy benzene (Anisole)

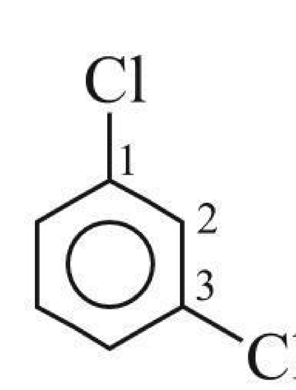
IV. 

Nitro benzene

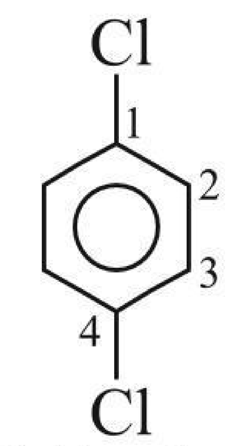
b. If the benzene ring is disubstituted, the substituents are located at the lowest number. In the trivial system of nomenclature, the terms *ortho* (*o*), *meta* (*m*), and *para* (*p*) are used as prefixes to indicate the relative positions 1,2- , 1,3-, and 1,4-, respectively, e.g.,

I. 

1,2-Dichloro benzene (*o*-Dichloro benzene)

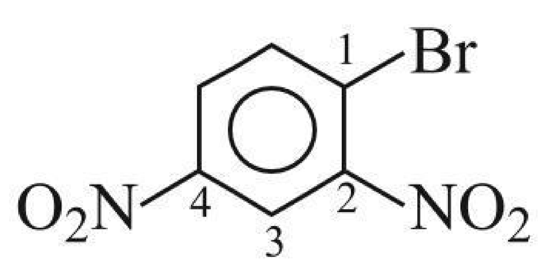
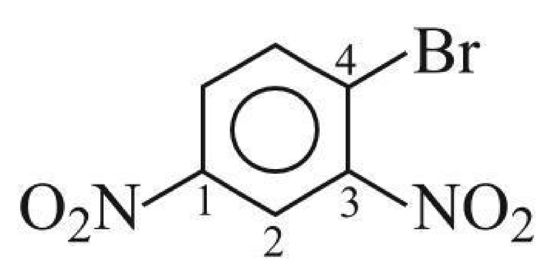
II. 

1,3-Dichloro benzene (*m*-Dichloro benzene)

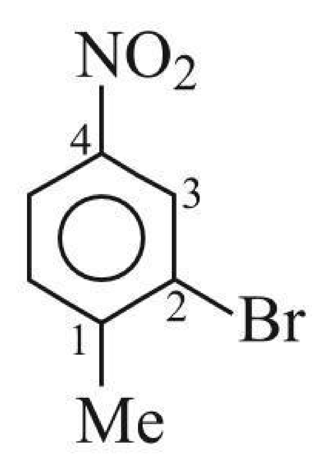
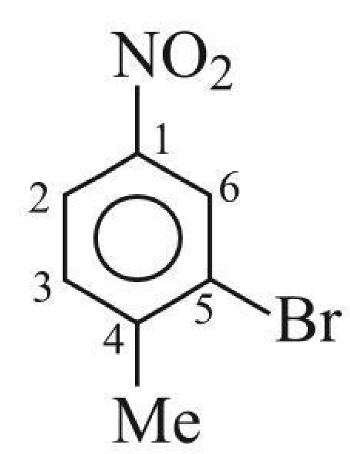
III. 

1,4-Dichloro benzene
(*p*-Dichloro benzene)

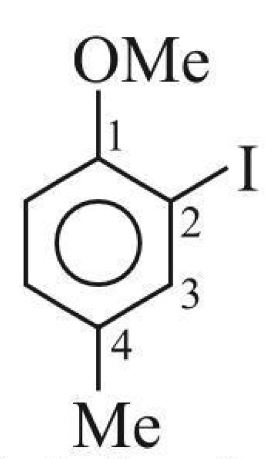
c. If the benzene ring is tri- or higher substituted, then the compounds are named by identifying the substituent position on the ring by following the lowest locant sum rule. The substituent of the base compound is given the number 1 and then the direction of the numbering is selected such that the next substituent gets the lowest number. The substituents are written in the name in alphabetical order, e.g.,

I.  (not) 

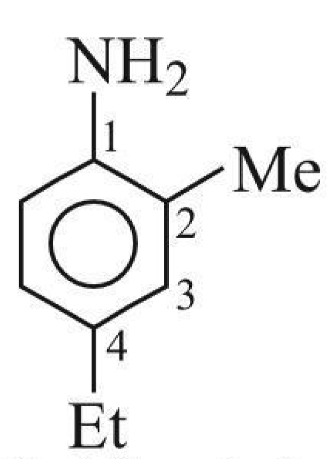
(1-Bromo-2,4-dinitro benzene) (Lowest sum = 1 + 2 + 4 = 7)
(not 4-Bromo-1,3-dinitro benzene) (Highest sum = 4 + 1 + 3 = 8)

II.  (not) 

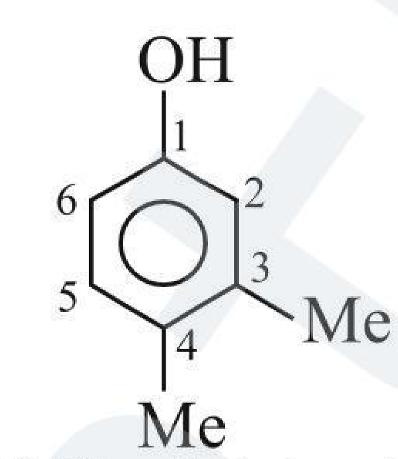
(2-Bromo-1-methyl-4-nitro benzene) (Lowest sum = 2 + 1 + 4 = 7)
(not 4-Methyl-5-bromo-nitro benzene) (Highest sum = 4 + 5 + 1 = 10)

III. 

2-Iodo-4-methyl anisole

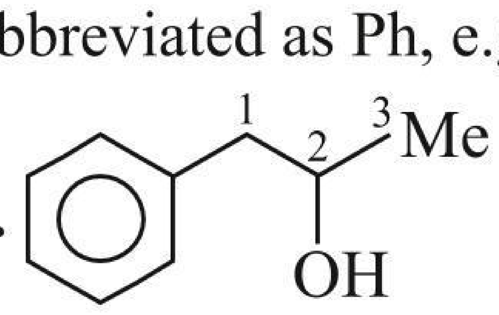
IV. 

4-Ethyl-2-methyl aniline

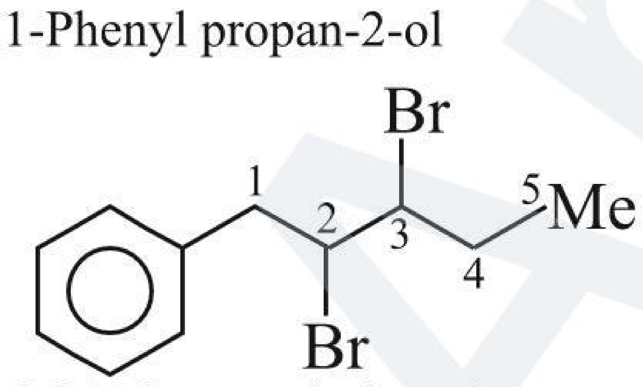
V. 

3,4-Dimethyl phenol

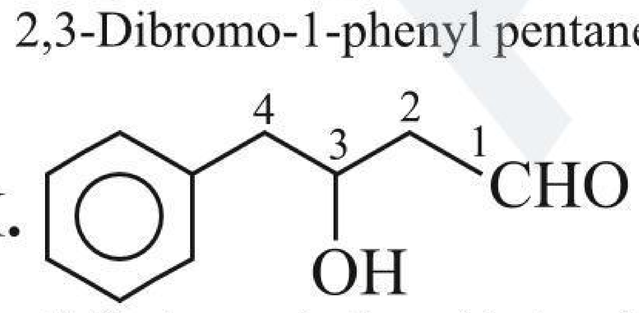
d. When a benzene ring is attached to an alkane with a functional group, it is considered as a substituent instead of a parent. The name for benzene as substituent is phenyl (C_6H_5—) also abbreviated as Ph, e.g.,

I. 

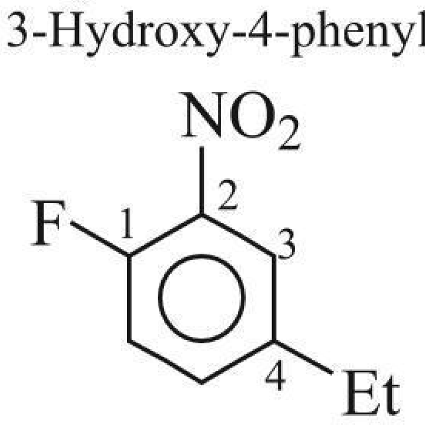
1-Phenyl propan-2-ol

II. 

2,3-Dibromo-1-phenyl pentane

III. 

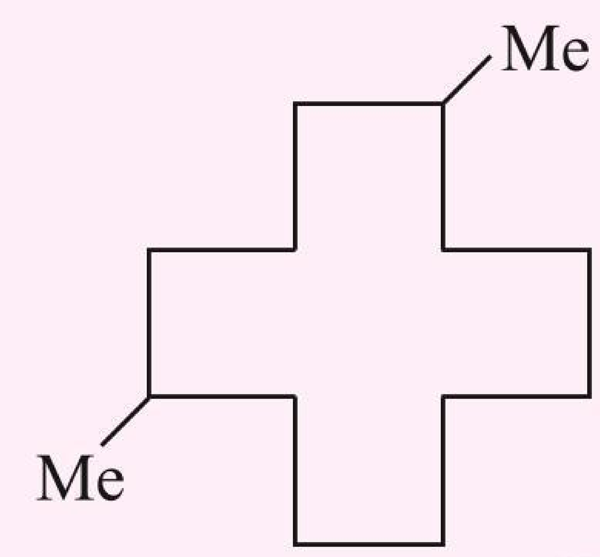
3-Hydroxy-4-phenyl butanal

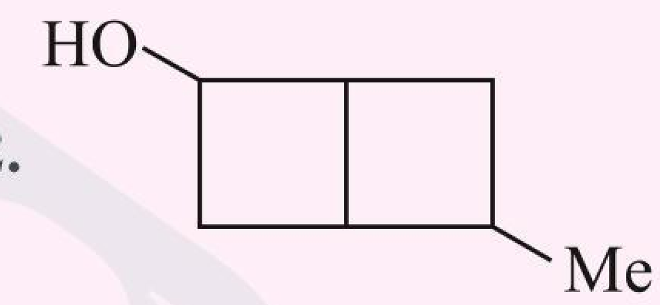
IV. 

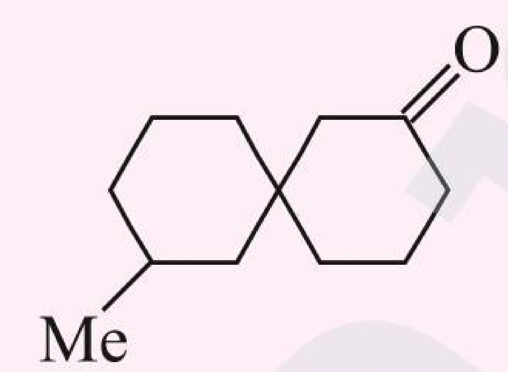
4-Ethyl-1-fluoro-2-nitro benzene

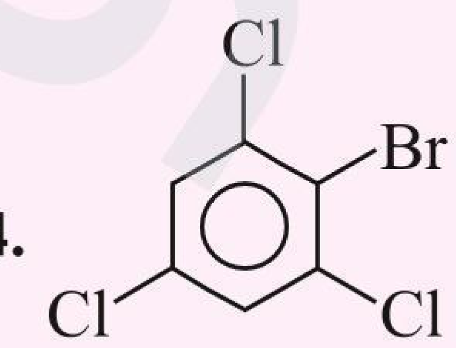
Concept Application Exercise 1.2

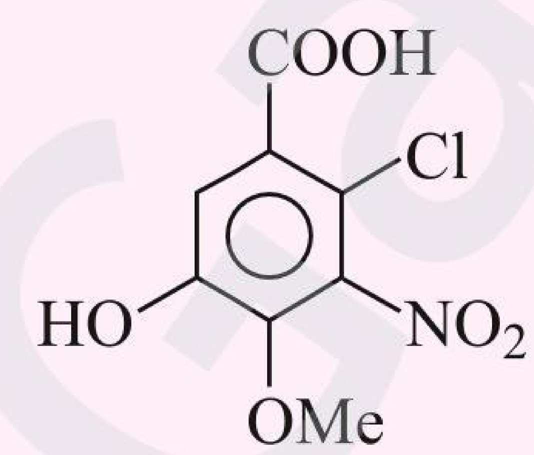
Write the IUPAC name of the following compounds:

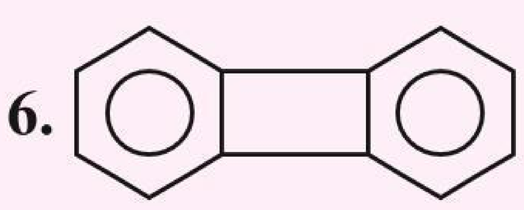
1. 

2. 

3. 

4. 

5. 

6. 

1.13 WRITING THE STRUCTURAL FORMULA FROM THE GIVEN IUPAC NAME

The IUPAC name of an organic compound consists of the following parts:

a. Root word
b. 1° suffix
c. 2° suffix
d. 1° prefix
e. 2° prefix

a. Root word indicates the longest chain. Thus, first locate the longest chain from the root word. Write the number of C atoms in a straight chain or in zig-zag manner (for bond line structure) and then number them from any end.

b. 1° suffix (-ane, -ene, or -yne) indicates the nature of the chain. Put the multiple bonds at a proper place in the chain.

c. 2° suffix indicates the principal functional group. Put it at a proper place in the chain.

d. Prefixes are the substituents or secondary functional groups. Put them at a proper place with the help of locants.

e. Add H atoms to satisfy valencies of each C atom if stick formula is used. If the structure is written in bond line, then there is no need of adding H atoms.

Illustration 1.11

Write the structural formula and bond line formula of 2-chloro-3 methyl hexanoic acid.

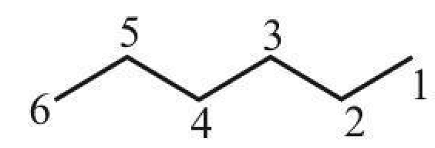
Sol. Structural formula of 2-chloro-3-methyl hexanoic acid

i. Root word = hex, 1° suffix = an (e).

ii. Stick formula

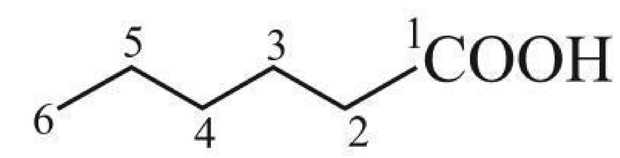
$\overset{6}{C}-\overset{5}{C}-\overset{4}{C}-\overset{3}{C}-\overset{2}{C}-\overset{1}{C}$

Bond line formula

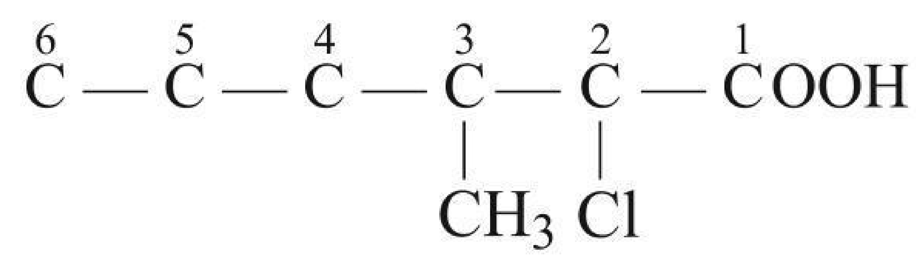


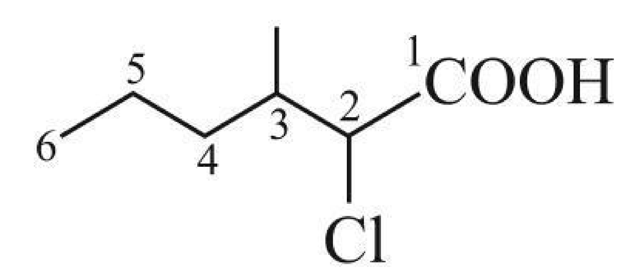
iii. Functional group = (—COOH) at C-1.

$\overset{6}{C}-\overset{5}{C}-\overset{4}{C}-\overset{3}{C}-\overset{2}{C}-\overset{1}{C}OOH$

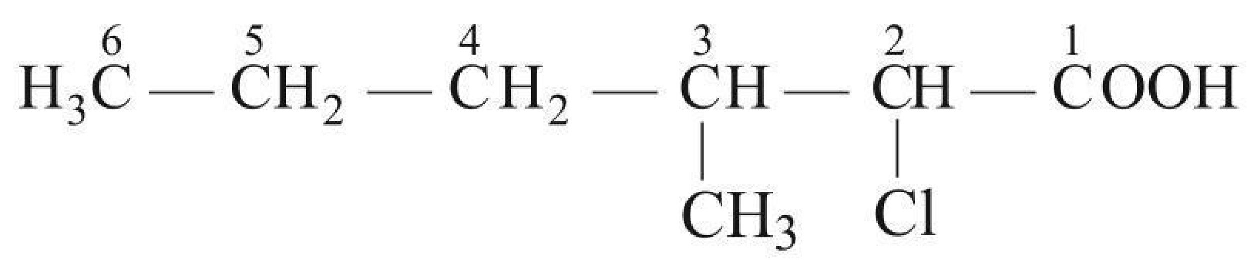


iv. Substituent = Cl at C-2 and methyl group at C-3





v. Add H atoms in the stick formula to structure.
Satisfy all the four valencies of H atoms.
The structural formula is



No need of adding H atoms in the bond line structure

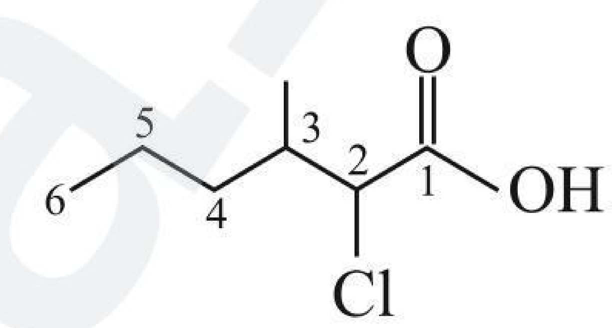
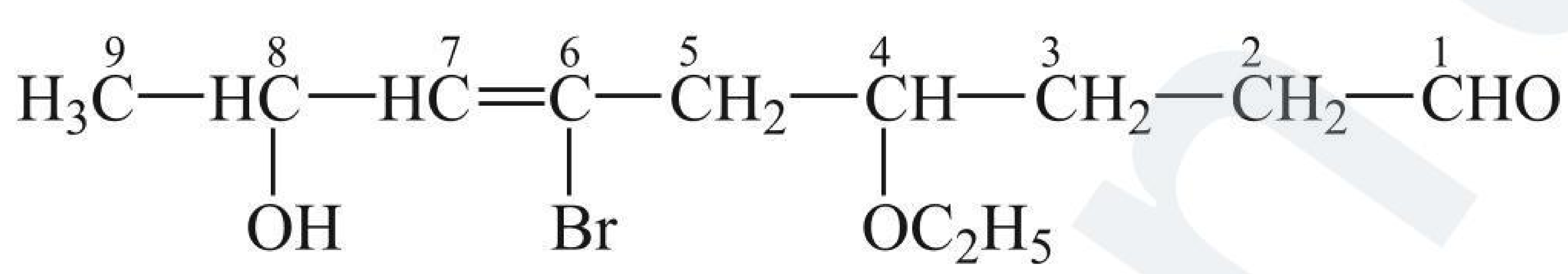


ILLUSTRATION 1.12

Write the structural formula and bond line formula of 6-bromo-8-hydroxy-4 ethoxy non-6-enal.

Sol. Structural formula of 6-bromo-8-hydroxy-4-ethoxy non-6-enal:

$\underbrace{\text{6-Bromo-8-hydroxy-4-ethoxy}}_{\substack{\text{Substituents with locants} \\ \text{2° prefix}}}\ \underbrace{\text{non}}_{\text{Root word}}\text{-}\underbrace{\text{6-en}}_{\substack{\text{(C=C) at C-6} \\ \text{1° suffix}}}\underbrace{\text{al}}_{\substack{\text{2° Suffix} \\ \text{(Functional group)}}}$



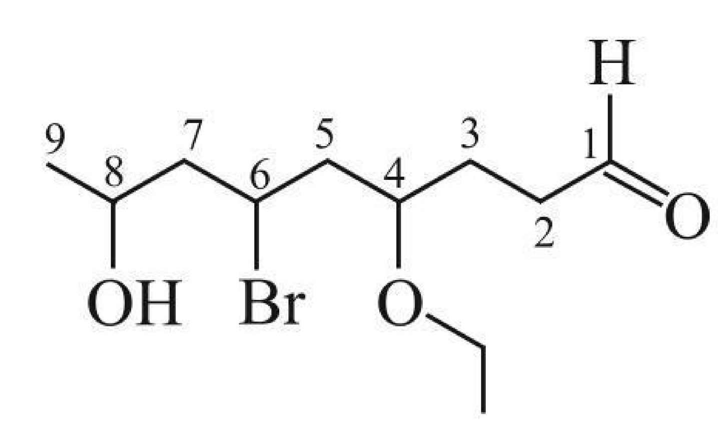


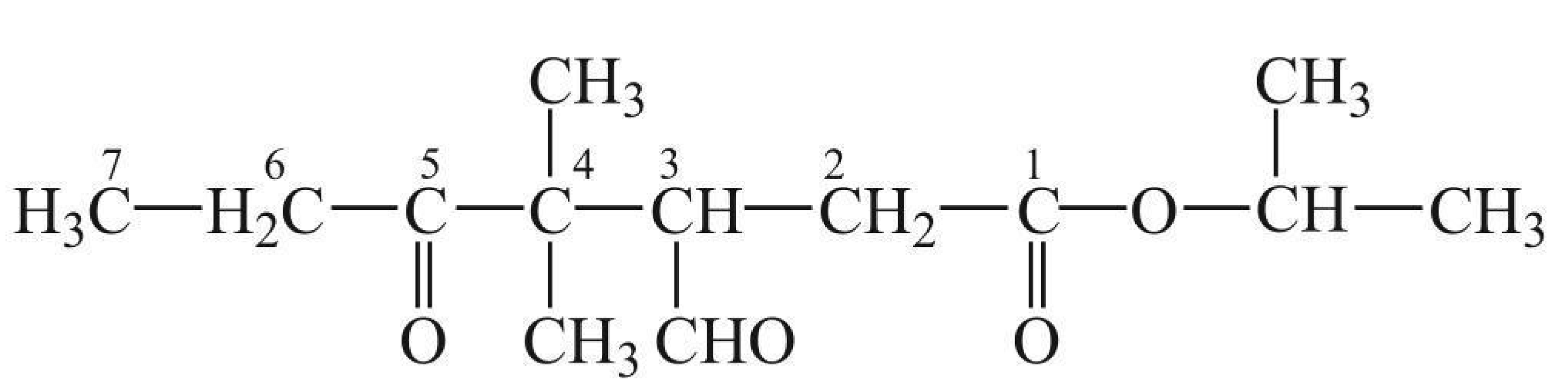
ILLUSTRATION 1.13

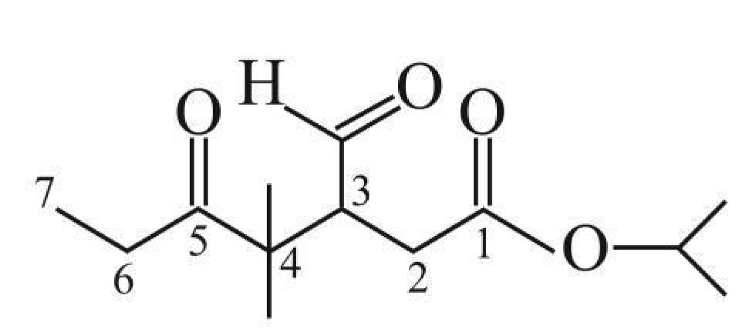
Write the structural formula and bond line formula of isopropyl-3-formyl-4, 4-dimethyl-5-oxoheptanoate.

Sol. Structural formula of isopropyl-3-formyl-4,4-dimethyl-5-oxoheptanoate:

Formyl = —CHO at C-3
Oxo = (C=O) at C-5
Two methyl = $2CH_3$ at C-4
Isopropyl = Me—CH— with ester
$\qquad\qquad\qquad\quad |$
$\qquad\qquad\qquad\ CH_3$

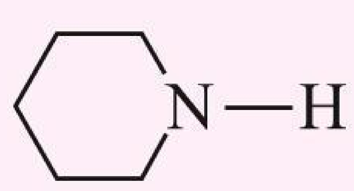
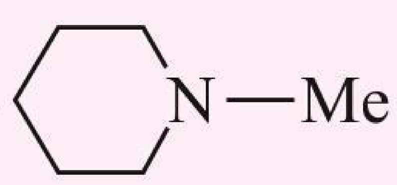
$\underbrace{\text{Isopropyl-3-formyl-4,4-dimethyl-5-oxo}}_{\substack{\text{Substituents with locants} \\ \text{2° prefix}}}\ \underbrace{\text{hept}}_{\text{Root word}}\ \underbrace{\text{an}}_{\text{1° suffix}}\ \underbrace{\text{oate}}_{\substack{\text{2° suffix} \\ \text{(Functional group)}}}$

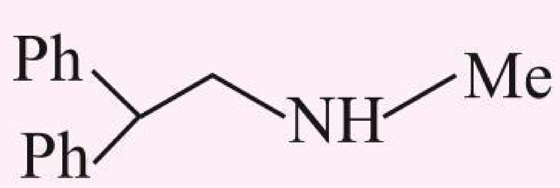


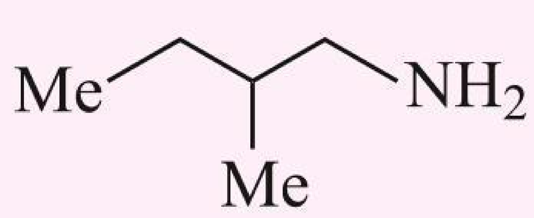
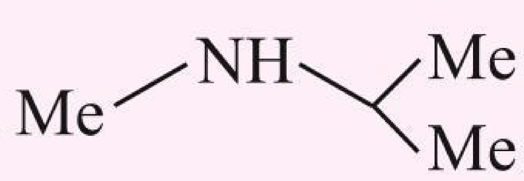


Concept Application Exercise 1.3

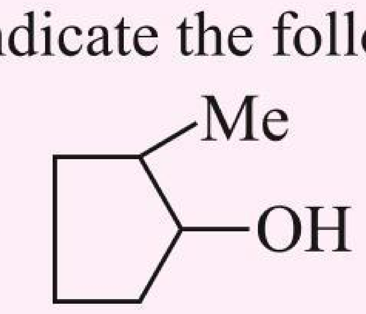
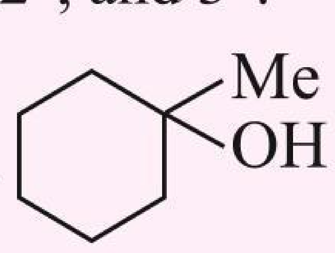
1. Write the structural formula for each of the following.
   - **a.** A 3° amine with the formula C_3H_9N.
   - **b.** Three ethers with the formula $C_4H_{10}O$.
   - **c.** A 3° alcohol with the formula C_4H_8O.
   - **d.** Three 2° alkyl halides with the formula $C_5H_{11}Cl$.
   - **e.** Three 1° alcohols with the formula C_4H_8O.
   - **f.** A 2° alcohol with the formula C_3H_6O.
   - **g.** Two esters with the formula $C_3H_6O_2$.
   - **h.** Four 1° alkyl halides with the formula $C_5H_{11}Cl$.
   - **i.** A 3° alkyl halide with the formula $C_5H_{11}Cl$.
   - **j.** Three aldehydes with the formula $C_5H_{10}O$.
   - **k.** Three ketones with the formula $C_5H_{10}O$.
   - **l.** A 2° amine with the formula C_3H_9N.
   - **m.** Two amides with the formula C_2H_5NO.
   - **n.** Two 1° amines with the formula C_3H_9N.
2. Indicate the following as 1°, 2°, and 3°.

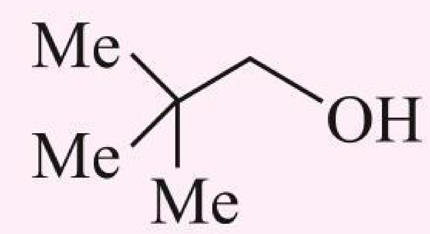
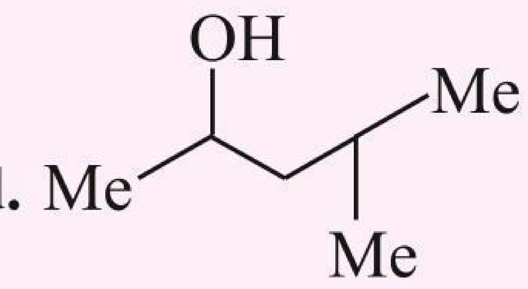
   a. 
   b. 

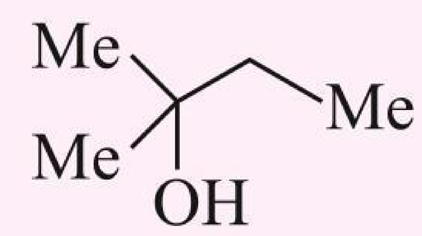
   c. 
   d. Et_3N

   e. 
   f. 

3. Indicate the following as 1°, 2°, and 3°.

   a. 
   b. 

   c. 
   d. 

   e. 

4. Write the structural formula for seven compounds with the formula C_3H_6O and identify the functional groups.
5. There are seven isomeric compounds with the formula $C_4H_{10}O$. Write their structures and identify their functional groups.
6. There are four alkyl chlorides with the formula C_4H_9Cl. Write their structures and identify them as 1°, 2°, and 3° alkyl chlorides.
7. There are four amides with the formula C_3H_7NO.
   - **a.** Write their structures.
   - **b.** Identify the amide which has lower melting point and boiling point than the other three.
8. Write the IUPAC name of the compound (A) which is a 2-methyl branched alkane having a molecular mass of 254. This compound is a sex-attractant and is isolated from female tiger moths.
9. Write the IUPAC name of the compound (A) which is a 2-methyl alkane with the molecular mass of 72.
10. Write the IUPAC name of the compound (A) in which the molar ratios of C, H, and O are equal having a molecular mass of 58.

Exercises

Single Correct Answer Type

1. The decreasing order of priority for the following functional groups is:
   I. —COOH **II.** —SO_3H
   III. —COOR **IV.** —COCl
   (1) (IV) > (III) > (II) > (I) (2) (I) > (II) > (III) > (IV)
   (3) (II) > (I) > (III) > (IV) (4) (IV) > (III) > (I) > (II)

2. The decreasing order of priority for the following functional groups is:
   I. —C≡N **II.** —$CONH_2$
   III. >C=O **IV.** —CHO
   (1) (II) > (I) > (IV) > (III) (2) (III) > (IV) > (I) > (II)
   (3) (I) > (II) > (IV) > (III) (4) (I) > (II) > (III) > (IV)

3. The decreasing order of priority for the following functional groups is:
   I. —OH **II.** —C≡C—
   III. >C=C< **IV.** —NH_2
   (1) (IV) > (I) > (II) > (III) (2) (IV) > (I) > (III) > (II)
   (3) (I) > (IV) > (III) > (II) (4) (II) > (III) > (IV) > (I)

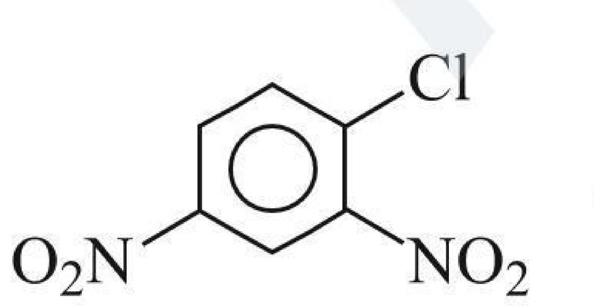
4. The number of 1°, 2°, and 3° H atoms in 2,5,6-trimethyl octane, respectively, is:
   (1) 16, 5, 3 (2) 15, 6, 3
   (3) 16, 6, 3 (4) 15, 5, 2

5. The number of 1°, 2°, and 3° H atoms in 3-ethyl-5-methyl heptane, respectively, is:
   (1) 12, 8, 1 (2) 14, 4, 2
   (3) 12, 6, 2 (4) 12, 8, 2

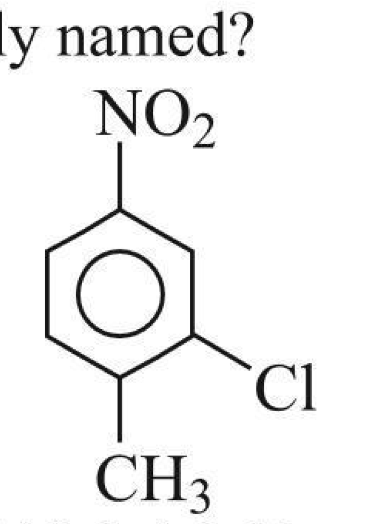
6. The number of σ- and π-bonds in hexane-2,4-diol, respectively, is:
   (1) 18, 2 (2) 17, 2
   (3) 17, 1 (4) 18, 2

7. The number of σ- and π-bonds in 5-oxohexanoic acid, respectively, is:
   (1) 18, 2 (2) 18, 1
   (3) 17, 2 (4) 17, 1

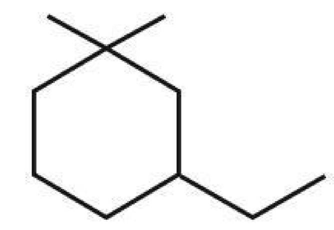
8. Which of the following is correctly named?

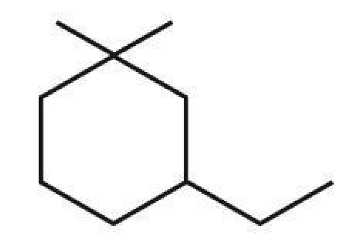
   (1) 

   (4-Chloro-1,3-dinitro benzene)

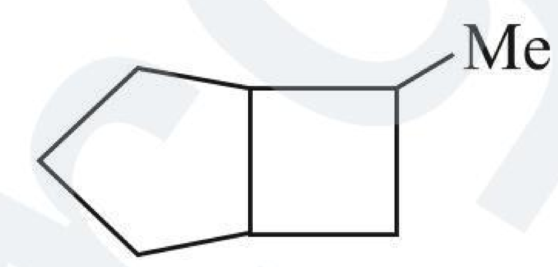
   (2) 

   4-Methyl-5-chloro-nitro benzene

   (3) 
   3-Ethyl-1,1-dimethyl cyclo hexane

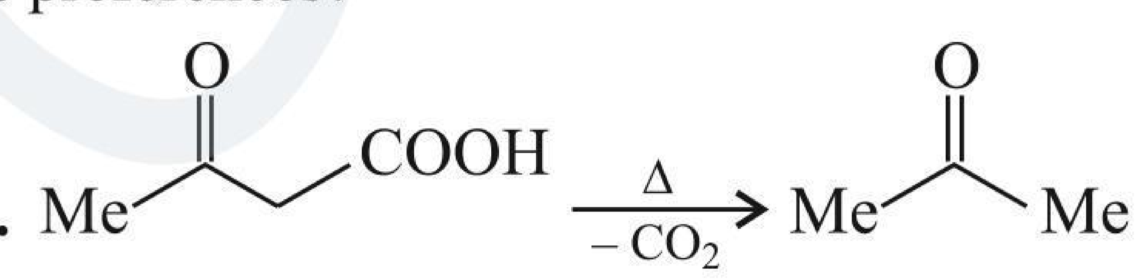
   (4) 
   1-Ethyl-3,3-dimethyl cyclohexane

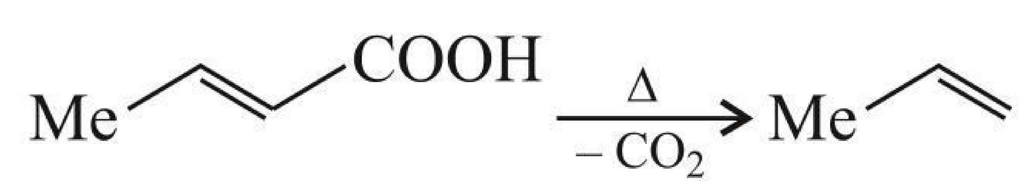
9. The systematic naming of the following cycloalkane is:

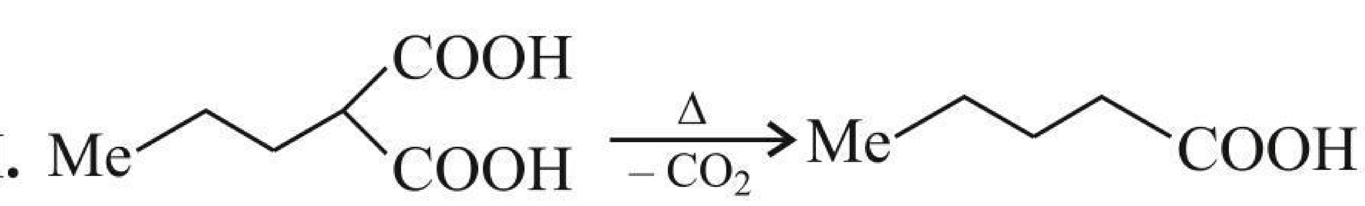


   (1) 6-Methyl bicyclo [3.2.0] heptane
   (2) 7-Methyl bicyclo [3.2.0] heptane
   (3) 2-Methyl bicyclo [3.2.0] heptane
   (4) 3-Methyl bicyclo [3.2.0] heptane

10. In which of the following reactions, the principal group loses its preferences?

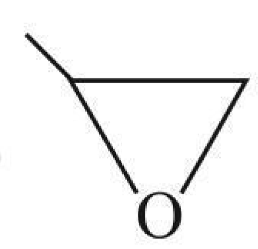
   I. 

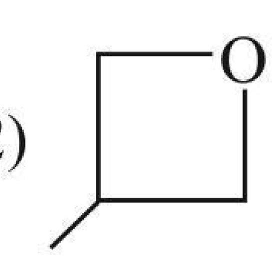
   II. 

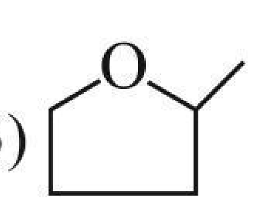
   III. 

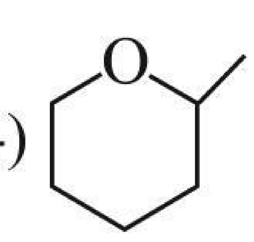
   (1) I (2) I, II
   (3) I, II, III (4) I, III

11. Which of the following is oxetane?

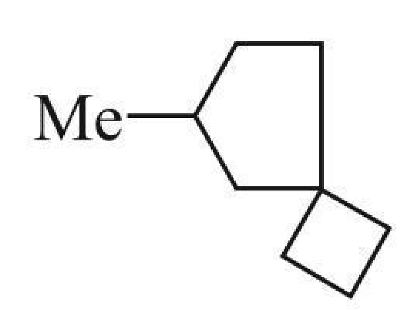
   (1) 

   (2) 

   (3) 

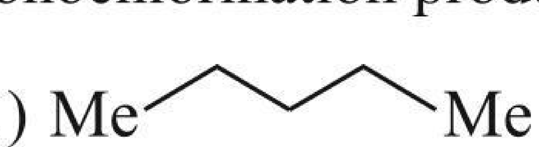
   (4) 

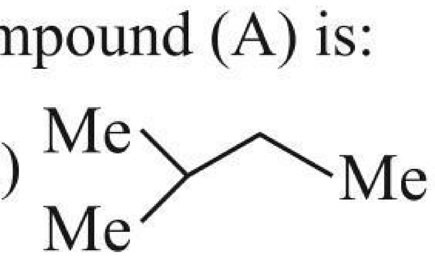
12. The systematic nomenclature of the following spiro-compound is:

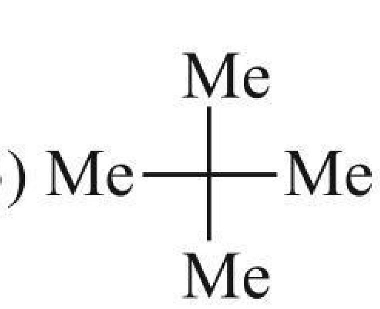


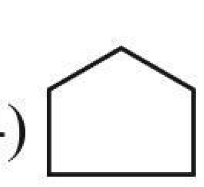
   (1) 2-Methyl spiro [3.4] octane
   (2) 3-Methyl spiro [3.4] octane
   (3) 6-Methyl spiro [3.4] octane
   (4) 7-Methyl spiro [3.4] octane

13. An alkane (A) having a molecular mass of 72 produces one monochlorination product. Compound (A) is:

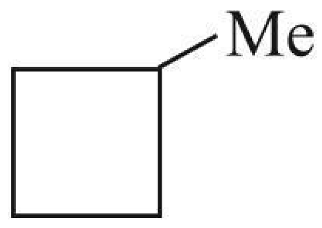
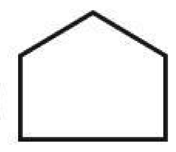
   (1) 

   (2) 

   (3) 

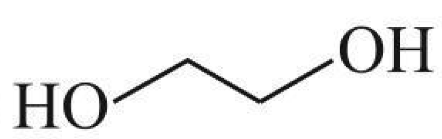
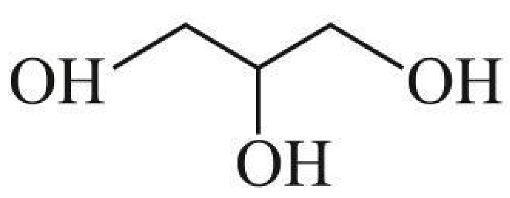
   (4) 

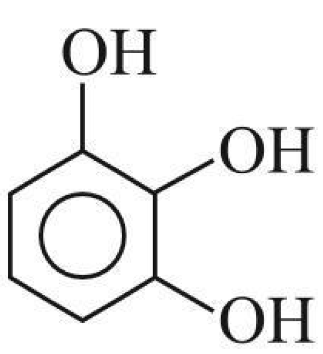
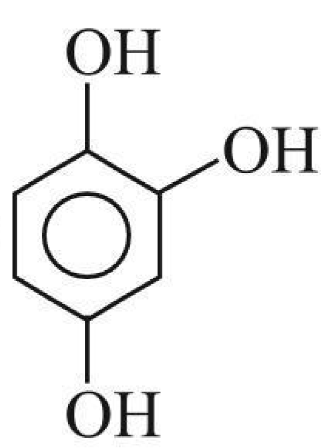
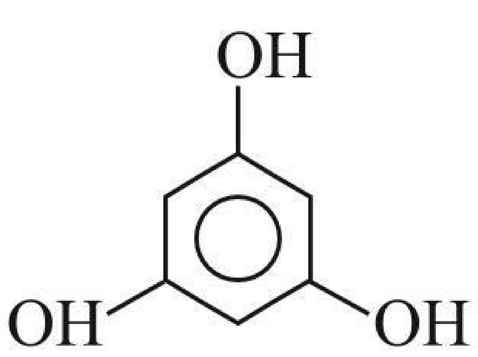
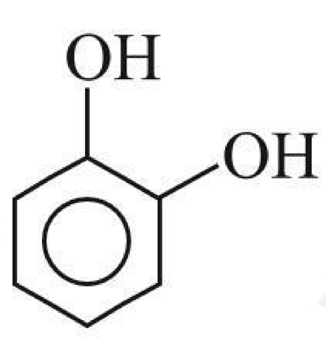
14. A compound (A) with molecular formula C_5H_{10} gives one monochlorination product. Compound (A) is:

(1) 
(2) 
(3) 
(4) 

15. Which of the following is a 3° amine?
(1) Propan-2-amine
(2) *N*-Methyl ethanamine
(3) Allyl amine
(4) *N,N*-Diethyl butan-1-amine

16. Which of the following is a 3° alcohol?
(1) *t*-Butyl carbinol
(2) 2-Methyl propan-2-ol
(3) 2-Methyl butan-1-ol
(4) Isoamyl alcohol

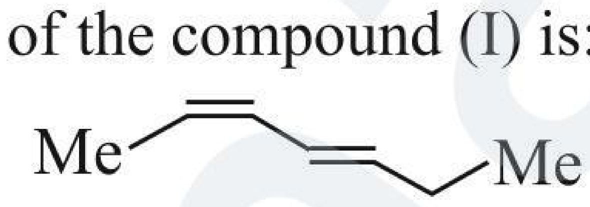
17. Which of the following is zerone?
(1) MeOH
(2) 
(3) 
(4) EtOH

18. Which of the following is pyrogallol?
(1) 
(2) 
(3) 
(4) 

19. In 3-chloro cyclohexanol, the primary prefix is:
(1) 3-Chloro
(2) Cyclo
(3) an(e)
(4) -ol

20. In 2-Chloro-3-methyl hexanoic acid, the primary suffix is:
(1) 2-Chloro-
(2) -3-Methyl
(3) an(e)
(4) oic acid

21. The correct name of the compound (I) is:

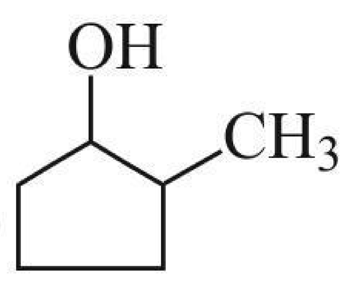
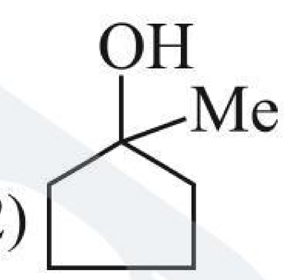
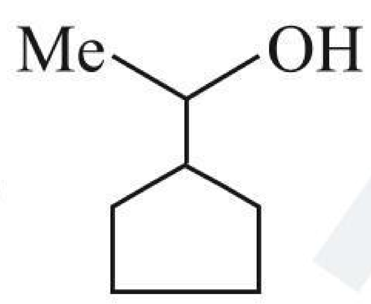
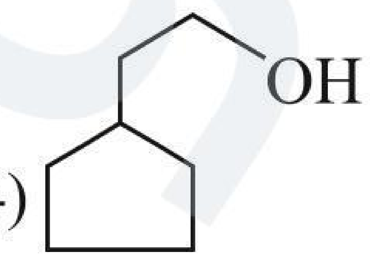


(1) (*E*-2), (*E*-4), Hepta-2,4-diene
(2) (*Z*-2), (*Z*-4), Hepta-2,4-diene
(3) (*E*-2), (*Z*-4), Hepta-2,4-diene
(4) (*Z*-2), (*E*-4), Hepta-2,4-diene

22. Which of the following is not a cumulated diene?
(1) Hexa-1,2-diene
(2) Hexa-2,3-diene
(3) Penta-2,3-diene
(4) Penta-1,3-diene

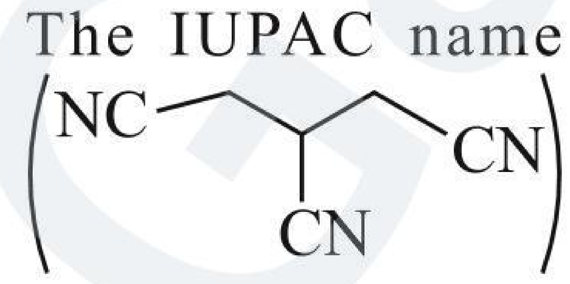
23. Which of the following has only 1° and 2° C atoms?
(1) 2-Methyl butane
(2) Butane
(3) 2,2-Dimethyl butane
(4) 2,2,3,3-Tetramethyl pentane

24. The IUPAC name of vinyl acetylene is:
(1) Pent-1-en-4-yne
(2) Pent-4-yn-1-ene
(3) But-1-en-3-yne
(4) But-1-yn-3-ene

25. Which of the following structures represents cyclopentyl methyl carbinol?
(1) 
(2) 
(3) 
(4) 

26. The IUPAC name of acrolein is:
(1) But-2-enal
(2) Prop-2-enal
(3) But-3-enal
(4) 2-Methyl prop-2-enal

27. The IUPAC name of the following compound is:

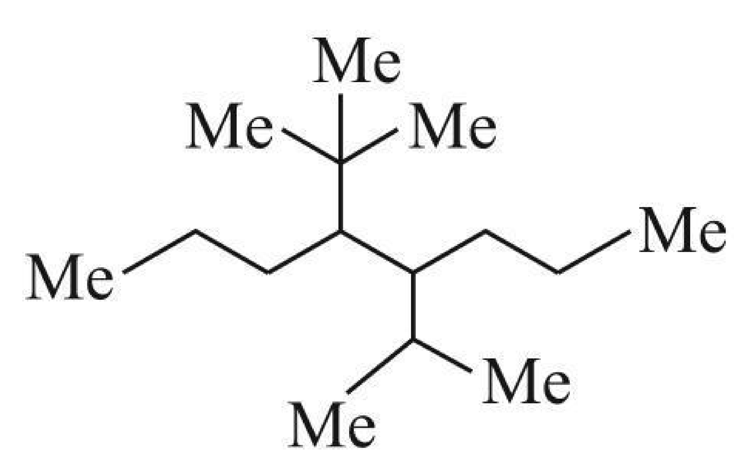


(1) Propane-1,2,3-tricarbonitrile
(2) 3-Cyanopetane-1,5-dinitrile
(3) Pentane-1,3,5-trinitrile
(4) All

28. Which of the following is not the name of CH_3NC?
(1) Methyl isocyanide
(2) Aceto isonitrile
(3) Methyl carbylamine
(4) Acetonitrile

29. The IUPAC name of PhCN is:
(1) Phenyl cyanide
(2) Benzonitrile
(3) Benzene nitrile
(4) All

30. Give the IUPAC name of:



(1) 2,2-Dimethyl-3-propyl-4-isopropyl heptane
(2) 4-Isopropyl-5-*t*-butyl octane
(3) 4-*t*-Butyl-5-isopropyl octane
(4) 2-Methyl-3-propyl-4-isopropyl heptane

31. Which of the following statements is wrong for a homologous series?
(1) All members have a general formula.
(2) All members have the same functional group.
(3) All members have the same chemical properties.
(4) All members have the same physical properties.

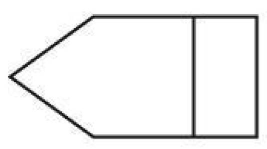
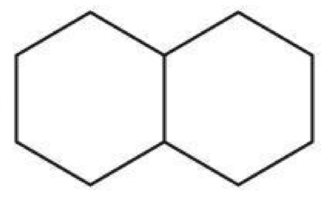
32. The alkane which has only 1° H atoms is:
(1) Neopentane
(2) Isopentane
(3) Pentane
(4) 2,2-Dimethyl butane

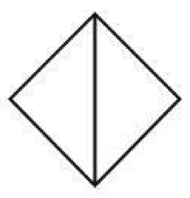
33. Which group is always taken as a substituent in the IUPAC system of nomenclature?

(1) —NO_2 (2) —C≡N

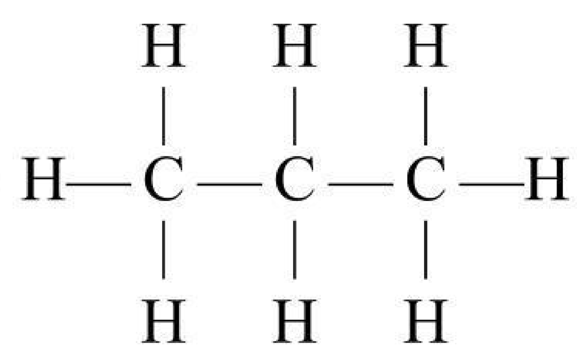
(3) >C=O (4) —NH_2

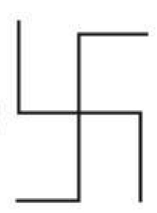
34. Which of the following is bicyclic compound?

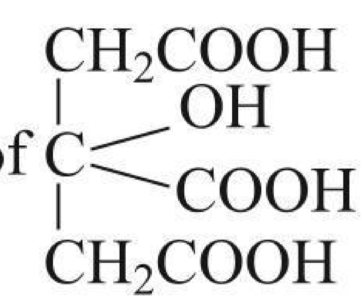
(1)  (2) 

(3)  (4) All of these

35. Which of the following is condensed structural formula?

(1)  (2) 

(3) CH_3—CH_2—CH_3 (4) 

36. The IUPAC name of  is:

(1) 3-Carboxy-3-hydroxypentanedicarboxylic acid

(2) 2-Hydroxypropane-1, 2, 3-tricarboxylic acid

(3) 2-Hydroxypropane-1, 2, 3-trioic acid

(4) 3-Hydroxypropane-1, 2, 3-tricarboxylic acid

37. Which of the following pairs is homologue?

(1) CH_3OH, CH_3—O—CH_3

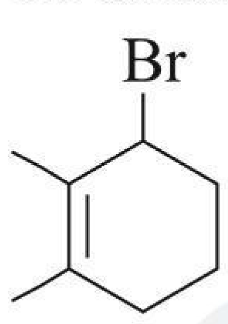
(2) CH_3OH, CH_3SH

(3) CH_3—O—CH_3, CH_3—CH_2—OH

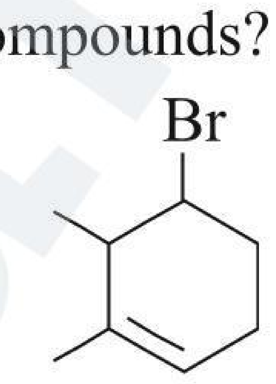
(4) CH_3—OH, CH_3CH_2OH

38. What is the sum of position assigned to bromine while numbering the parent chain in the below compounds?


(I)


(II)

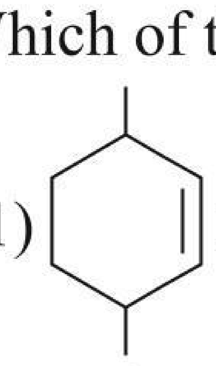

(III)

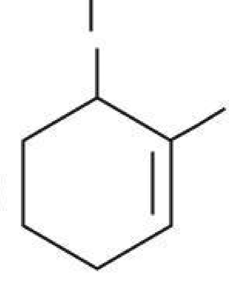

(IV)

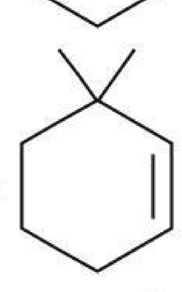
(1) 13 (2) 14

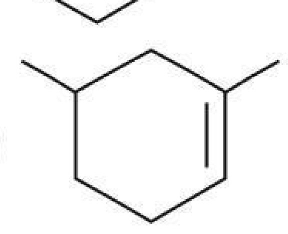
(3) 15 (4) 16

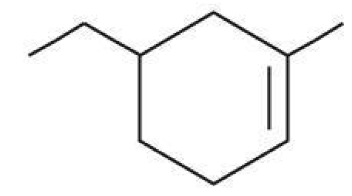
39. Which of the following name will be incorrect?

(1)  3, 6-Dimethyl cyclohexene

(2)  1, 6-Dimethyl cyclohexene

(3)  6, 6-Dimethyl cyclohexene

(4)  1, 5-Dimethyl cyclohexene

40. 

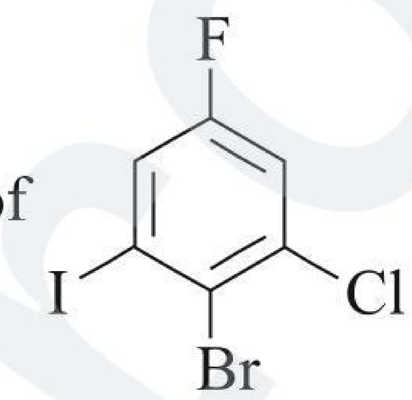
Correct IUPAC name

(1) 1-Methyl-3-ethyclohexene

(2) 5-Ethyl-1-methylcyclohexene

(3) 2-Ethyl-4-methylcylohexene

(4) 3-Ethyl-1-methlycylohexne

41. The IUPAC name of  is:

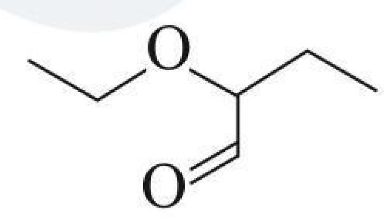
(1) 1-Bromo-2-chloro-3-fluoro-6-iodo benzene

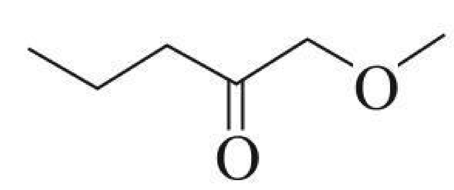
(2) 2-Bromo-1-choloro-5-fluoro-3-iodo benzene

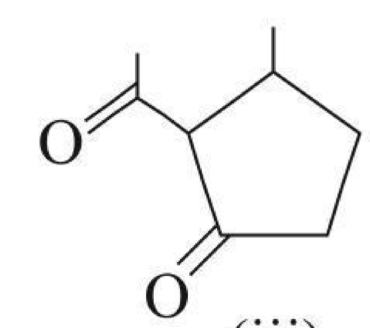
(3) 4-Bromo-2-chloro-5-iodo-1-fluoro benzene

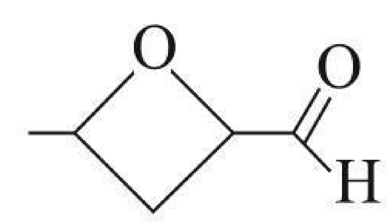
(4) 2-Bromo-3-chloro-1-iodo-5-fluoro benzene

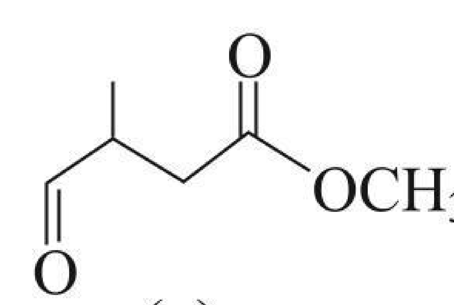
42. Many organic compounds contain more than one functional group. Which of the following is both an aldehyde and an ether?


(i)


(ii)


(iii)

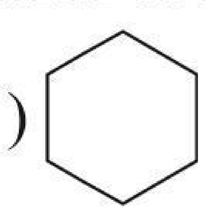
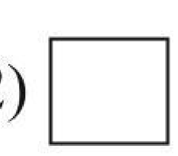

(iv)

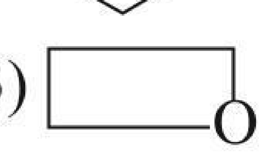

(v)

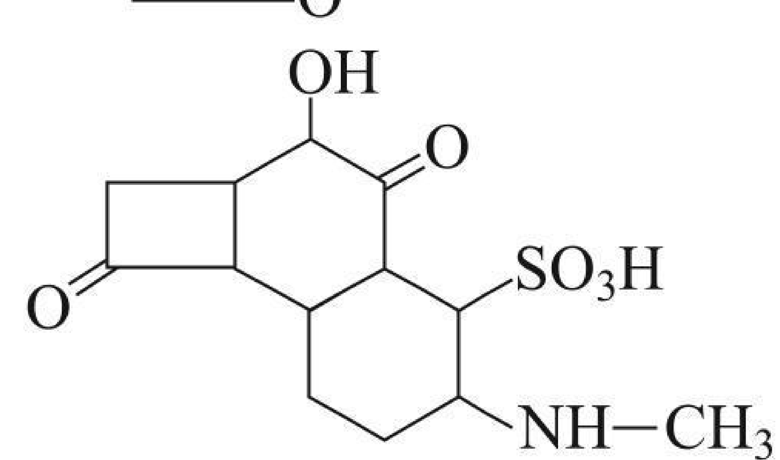
(1) (i) only (2) (i) and (iv)

(3) (ii) and (v) (4) (iii) and (iv)

43. Which of the following is homocyclic compound?

(1)  (2) 

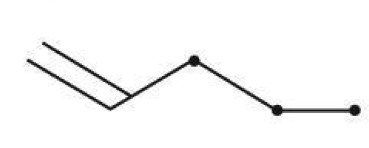
(3)  (4) Both (1) and (2)

44. 

Which functional group is present in above compound?

(1) Carboxylic acid (2) Aldehyde

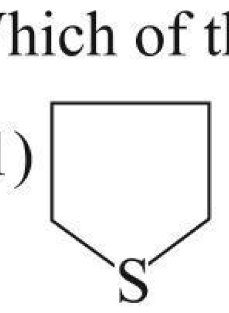
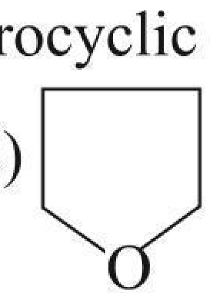
(3) Thiol (4) Alcohol

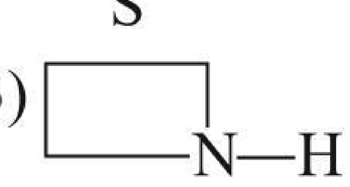
45.  ≡CH IUPAC name will be:

(1) Hex-5-en-1-yne (2) Hex-1-en-5-yne

(3) Hex-6-en-1-yne (4) Hex-1-en-6-yne

46. Which of the following is heterocyclic compound?

(1)  (2) 

(3)  (4) All of these

Multiple Correct Answers Type

1. Which of the following statements is/are wrong?
(1) C_nH_{2n} is the general formula of alkanes.
(2) In homologous series, all members have the same physical properties.
(3) IUPAC means International Union of Physics and Chemistry.
(4) Butane contains two 1° C atoms and two 2° C atoms.

2. Which of the following statements is/are correct?
(1) Homologous series can be represented by a general formula.
(2) The chemical properties of an organic compound depend on the functional group.
(3) Groups obtained by the removal of one H atom from the alkane are called alkyl groups.
(4) Alkynes consist of one double-bond in their molecules.

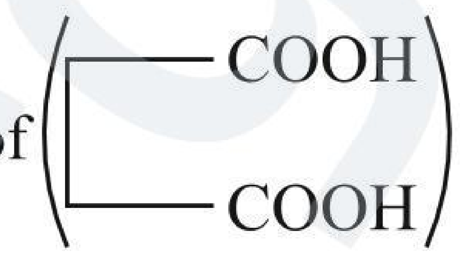
3. Which of the following statements is/are wrong?
(1) Acetic acid is the systematic name of vinegar.
(2) $Me—\overset{\overset{\displaystyle O}{||}}{C}—OH$ is an unsaturated compound.
(3) Prefixes like *n*-, iso, *sec*-, *tert*, neo-, etc., are used in IUPAC system.
(4) The systematic names of acids are formed by dropping -e of the name of parent alkane and adding -oic acid.

4. Which of the following statements is/are correct?
(1) $R—\overset{\overset{\displaystyle O}{||}}{C}—O—\overset{\overset{\displaystyle O}{||}}{C}—R$ is an unsaturated compound.
(2) Neohydrocarbons contain a 3° C atom.
(3) The IUPAC name of isopropyl alcohol is propan-2-ol.
(4) The IUPAC name of (CH_3CN) is ethanenitrile.

5. Which of the following statements is/are correct?
(1) Methane was named as fire damp as it forms explosive mixture with air.
(2) Primary suffixes are added to the root word to show saturation or unsaturation in a C atom.
(3) The IUPAC name of valeric acid is pentanoic acid.
(4) The common name of hexanoic acid is caproic acid.

6. Which of the following statements is/are correct?
(1) The IUPAC name of amyl alcohol is pentanol.
(2) The IUPAC name of isoamyl alcohol is 3-methyl butanol.
(3) Wood spirit is methanol.
(4) Methyl alcohol is also called carbinol.

7. Which of the following statements is/are correct?
(1) The trivial names of organic compounds are called common names.
(2) The systematic names of organic compounds are obtained from the IUPAC system.
(3) The systematic names of alkanes are based on the number of C atoms in the longest continuous chain of C atoms.
(4) The maximum number of functional groups must be included in the C atom chain selected even if it does not satisfy the longest chain rule.

8. Which of the following statements is/are correct?
(1) The common name of ($HOOC—CH_2—COOH$) is malonic acid.
(2) The common name of  is succinic acid.
(3) The IUPAC name of ($CH_2{=}CH—OCOCH_3$) is vinyl acetate.
(4) The IUPAC name of acrylonitrile is pro-2-ene-nitrile.

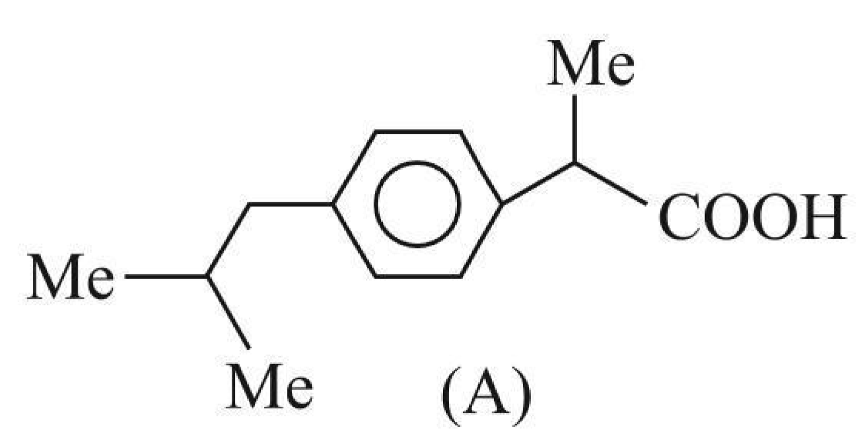
9. Which of the following statements is/are correct?
(1) The common name of benzene-1,2-diol is catechol.
(2) The common name of benzene-1,3-diol is resorcinol.
(3) The common name of benzene-1,4-diol is quinol.
(4) The common name of benzene-1,4-diol is hydroquinone.

10. Which of the following statements is/are correct?
(1) The common name of benzene-1,2,3-triol is pyrogallol.
(2) The common name of benzene-1,2,4-triol is hydroxyquinol.
(3) The common name of benzene-1,3,5-triol is phloroglucinol.
(4) The common name of ($CH_2{=}CH—Ph$) is styrene.

Linked Comprehension Type

Paragraph 1

The analgesic drug ibuprofen (A) is chiral and exists in (+) and (–) forms. One enantiomer is physiologically active, while the other is inactive. The structure of ibuprofen is given below:



(A)

1. The principal functional group in (A) is:
(1) Phenyl (2) —COOH group
(3) Isopropyl (4) Both (1) and (2)

2. The IUPAC name of (A) is:
(1) 3-(*p*-Isobutyl phenyl) propanoic acid
(2) 2-(*p*-Isobutyl phenyl) propanoic acid
(3) 3-(*p*-*sec*-Butyl phenyl) propanoic acid
(4) 2-(*p*-*sec*-Butyl phenyl) propanoic acid

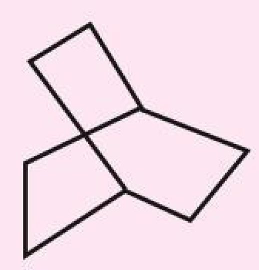
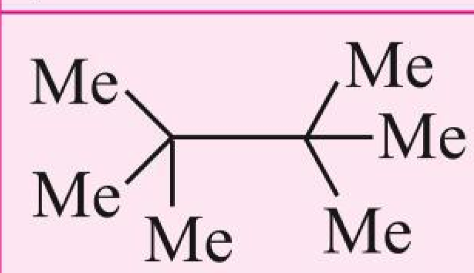
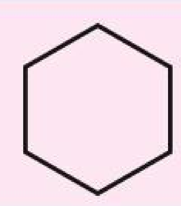
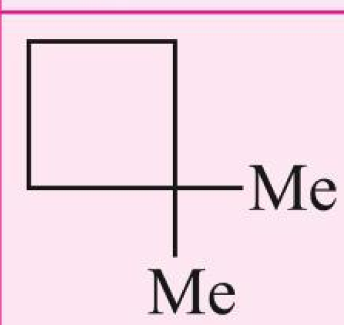
3. The number of π-bonds in (A) is:
(1) 2 (2) 3
(3) 4 (4) 5

4. The number of σ-bonds in (A) is:
(1) 30 (2) 31
(3) 32 (4) 33

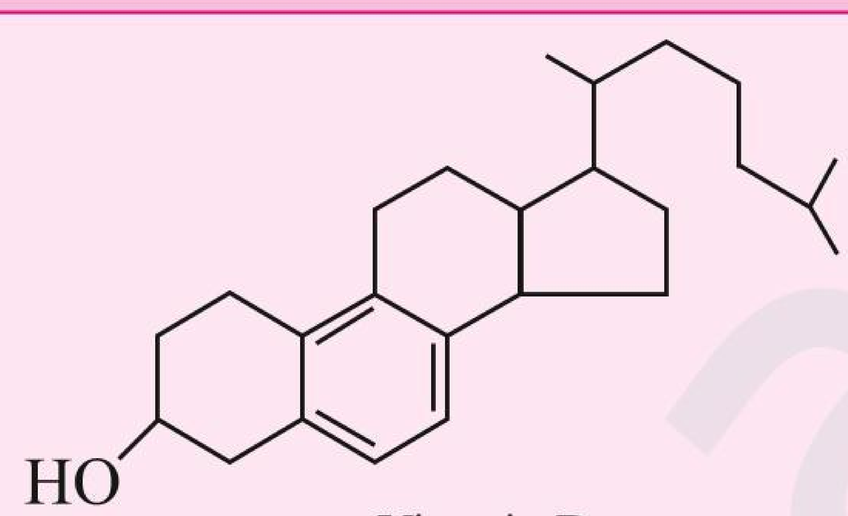
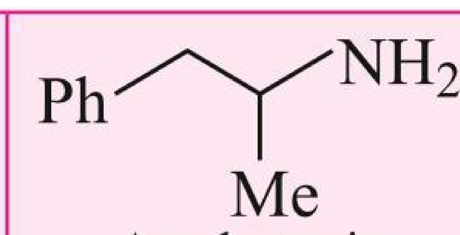
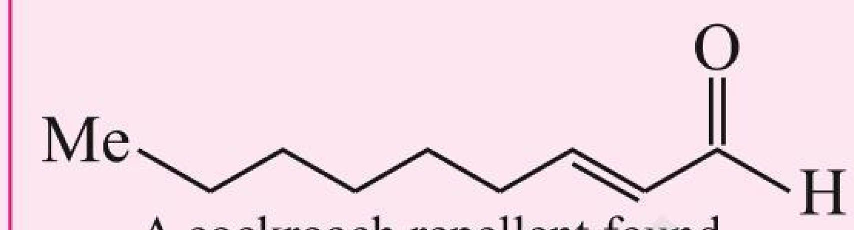
Matrix Match Type

Match the compounds in column I with their structure(s)/characteristic(s)/test(s)/reaction(s)/stereochemistry, etc., given in column II. Matching can be one or more than one.

1.

	Column I		Column II
	Compound		**Structure**
a.	C_8H_{18} with only 1° H atoms	p.	
b.	C_6H_{12} with only 2° H atoms	q.	
c.	C_6H_{12} with only 1° and 2° H atoms	r.	
d.	C_8H_{14} with 12 secondary and 2 tertiary H atoms	s.	

2.

	Column I		Column II
	Compound		**Containing all the functional groups**
a.	 Vitamin D_3	p.	1° amine
b.	 Amphetamine	q.	2° alcohol
c.	 A cockroach repellent found in cucumbers	r.	Triene
		s.	Aldehyde and ene

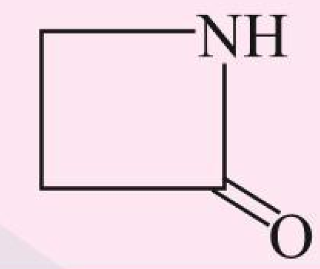
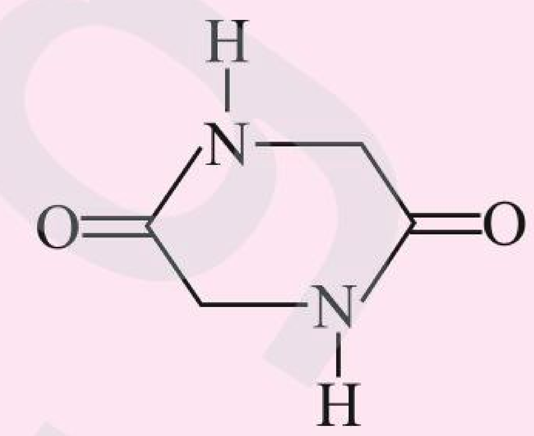
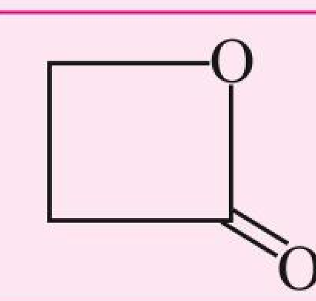
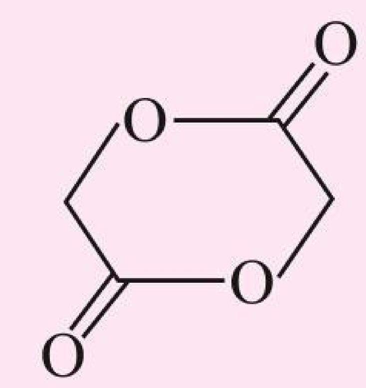
3.

	Column I		Column II
	Group		**Name**
a.	$CH_3-CH(CH_3)-$	p.	*sec*-Butyl
b.	$CH_3-CH_2-CH(CH_3)-$	q.	*tert*-Butyl
c.	$CH_3-CH(CH_3)-CH_2-$	r.	Isopropyl
d.	$CH_3-C(CH_3)_2-$	s.	Neopentyl
e.	$CH_3-C(CH_3)_2-CH_2-$	t.	Isobutyl

4. Match the items given in column I with that in column II and III.

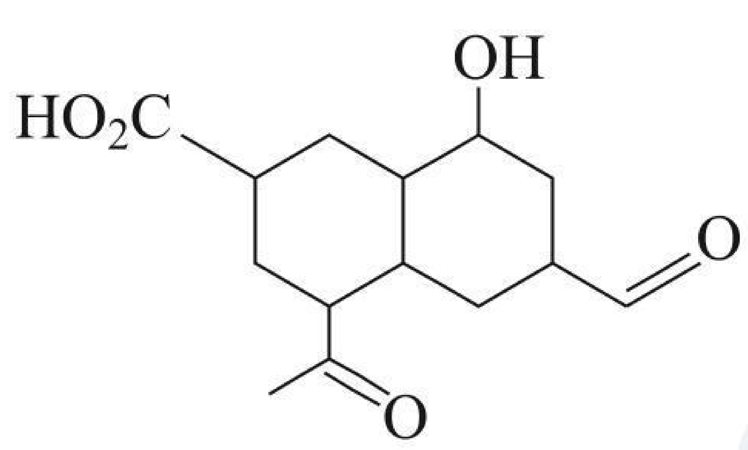
	Column I		Column II		Column III
	Common name		**Characteristics (I)**		**Characteristics (II)**
a	Decalin	i	Aromatic	p	Contains two 3°C-atoms
b	Malic acid	ii	Bicyclo compound	q	Gives colour with $FeCl_3$ solution
c	Maleic acid	iii	Contains 4–C	r	Saturated
d	Carbolic acid	iv	Dibasic acid	s	Decolourises Br_2/CCl_4 solution

5. Match the items given in column I with that in column II and III.

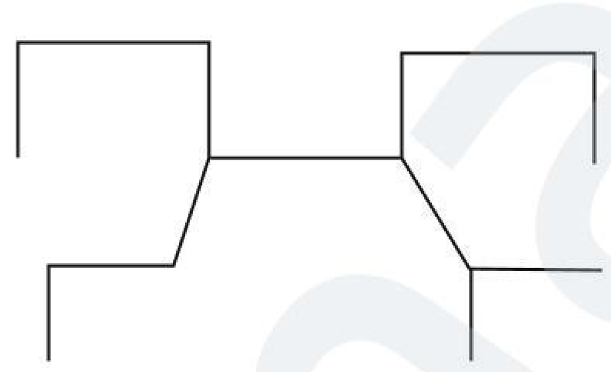
	Column I		Column II		Column III
	Common name		Characteristics (I)		Structure
a.	Piperazine	i.	Cyclic esters	p.	
b.	Lactide	ii.	Cyclic imide	q.	
c.	Lactam	iii.	Cyclic diester	r.	
d.	Lactones	iv.	Cyclic di-imide	s.	

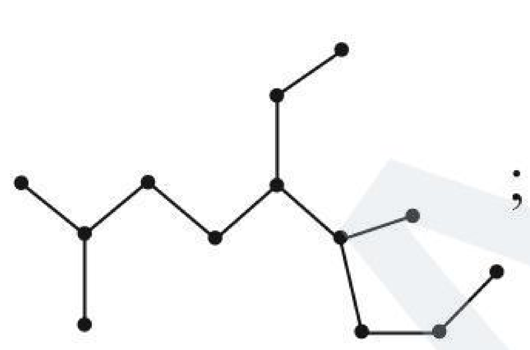
Numerical Value Type

1. How many different functional groups are present in given compound?

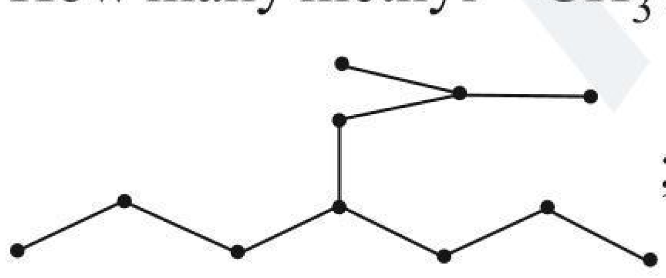


2. How many carbon atoms are present in parent carbon chain in the given following compound?

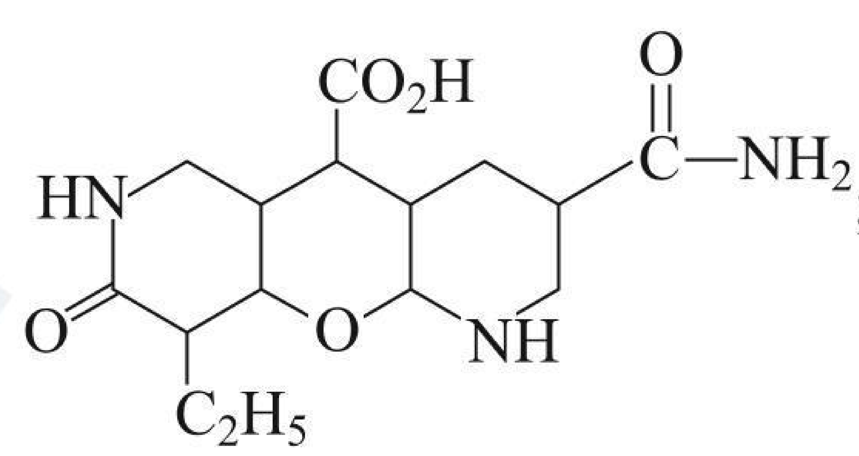


3.  ;

How many methyl—CH_3 group are present in given alkane?

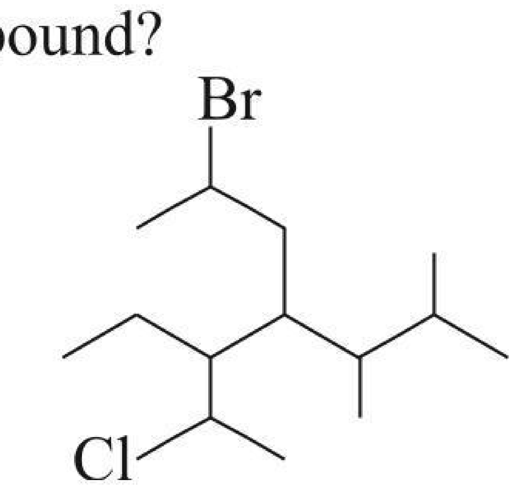
4.  ; How many 2° carbon present in above compound?

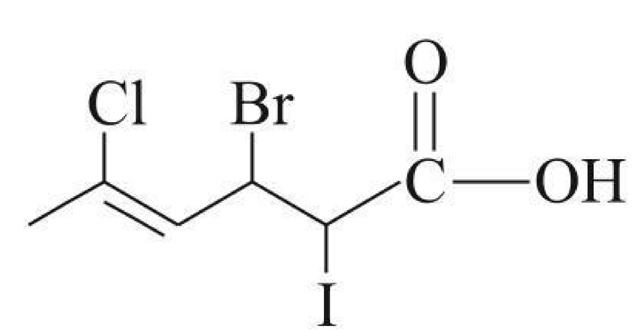
5. How many 1° carbon atom will be present in a simplest hydrocarbon having two 3° and one 2° carbon atom?

6.  ;

Number of functional groups in the above compound is:

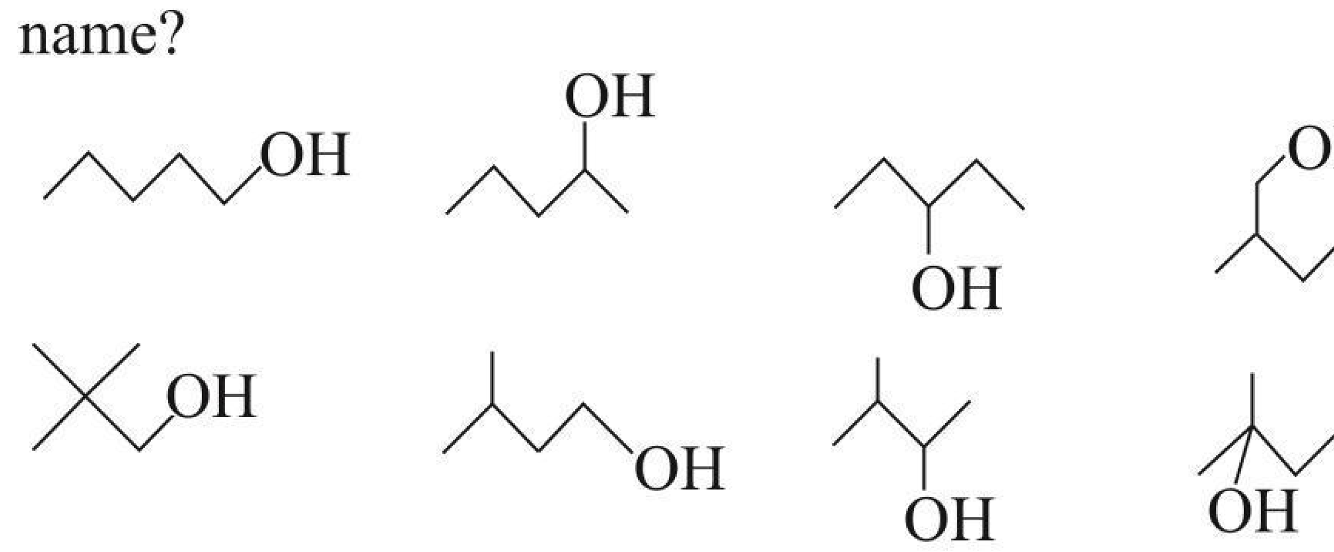
7. How many total number of substituent are present in the following compound?

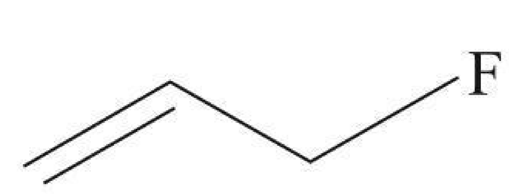


8. 

Total number of substituent present in the above compound:

9. How many number of compounds, which have same IUPAC name?



10. 

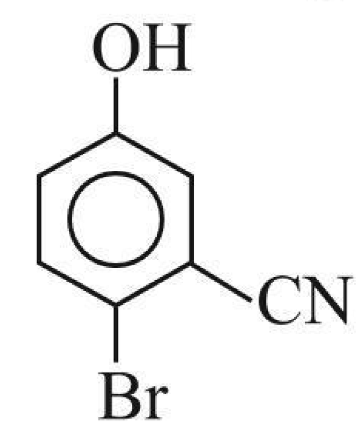
When IUPAC name of following compound is given, then double bond and substituent gets respectively (x and y) number so the sum of ($x + y$) will be:

Archives

JEE ADVANCED

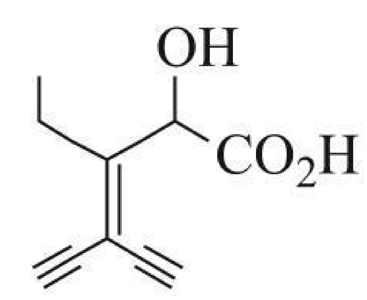
Single Correct Answer Type

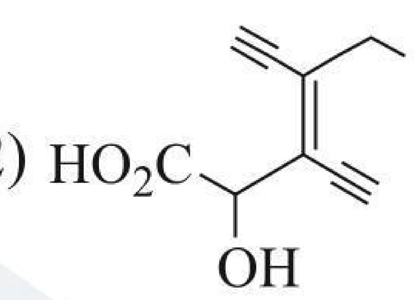
1. The IUPAC name of the following compound is:

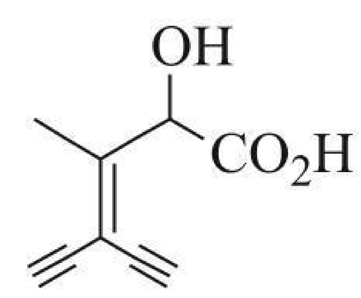


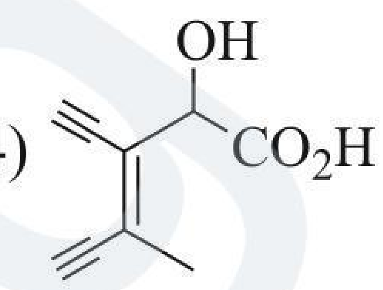
(1) 4-Bromo-3-cyanophenol
(2) 2-Bromo-5-hydroxy benzo nitrile
(3) 2-Cyano-4-hydroxy bromo benzene
(4) 6-bromo-3-hydroxy benzonitrile **(IIT-JEE 2009)**

2. Which one of the following structures has the IUPAC name 3-ethynyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid?

(1) 

(2) 

(3) 

(4) 

(JEE Advanced 2020)

Multiple Correct Answers Type

1. The IUPAC name of $(CH_3)_3C—CH{=}CH_2$ is:
(1) 2,2-Dimethyl but-3-ene
(2) 2,2-Dimethyl pent-4-ene
(3) 3,3-Dimethyl but-1-ene
(4) Hex-1-ene **(JEE Advanced 2017)**

Answers Key

EXERCISES

Single Correct Answer Type

1. (2) **2.** (1) **3.** (3) **4.** (2) **5.** (4)
6. (2) **7.** (1) **8.** (3) **9.** (1) **10.** (2)
11. (2) **12.** (3) **13.** (3) **14.** (2) **15.** (4)
16. (2) **17.** (1) **18.** (1) **19.** (2) **20.** (3)
21. (4) **22.** (4) **23.** (2) **24.** (3) **25.** (3)
26. (2) **27.** (1) **28.** (4) **29.** (3) **30.** (3)
31. (4) **32.** (1) **33.** (1) **34.** (4) **35.** (3)
36. (2) **37.** (4) **38.** (3) **39.** (3) **40.** (2)
41. (2) **42.** (2) **43.** (4) **44.** (4) **45.** (2)
46. (4)

Multiple Correct Answers Type

1. (1, 2, 3) **2.** (1, 2, 3) **3.** (1, 2, 3) **4.** (3, 4)
5. (1, 2, 3, 4) **6.** (1, 2, 3, 4) **7.** (1, 2, 3, 4) **8.** (1, 2, 4)
9. (1, 2, 3, 4) **10.** (1, 2, 3, 4)

Linked Comprehension Type

1. (2) **2.** (2) **3.** (3) **4.** (4)

Matrix Match Type

Q.No.	a	b	c	d	e	f
1.	q	r	s	p	—	—
2.	r, q	p	s	—	—	—
3.	r	p	t	q	s	—
4.	ii, p	iii, iv, r	iii, iv, s	i, q	—	—
5.	iv, q	iii, s	ii, p	i, r	—	—

Numerical Value Type

1. (4) **2. (8)** **3. (2)** **4. (5)** **5. (4)**
6. (5) **7. (4)** **8. (3)** **9. (0)** **10. (4)**

ARCHIVES

JEE Advanced

Single Correct Answer Type

1. (2) **2.** (4)

Multiple Correct Answers Type

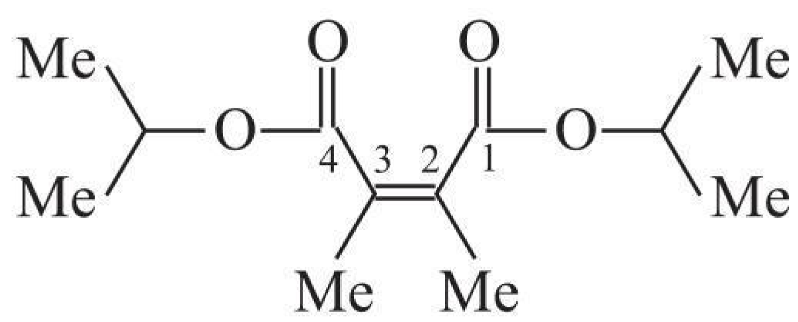
1. (1, 2, 3)

Solutions

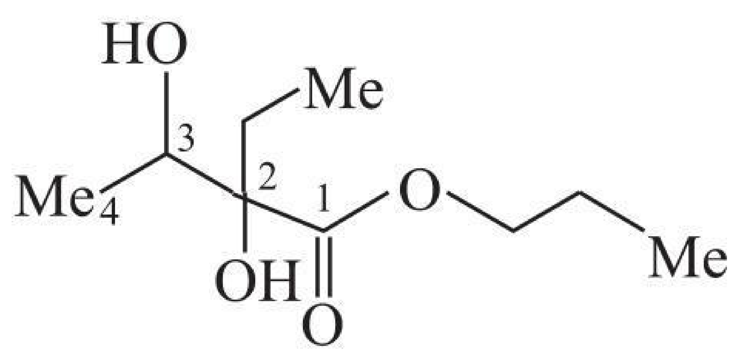
Chapter 1

Concept Application Exercises

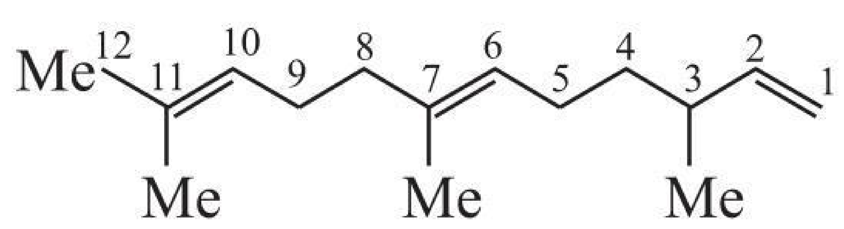
Exercise 1.1

1. 

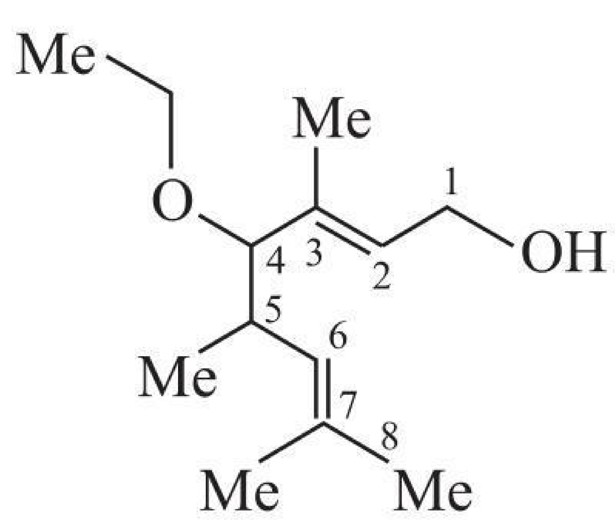
Diisopropyl-*cis*-2,3-dimethyl but-2-en-1,4-dioate (synthetic cockroach repellent)

2. 

Propyl-2,3-dihydroxy-2-methyl butanoate

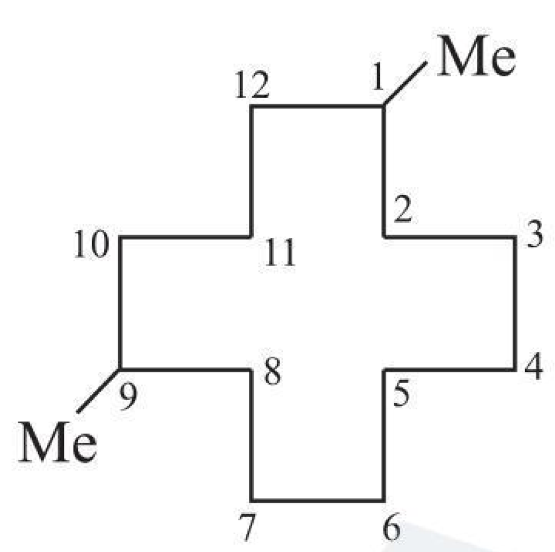
3. 

3,7,11-Trimethyl dodec-1,6,10-triene (common name: Farnesene). Found in waxy coating of apples.

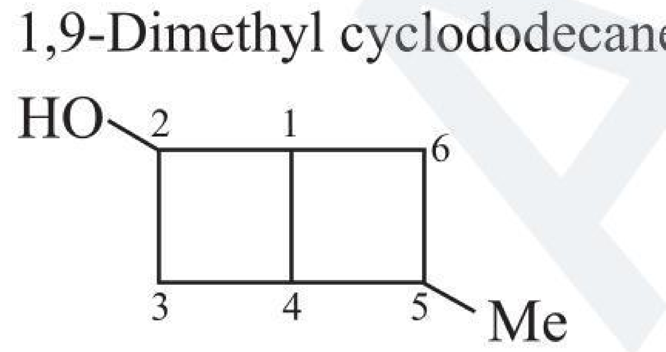
4. 

4-Ethoxy-3,5,7-trimethyl oct-2,6-diene-1-ol

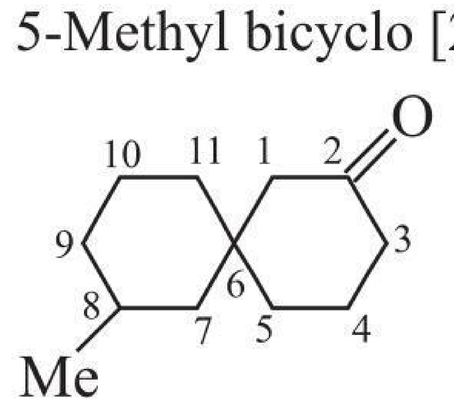
Exercise 1.2

1. 

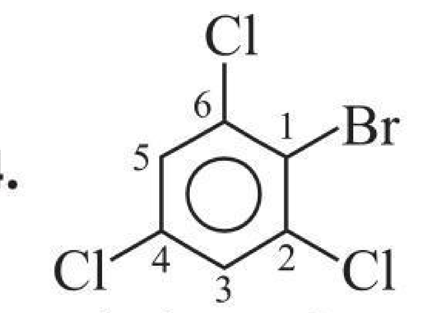
1,9-Dimethyl cyclododecane

2. 

5-Methyl bicyclo [2.2.0] hexan-2-ol

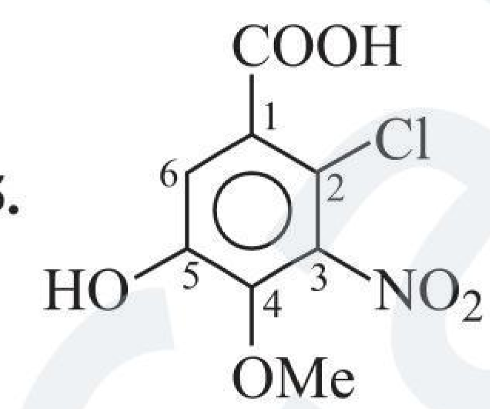
3. 

8-Methyl spiro [5.5] undecan-2-one

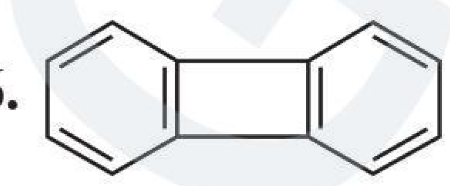
4. 

Br is given preference over Cl.

2,4,6-Trichloro-1-bromo benzene

5. 

2-Chloro-5-hydroxy-4-methoxy-3-nitro benzoic acid

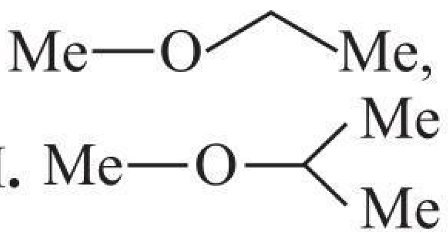
6. 

Dibenzocyclo butadiene (Double bond is common in central ring from both sides of benzene ring.) (See Table 1.4.)

Exercise 1.3

1. **a.** $(CH_3)_3N$

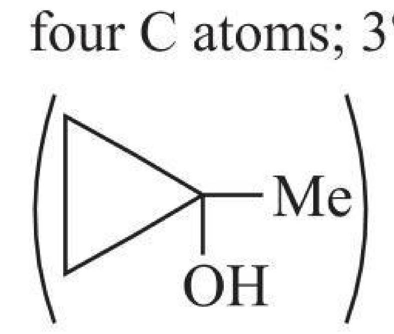
b. **I.** , **II.** ,

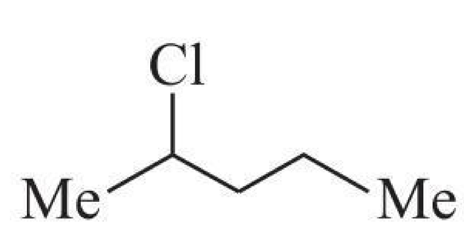
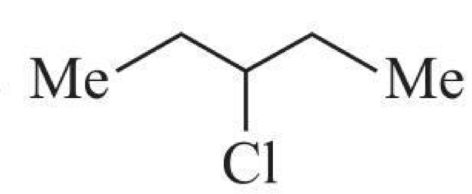
III. 

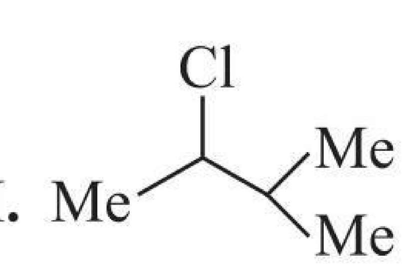
c. Degree of unsaturation:

$$\frac{(2n_C + 2) - n_H}{2} = \frac{10 - 8}{2} = 1°$$

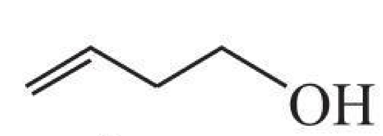
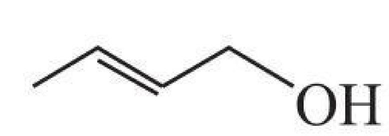
1° D.U. suggests either unsaturated alcohol or cyclic 3° alcohol four C atoms; 3° alcohol cannot be unsaturated, so it is cyclic.

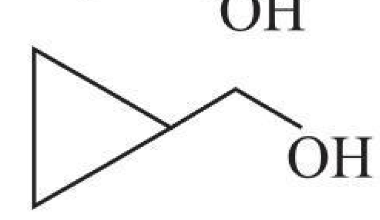


d. **I.**  **II.** 

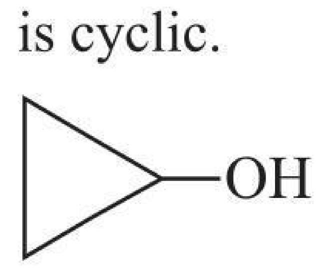
III. 

e. D.U. = 1°, 1 D.U. suggests unsaturated alcohol or cyclic 2° alcohol.

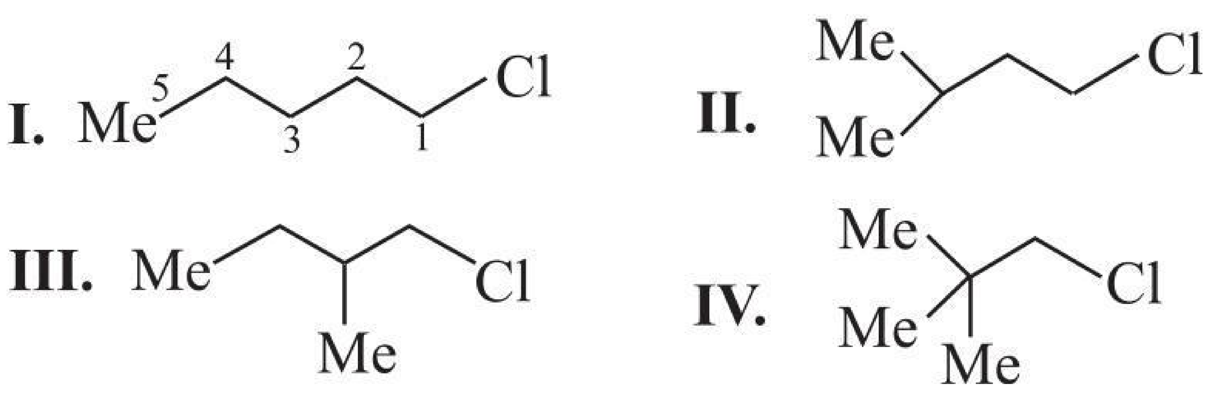
I.  **II.** 

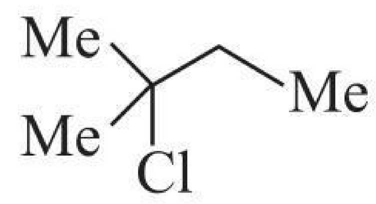
III. 

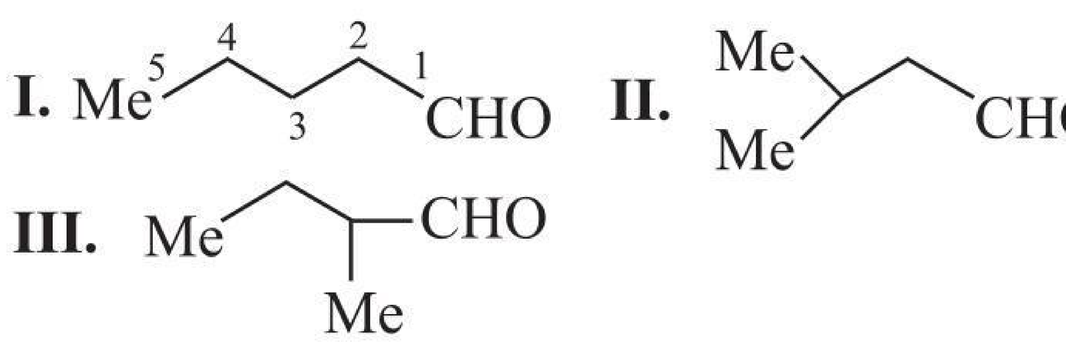
f. D.U. = 1°, 1 D.U. suggests unsaturated alcohol or cyclic 2° alcohol three C atoms; 2° alcohol cannot be unsaturated, so it is cyclic.

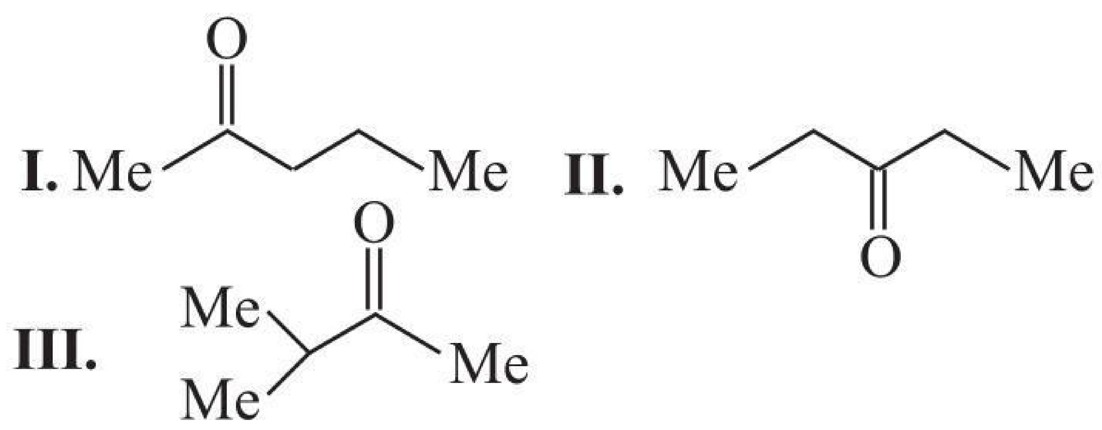


g. 

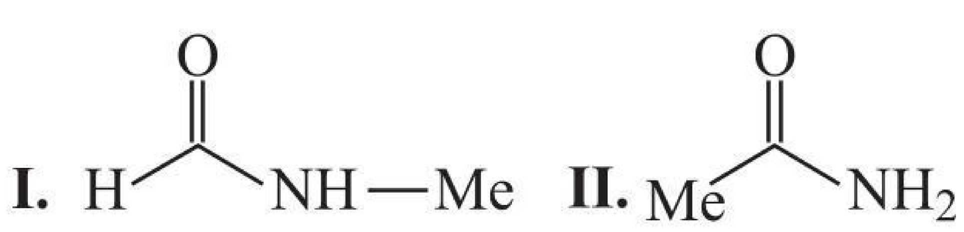
h. 

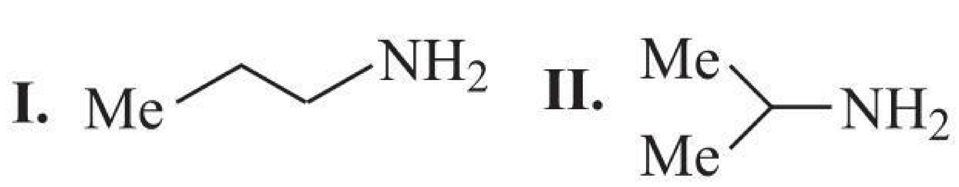
i. 

j. 

k. 

l. 

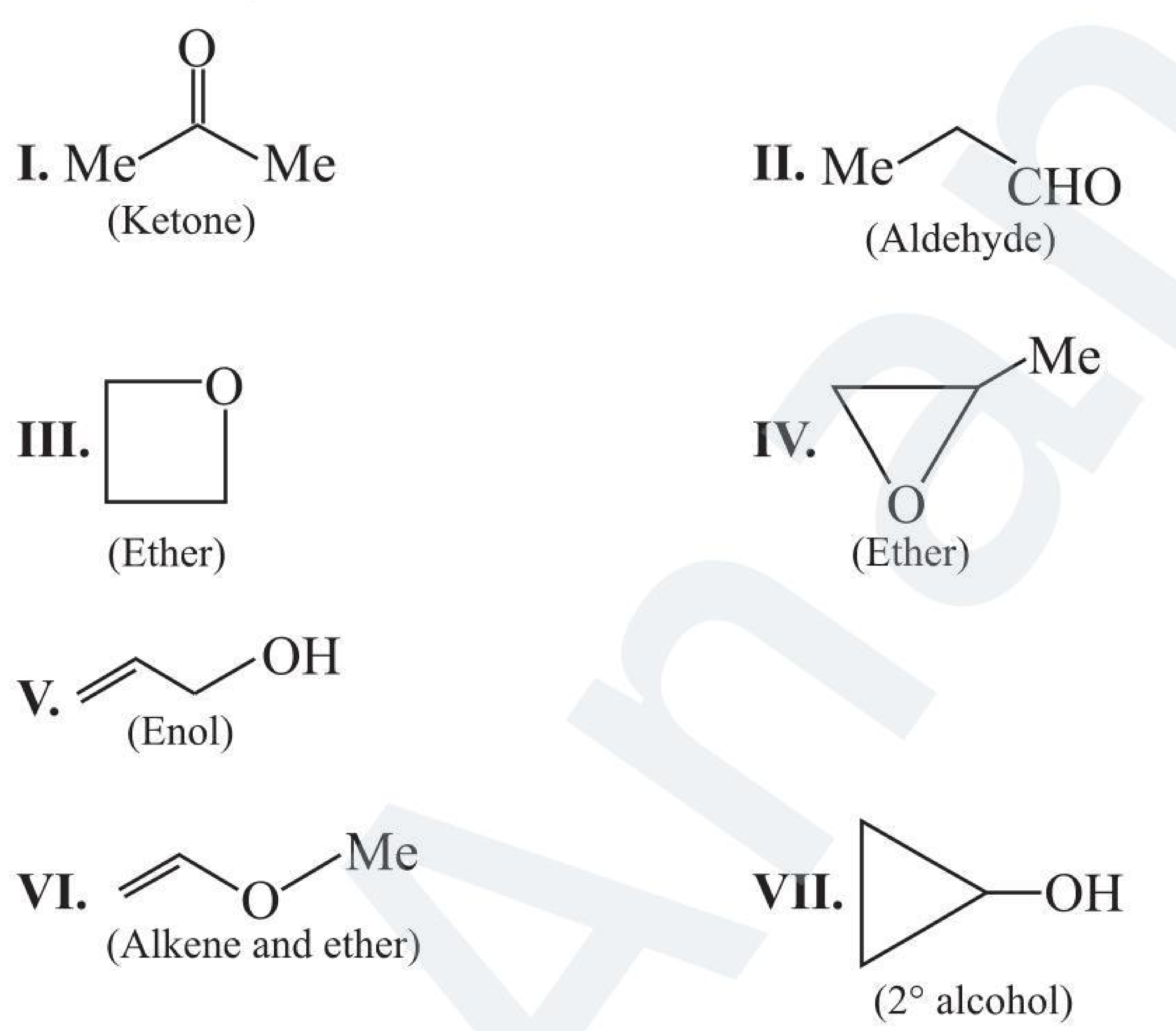
m. 

n. 

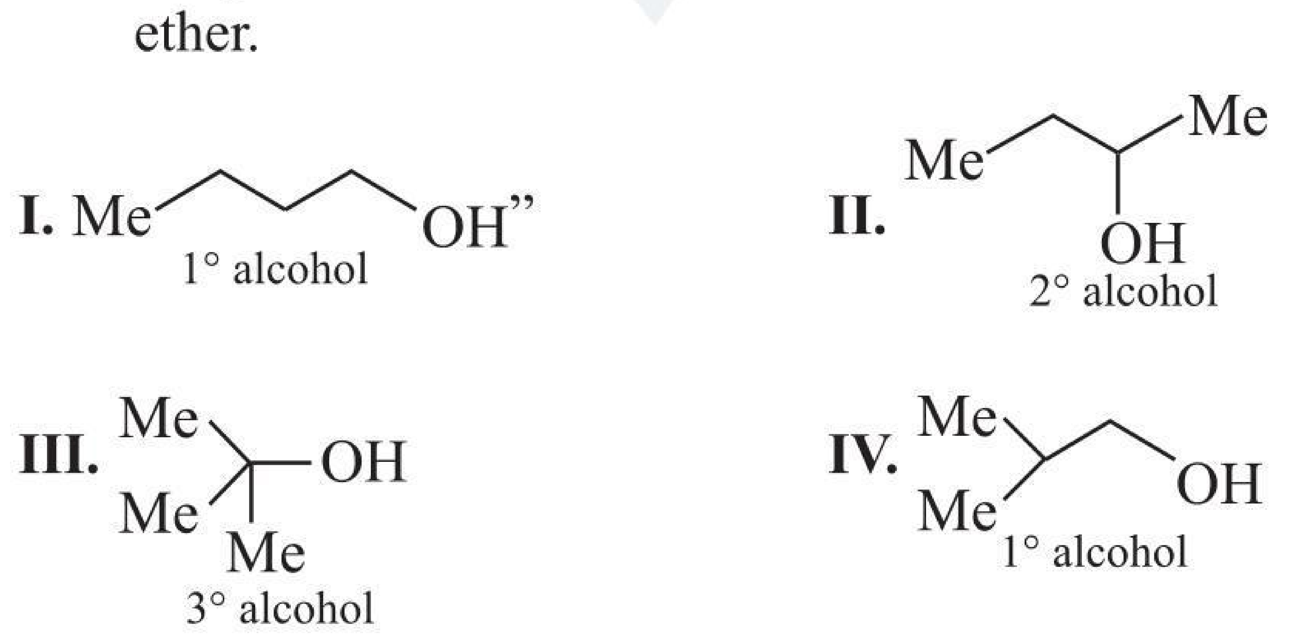
2. a. 2° **b.** 3° **c.** 2° **d.** 3° **e.** 1° **f.** 2°

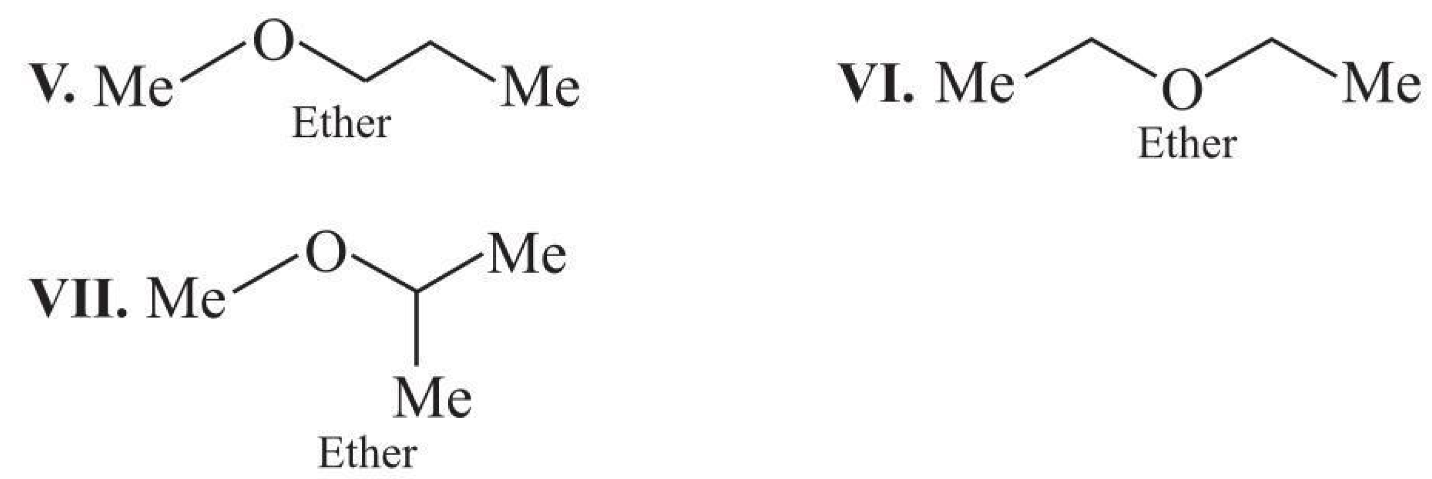
3. a. 2° **b.** 3° **c.** 1° **d.** 2° **e.** 3°

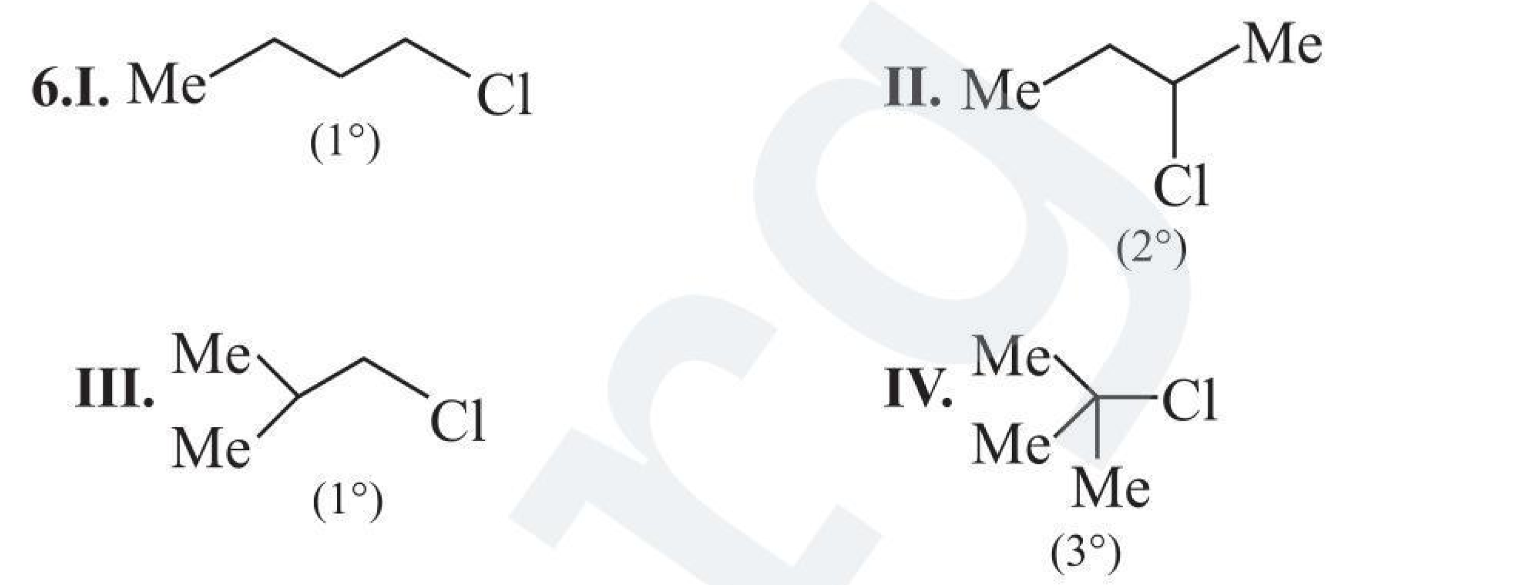
4. 1 D.U. suggests either aldehyde or ketone or unsaturated alcohol or ether, or cyclic ether or alcohol.

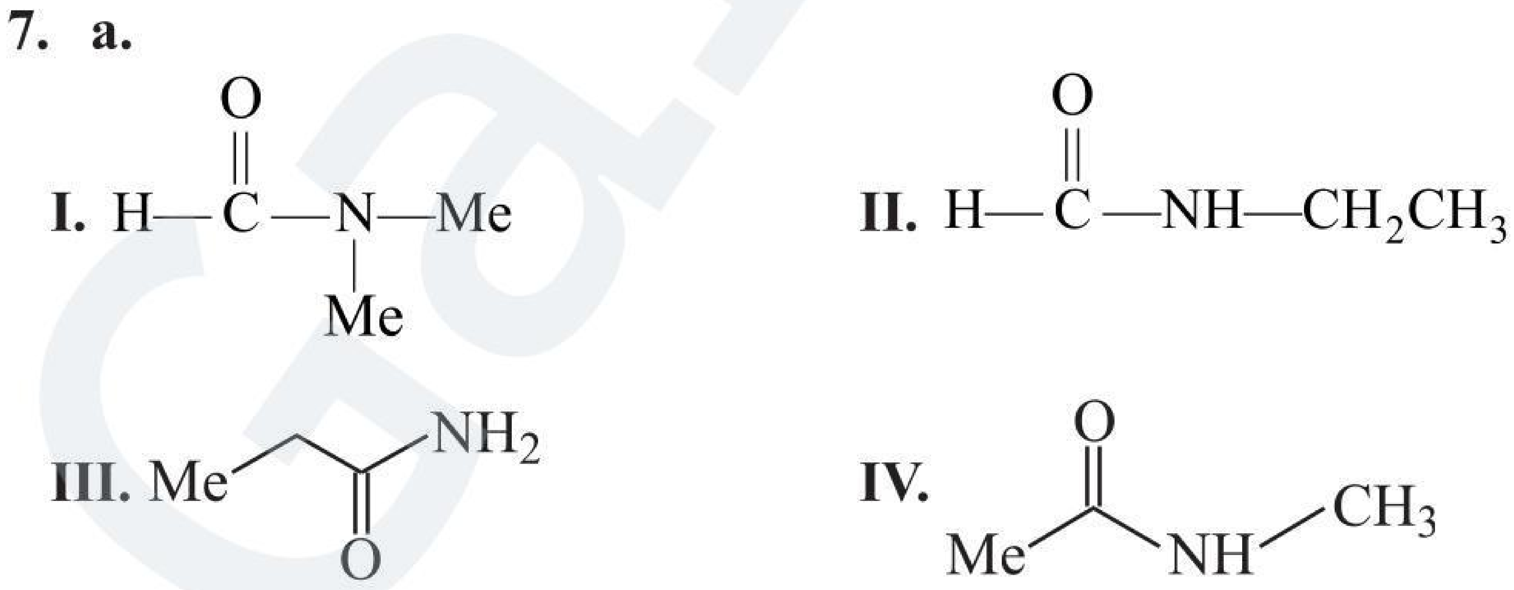


5. The compound is saturated with zero D.U., so it can be alcohol or ether.





6. 

7. a. 

b. The compound (I) has the lowest boiling point and melting point because it does not have an H atom covalently bonded to an N atom and, therefore, it cannot form H-bonds with each other. On the other hand, the compounds (II), (III), and (IV) have H atoms covalently bonded to N atom and, therefore, form H-bonding.

8. The number of C atoms in compound (A)

$= C_nH_{2n+2}$ (since it is an alkane)

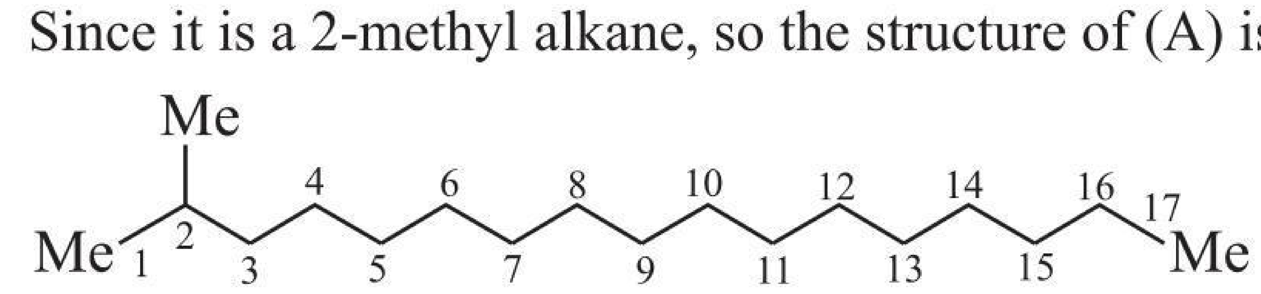
$12n + 2n + 2 = 254$

$\Rightarrow\ 14n = 252$

$\Rightarrow\ n = 18$

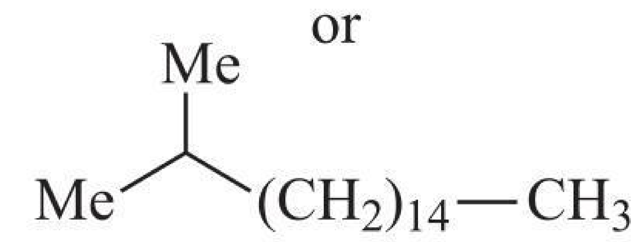
The compound is an 18-C alkane.

Since it is a 2-methyl alkane, so the structure of (A) is



(2-Methyl hepta decane)

or



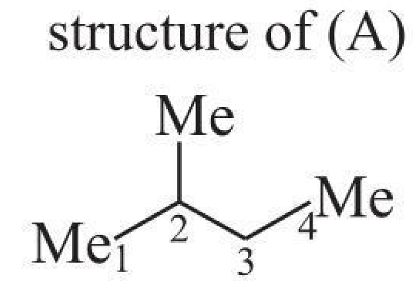
(Total C atoms = 18)

9. Since it is an alkane (C_nH_{2n+2}),

$\therefore\ 12n + 2n + 2 = 72$

$\Rightarrow\ n = 5$

The compound (A) is 5C-atom alkane with (Me) group at C-2. The structure of (A) is



(2-Methyl butane)

10. Molecular formula of (A) = $C_xH_xO_x$.

$\therefore\ 12x + x + 16x = 58$

$\Rightarrow\ x = 2$

Formula = $C_2H_2O_2$. It can be O=CH—CH=O

Ethane dial or Glyoxal

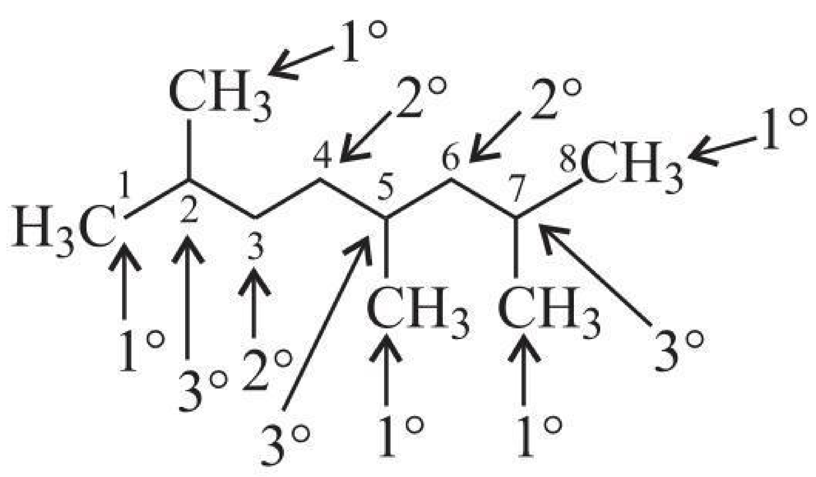
Exercises

Single Correct Answer Type

1. (2) **2. (1)**

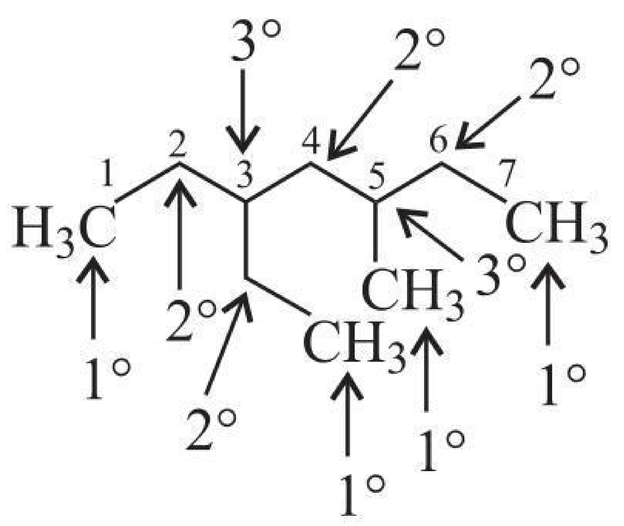
3. (3) (For priority order, see Section 1.5.6.)

4. (2)



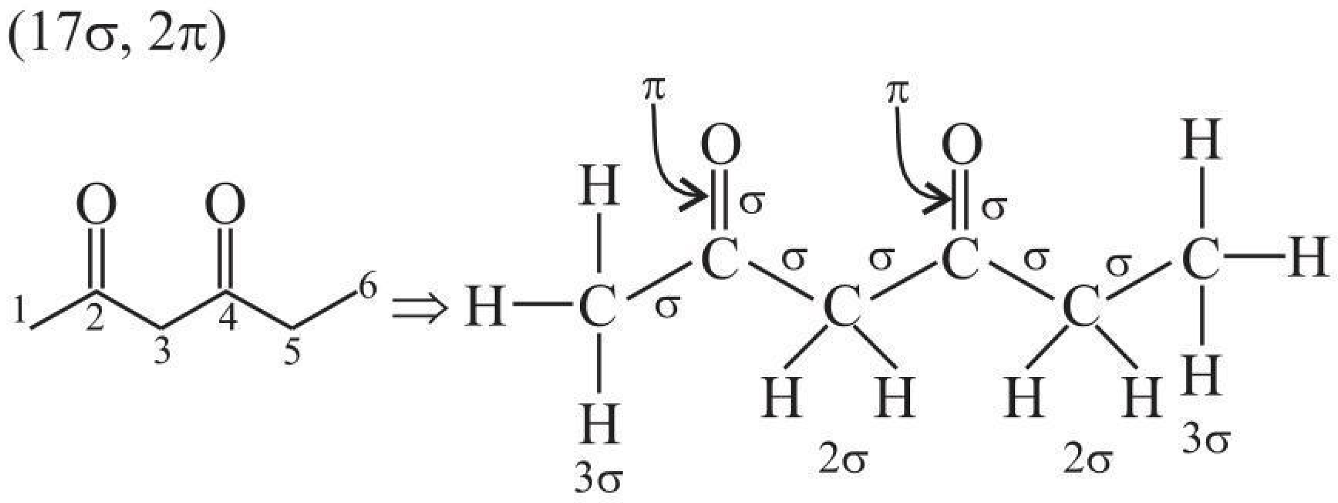
1° H = 15
2° H = 6
3° H = 3

5. (4)

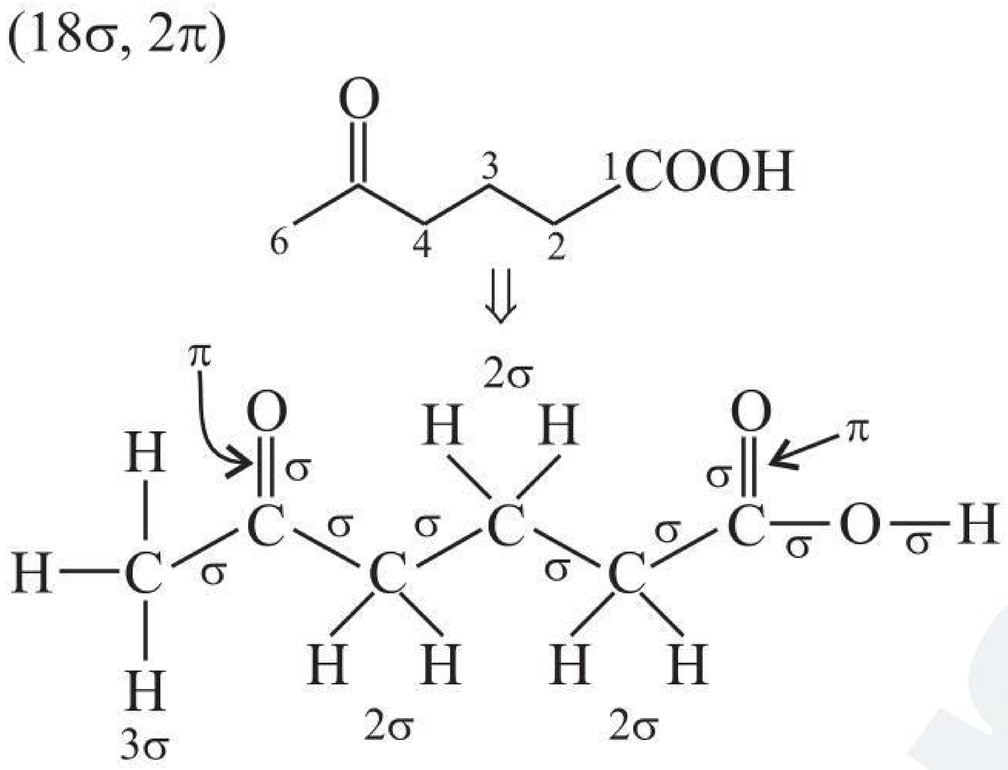


1° H = 12
2° H = 8
3° H = 2

6. (2) (17σ, 2π)

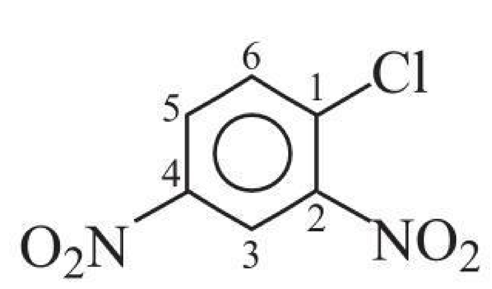


7. (1) (18σ, 2π)



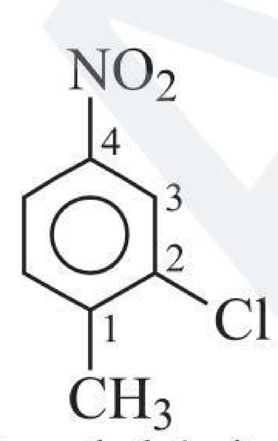
8. (3)

(1)



1-Chloro-2,4-dinitro benzene
(Lowest locant = 1 + 2 + 4 = 7)
(Correct name)
In 4-chloro-1,3-dinitro benzene
(Highest locant = 4 + 1 + 3 = 8)
(So incorrect)

(2)



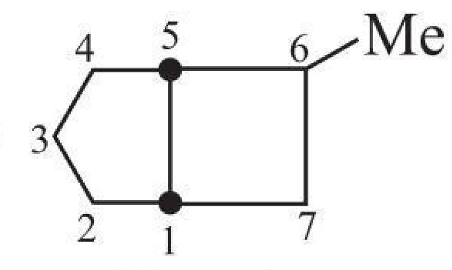
2-Chloro-1-methyl-4-nitrobenzene
(Correct name)
(Lowest locant = 2 + 1 + 4 = 7)
In 4-methyl-5-chloro-1-nitro benzene
(Highest locant = 4 + 5 + 1 = 10)
(So incorrect)

(3) In (3) lowest locant is 3 + 1 + 1 = 5

(4) In (4) highest locant is 1 + 3 + 3 = 7

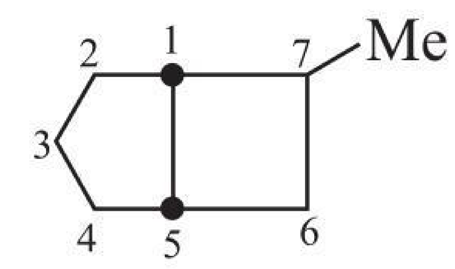
So (3) is correct.

9. (1)



(Correct)
Locant at C-6

In bicyclo compounds, numbering starts from the bridge-head to the larger ring ending towards the smaller ring, following the lowest locant rule. So the correct name is 6-methyl bicyclo [3.2.0] heptane.



(Incorrect)
Locant at C-7

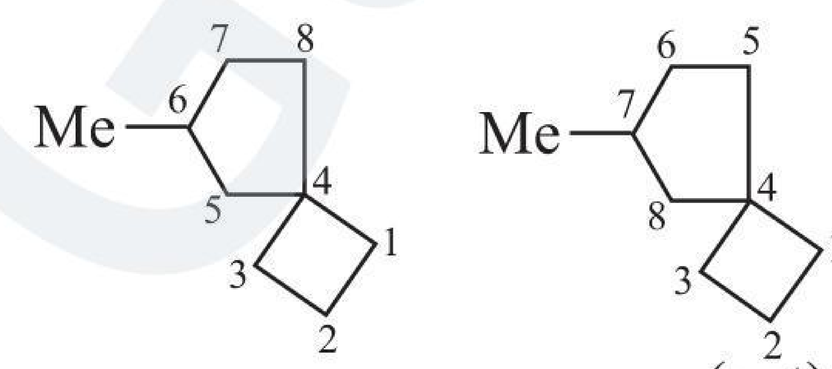
10. (2) **I.** The principal group (—COOH) is lost.

II. Same as is (I).

III. One (—COOH) group is lost, but still one (—COOH) group is left in the product.

11. (2) For cyclic ethers, see Section 1.10.

12. (3) In spiro compounds, numbering starts from the next C atom from the single-fused point to smaller ring ending in the larger ring, following the lowest locant rule.



(not)

So the correct name is 6-methyl spiro [3.4] octane.

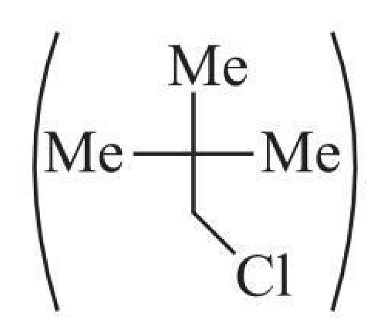
13. (3) Since it is an alkane, its formula is C_nH_{2n+2}.

$\therefore\ 12n + 2n + 2 = 72$

$\Rightarrow n = 5$

(A) has the molecular formula C_5H_{12}.

Only (3) gives one product on monochlorination

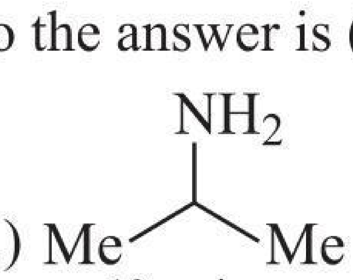


14. (2) D.U. in (A) $= \dfrac{(2n_C + 2) - n_H}{2} = \dfrac{12 - 10}{2} = 1°$.

1 D.U. suggests that either it is an alkene or a cyclopentane (C_5H_{10}). Only cyclopentane gives one product on monochlorination.

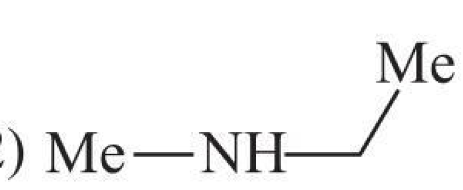
So the answer is (2).

15. (4) (1)



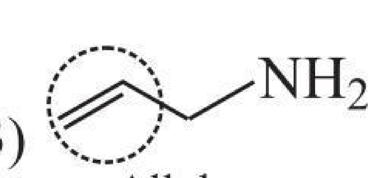
1° amine

(2)



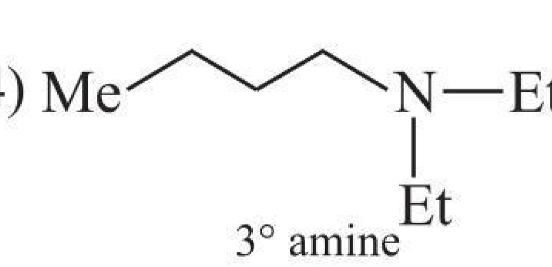
2° amine

(3)



Allyl group
1° amine

(4)

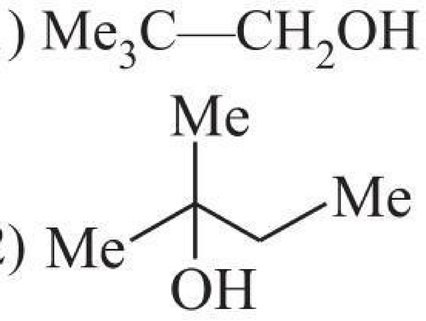


3° amine

16. (2) Carbinol is (MeOH).

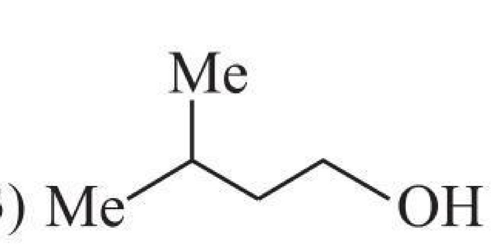
(1) $Me_3C—CH_2OH$ (1° alcohol)

(2)

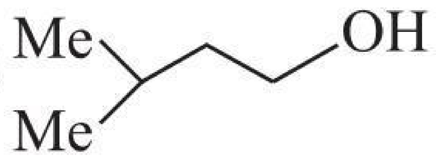


(3° alcohol)

(3)



(1°)

(4)  (1°) Isoamyl alcohol

17. (1) CH_3OH is also called zerone.

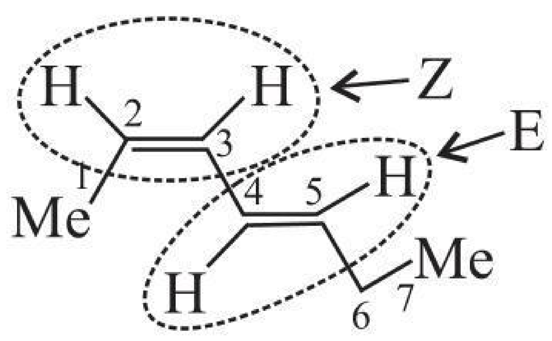
18. (1)

19. (2) 3-Chloro-cyclo-hex-an-ol

2° prefix, 1° prefix, Word root, 1° suffix, 2° suffix

20. (3) 2-Chloro-3-methyl-hex-an-oic acid

2° prefix, Word root, 1° suffix, 2° suffix

21. (4) 

Hence the correct name is (d).

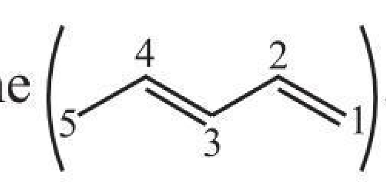
22. (4) Cumulated diene is (C═C═C).

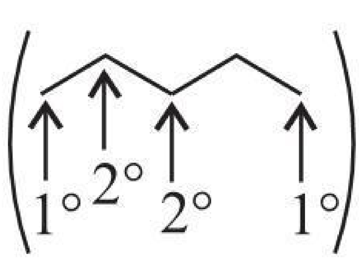
(1) 

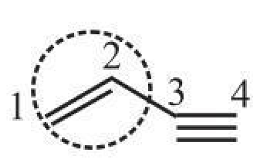
(2) 

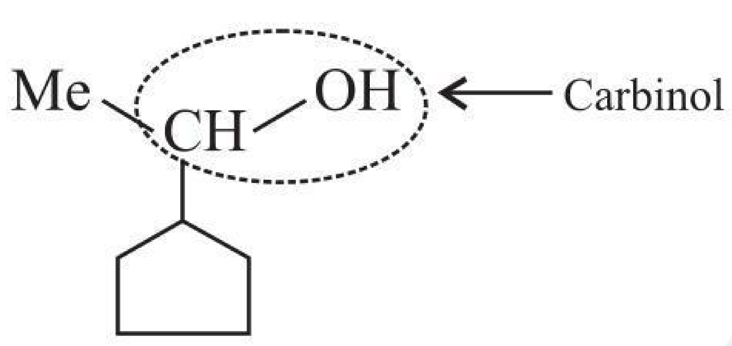
(3) 

(All cumulated diene)

(4) It is not a cumulated diene .

23. (2) 

24. (3)  (Vinylacetylene) (But-1-en-3-yne)

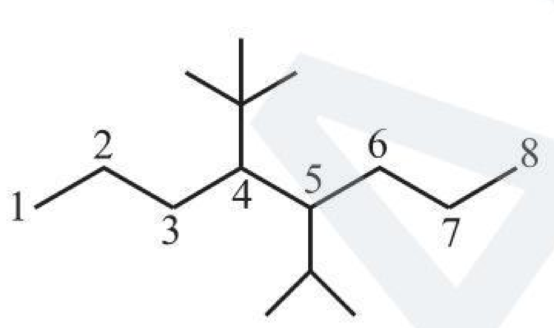
25. (3) 

26. (2) Acrolein is $\overset{3}{C}H_2 = \overset{2}{C}H - \overset{1}{C}HO$ (Prop-2-enal)

27. (1)

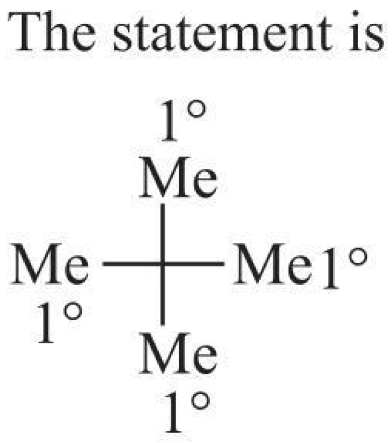
28. (4)

29. (3)

30. (3) 

Numbering is started from that side of the chain where the complex substituent is at a lower position.

31. (4) The statement is self-explanatory.

32. (1) 

33. (1) Self-explanatory.

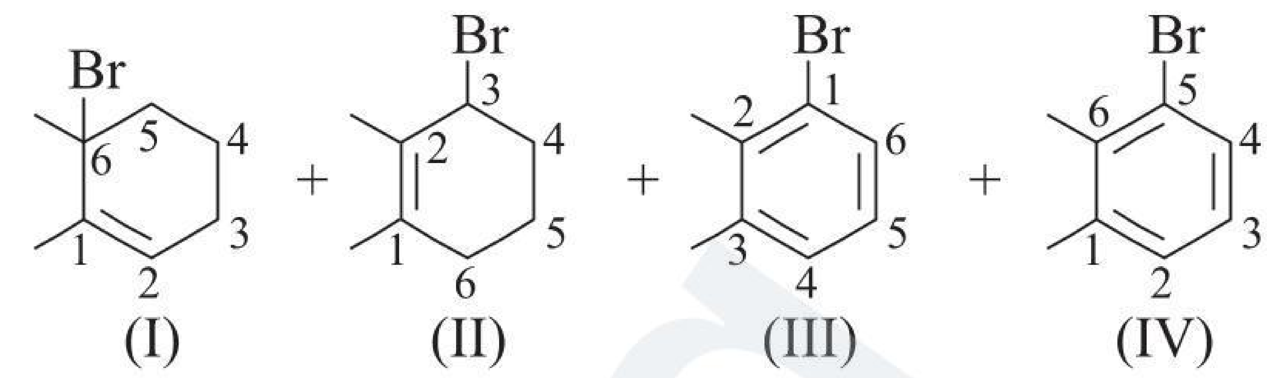
34. (4)

35. (3)

36. (2)

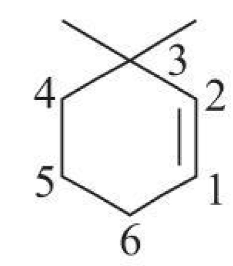
37. (4) (Homologues contain same functional groups with less or more C-atoms).

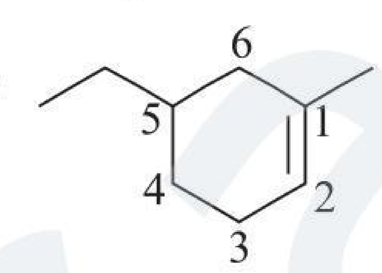
38. (3)

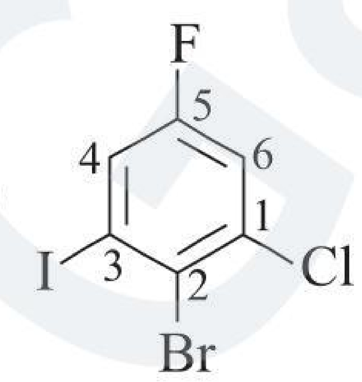


(6 – Br) + (3 – Br) + (1 – Br) + (5 – Br) = 15

39. (3) Correct name

 3, 3-Dimethylcyclohexene

40. (2) 

41. (2) 

42. (2)

i. and **iv.** have both aldehyde and ether groups.

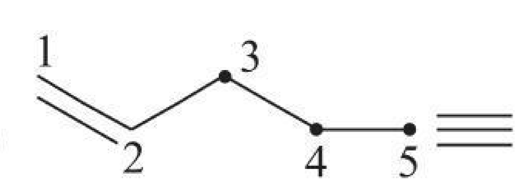
ii. have keto and ether groups.

iii. have diketo groups

v. have aldehyde and ester groups

43. (4)

44. (4)

45. (2)  ≡CH (Hex-1-en-5-yne)

46. (4)

Multiple Correct Answers Type

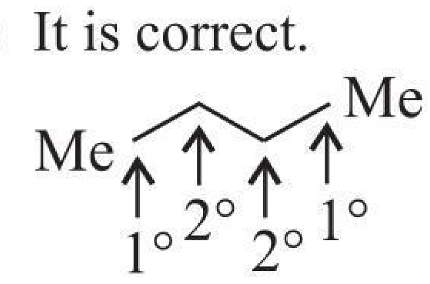
1. (1, 2, 3)

(1) The general formula of alkane is C_nH_{2n+2}.

(2) They have different physical properties, but same chemical properties.

(3) International Union of Pure and Applied Chemistry.

(4) It is correct.



2. (1, 2, 3)

(1), (2), and (3) are self-explanatory.

(4) is wrong; alkyne consists of one triple-bond.

3. (1, 2, 3)

(1) It is the common name.

(2) It is a saturated compound since it does not have (C═C) or (C≡C) bonds.

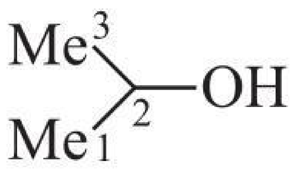
(3) They are used in trivial system.

(4) It is correct.

4. (3, 4)

(1) It is a saturated compound.

(2) It contains one 4° C atom.

(3)  (Propan-2-ol) (correct)

(4) $\overset{2}{C}H_3-\overset{1}{C}\equiv N$ (Ethane nitrile) (correct)

5. (1, 2, 3, 4)
All statements are self-explanatory.

6. (1, 2, 3, 4)
All statements are self-explanatory.

7. (1, 2, 3, 4)
All statements are self-explanatory.

8. (1, 2, 4)
(3) Vinyl ethanoate.

9. (1, 2, 3, 4)

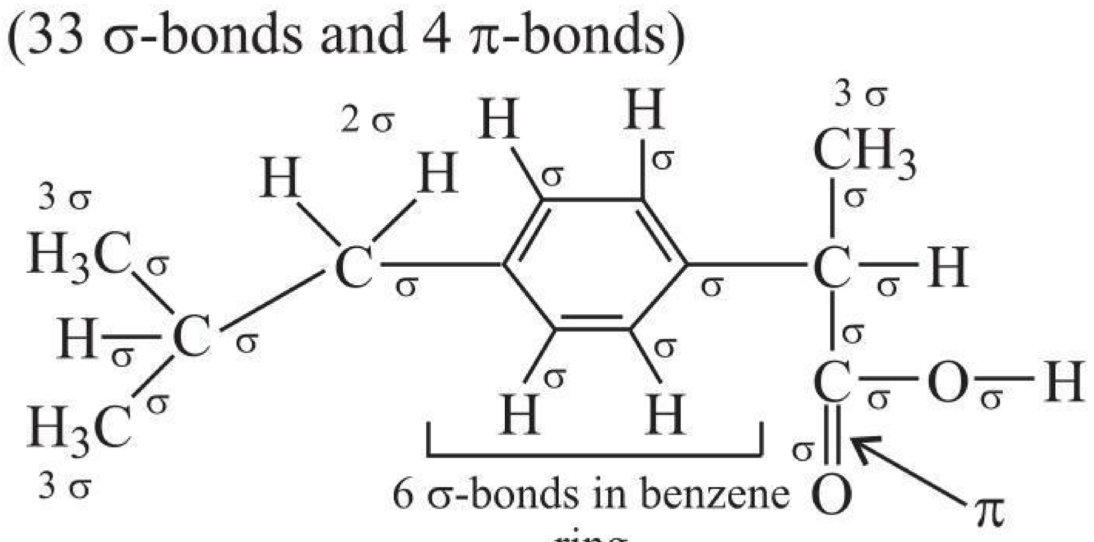
10. (1, 2, 3, 4)

Linked Comprehension Type

Paragraph 1

1. (2) **2. (2)** **3. (3)** **4. (4)**

(33 σ-bonds and 4 π-bonds)

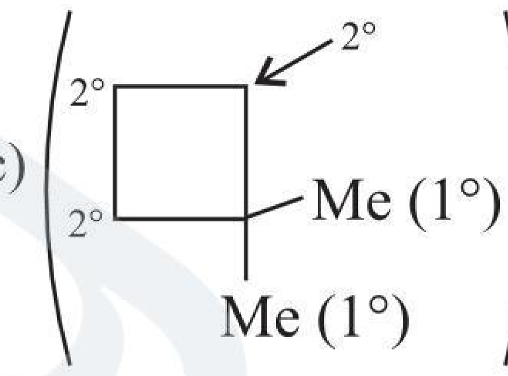


Matrix Match Type

1. (a → q; b → r; c → s; d → p)

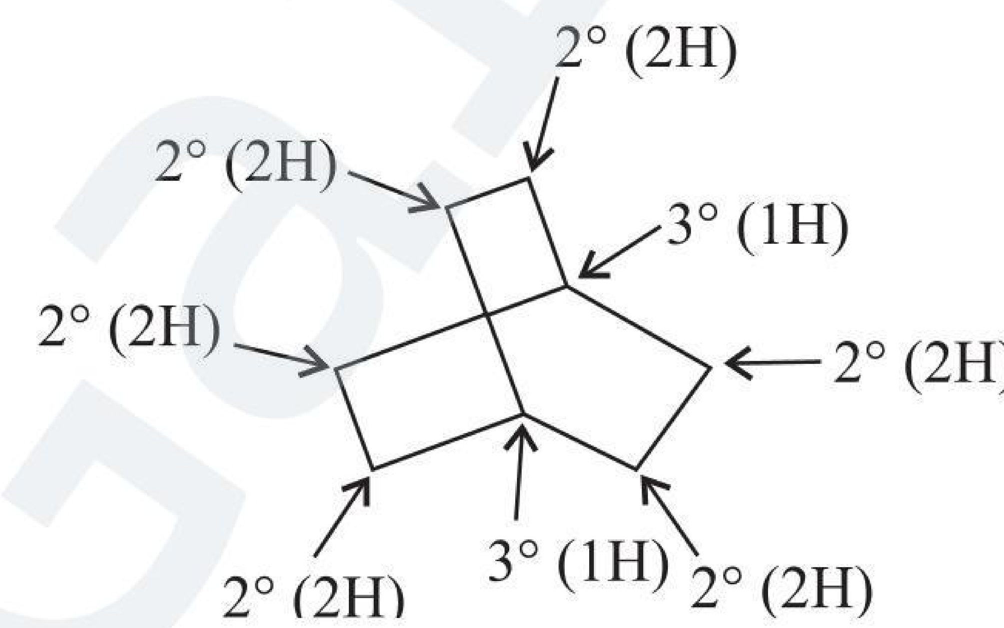
(a → q) C_8H_{18}, saturated alkane.

(b → r) C_6H_{12} (1 D.U. means alkene or cyclic.) It can be only (r).

(c → s) C_6H_{12} (1 D.U., cyclic) 

(d → p) C_8H_{14} (1 D.U., cyclic)

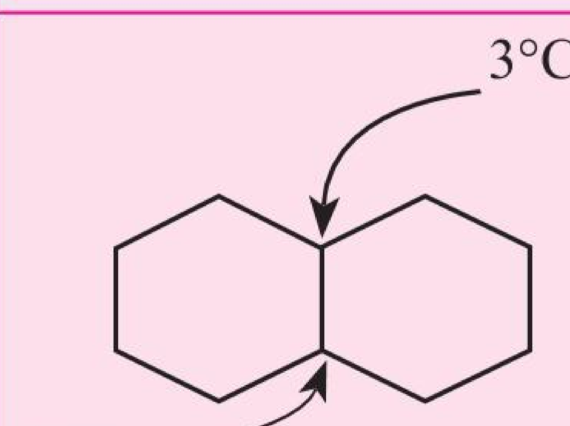
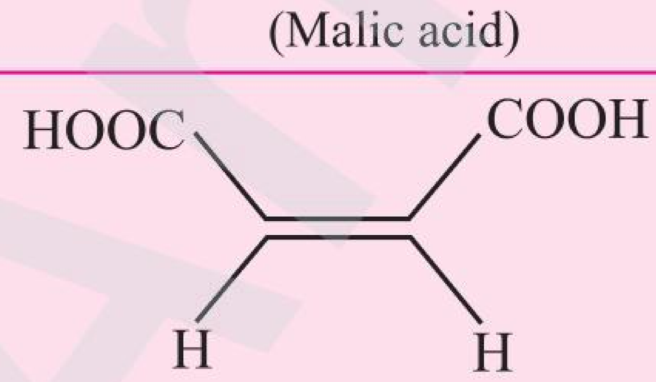
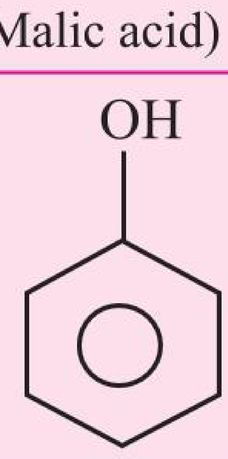
2°(12H) ; 3° (2H)



2. (a → r, q; b → p; c → s)
All are self-explanatory.

3. (a → r; b → p; c → t; d → q; e → s)

4. (a → ii, p; b → iii, iv, r; c → iii, iv, s; d → i, q)

	Compound	Explanation
a	 , Bicyclo [4.4.0] decane (Decalin)	Bicyclo, Two 3°C-atom
b	$HOOC-CH_2-CH(OH)-COOH$, (Malic acid)	4 – C, dibasic saturated acid, contains alcoholic group
c	 (Malic acid)	Cis, 4 – C, dibasic, unsaturated, decolourises Br_2 solution.
d		Phenol is carbolic acid, aromatic. Gives colour with neutral $FeCl_3$ solution (test for phenolic compound)

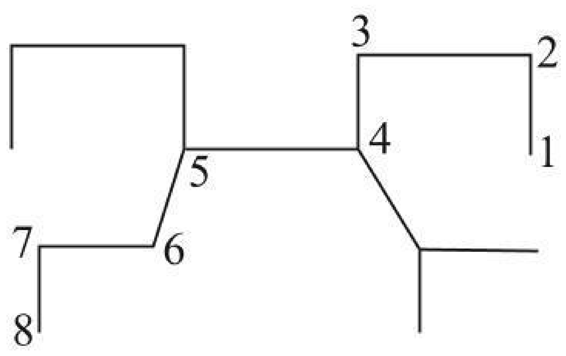
5. (a → iv, q; b → iii, s; c → ii, p; d → i, r)

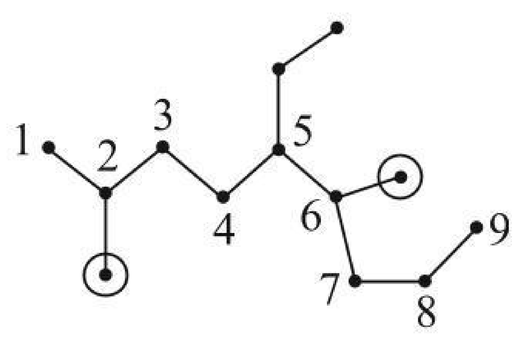
Numerical Value Type

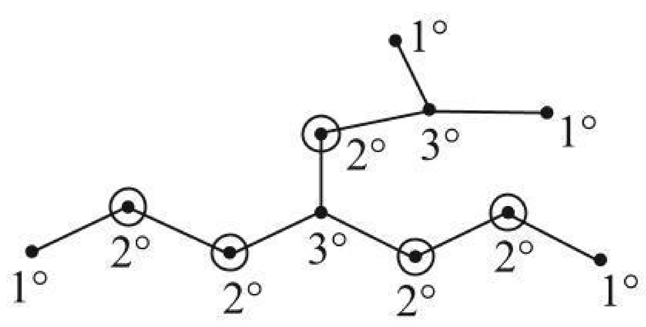
1. (4)

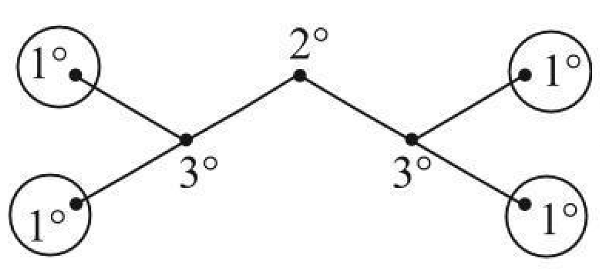
i. Acid (–COOH), **ii.** Keto ($>\!=\!0$),

iii. Aldehyde $\left(\begin{smallmatrix}R\\H\end{smallmatrix}\!\!>C{=}O\right)$, **iv.** Alcohol (–OH)

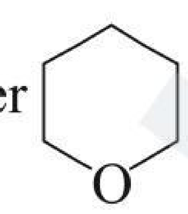
2. (8)  (8–C–atoms)

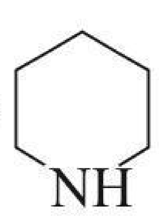
3. (2)  Two $-CH_3$ groups at 2 and 6-positions

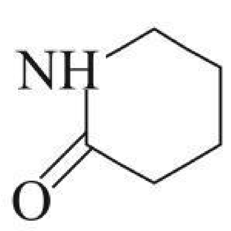
4. (5)  Give 2° carbon as shown in circle.

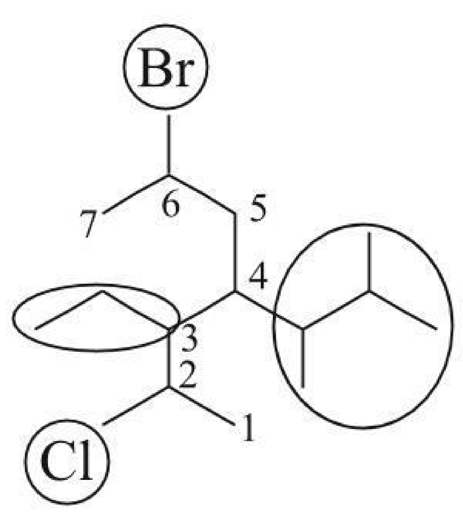
5. (4)  Four 1° carbon as shown in circle.

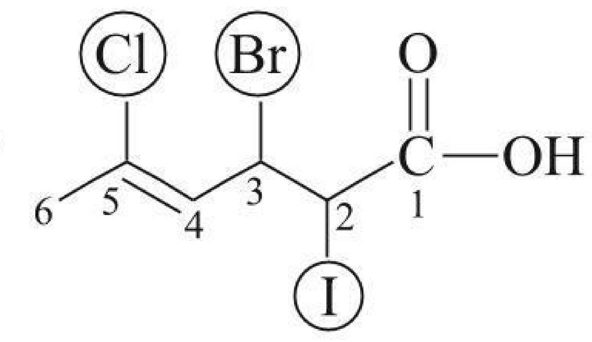
6. (5)

i. Acid (–COOH) **ii.** Cyclic ether 

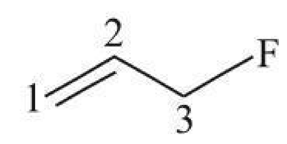
iii. Cyclic 2° amine  **iv.** Amide $\left(-\overset{\displaystyle O}{\overset{\|}{C}}-NH_2\right)$

v.  (Lactam) (cyclic imide)

7. (4)  Four substituents as shown in circle.

8. (3)  Three substituents as shown in circle. (–COOH) group are functional group.

9. (0) All have different IUPAC name.

10. (4) 

Double bond is in 1-position, $\therefore x = 1$.

Substituent (–F group) is present at 3-position, $\therefore y = 3$.

$\therefore x + y = 1 + 3 = 4$

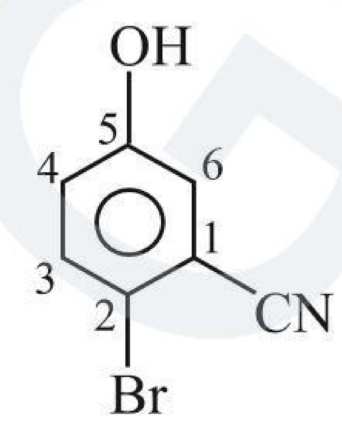
Archives

JEE Advanced

Single Correct Answer Type

1. (2)

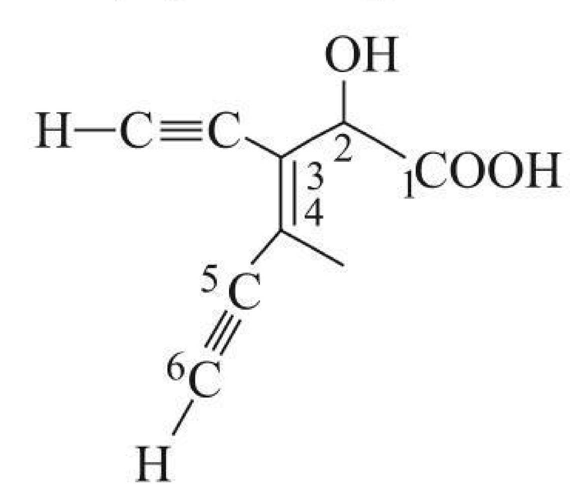
(CN) group is given the highest priority.

 (2-Bromo-5-hydroxy benzonitrile)

2. (4) In (1), Ethynyl is at 4 position.

In (2), at 4 – position there is ethyl.

In (3), there is ethynyl at 4 – position.



3-ethynyl-2-hydroxy-4-methyl-hex-3-en-5-ynoic acid.

Multiple Correct Answers Type

1. (1, 2, 3)